Norwegian Driving license theory

Contents

Chapter 1- The Driving License

Introduction

In the first chapter we will look closer at the rules for practice driving, the requirements for obtaining a driving license and what the different categories of licenses will allow you to drive.

The student

The most important rules regarding practice driving with a car:

You must be at least 16 years old.

You must have completed the basic traffic course.

You must bring photo ID.

You must be attentive, careful, sober and alert.

If you violate any of these rules the police will give you a practice driving suspension and you will have to postpone any further practice driving.

You do not have to take the basic traffic course if:

You already have a driving licence in another class. Remember to bring your driving licence as proof.

You are 25 or older. Then you only need the first aid part and the driving in the dark part.

You can start practice driving the day you turn 16.

Supervising driver

The most important rules regarding the supervising driver:

The supervising driver should be 25 years old or older and have held a category B driving licence continuously for the past 5 years.

The supervising driver should be attentive, careful, sober and alert.

Who is responsible?

The person supervising you is legally deemed responsible for the driving.

Your supervising driver must have held a category B driving licence continuously for the past 5 years.

The car

The most important rules regarding the car:

The car must be equipped with a large red letter L on white background in the back. This sticker should be removed when the car is not used for practice driving.

The car should be equipped with an extra rear-view mirror, so that the person supervising may easily see what is happening behind the car.

Where can you practice?

Practice driving must not endanger or unnecessarily inconvenience other road users. The police may prohibit or restrict practice driving in certain areas or at certain times on account of traffic conditions.

Remember to remove the red L when you are finished practice driving.

Conditions for getting a driving license

Age

You can take the theory test up to 6 months before turning 18

You can take the practical driving test once you have turned 18.

Health

When you apply for a driving licence you will have to submit a self-completed health certificate stating that you meet the health- and mobility requirements.

In addition you will have to pass a simple vision test conducted at the traffic station.

Mandatory training

You have to complete all mandatory training before the practical driving test. (The exception is the training for driving in the dark, if you take the practical driving test between March 16th and October 31st.

You have to apply for a driving licence before you take the theory test. This can be done on Statens vegvesen's web page.

If you drive a car with automatic transmission on your practical driving test you will not be allowed to drive a car with manual transmission.

Practical driving test during the summer.

If you haven't completed the training for driving in the dark before get your licence, you will have to complete the training before January 31th the following year.

Suspension period

A suspension period means you will not be able to obtain a driving licence in that period. You may be placed under suspension if you commit a crime (e.g. illegal practice driving.)

The most important rules regarding suspension periods:

If you are not yet 18 years old, the suspension period begins to run when you turn 18.

Driving when a suspension period has incurred is not allowed unless you get a special permission from the police

Example

If you get a 6 month suspension period when you are 16 years old, you can't practice drive or take the practical driving test before you are at least 18 years and 6 months old.

Probation period

When you get your driving license for the first time you will be under a two year probation period. If your license is revoked during the two year period you will have to complete both the theory test and the practical driving exam again. In addition you will get another two year probation period once your new license has been issued.

During the probation period you will receive twice as many penalty points for traffic transgressions.

Chapter 2- The Car

Introduction

As a driver, it is your responsibility to make sure the car you are driving is safe. You therefore need to be familiar with the different parts of the car and their functions, the safety systems present in most cars today, and how to perform necessary safety checks of the car.

The rules concerning driving with a trailer will also be explained in this chapter.

Vehicle condition

As stated in the Road Traffic Act, you are responsible for ensuring that your car complies with all vehicle regulations and is safe to drive.

According to regulation

A car that is in accordance with regulations, complies with all vehicle regulations and standards. This chapter will go through which of these you are expected to check yourself.

Safe to drive

A car that is safe to drive should not pose any danger to the driver, passengers or other road users. If the car is not safe to drive it may not be used, and you will need to pull over and stop the car if you are driving.

Examples

While driving the light above the license plate stops working. The car is no longer according to regulation, but is still considered safe to drive. You are therefore not required to stop and fix it.

While driving in the dark the rear lights (tail lights) stop working. As other road users will not be able to see you, the car is no longer considered safe. You will therefore need to pull over at the earliest opportunity and fix the light, or request assistance.

Safety checks

There are several safety checks you can perform to inspect the different parts of the car – these will be explained in more detail throughout the chapter.

In general, you should always take a closer look or contact a mechanic if the car's handling characteristics change, or if you notice any unfamiliar noises while driving. These noises often indicate that a part needs to be repaired or replaced.

Dashboard instruments and warning lights

To make it easier to discover faults with the car, all cars today are equipped with instruments and warning lights that notify the driver if anything is wrong.

The instruments on the dashboard show your speed, RPM, fuel level, etc. — information of relevance to your driving.

In addition there are several warning lights that light up when something is not working properly. It is important that you are familiar with what these warning lights indicate, and how to respond when they light up either when starting the car or while driving.

The warning lights will differ in design and colour from car to car, so find out what the different lights in your car indicate by reading the owner's manual.

All the warning lights light up

If the warning lights are functioning properly, they will light up for a few seconds when starting the engine, and then turn off again automatically.

Brake pedal and parking brake

All cars are equipped with a brake pedal and a parking brake.

Brake pedal

The brake pedal must comply with certain requirements:

The brakes should apply fully on the first press.

The harder you push down on the brake pedal, the more braking force should be applied. Full braking force should be reached before the pedal is depressed 2/3 of the way.

The brake pedal should withstand a force of at least 100 kp (approximately 100 kg).

To check that your brake pedal can withstand the required force, depress the pedal as far as possible for five seconds while the car is standing still. The pedal should not give out. If the pedal gives out when performing a pressure test, this indicates a brake fluid leak.

Parking brake/hand brake

As a driver, you are responsible for ensuring that the car does not move after you have left it. The parking brake (hand brake) prevents the car from rolling when parked. When the parking brake is applied, it should prevent the vehicle from moving when the engine starts, and keeps it stationary when standing on a hill, even when you are not present.

Braking system

Brake circuits

The braking system consists of two independent brake circuits. If there is a fault with one of the brake circuits, you will then still have sufficient braking force to stop the car in a safe manner.

Brake booster

The brake booster makes it easier to reach full braking force when braking. If the brake booster is not working properly, the brakes are less effective, and the brake pedal will be harder to depress.

You can check that the brake booster is working by pumping the brake pedal 5–6 times until it becomes "hard". Depress the brake pedal firmly and hold it in while starting the engine. The pedal should drop slightly when the engine starts.

Driving safety

It is not safe to drive if the brakes are not functioning properly. If there is a fault with one of the brake circuits or the brake booster stops working, you must immediately pull over and stop the car.

Brake fluid

As mentioned above, if the brake pedal gives out when performing a pressure test, this indicates a brake fluid leak.

You can check the brake fluid level of the brake fluid reservoir, located under the hood of your car. The level in the reservoir should be between MIN and MAX. Low levels of brake fluid may be caused by worn brake pads or a leak in the braking system, and the braking system should be checked by a mechanic.

Brake force distribution

The brake system should apply more force to the front wheels, than the rear wheels. Brakes should be applied evenly to wheels on the same axle, i.e. both front wheels or both rear wheels, so the car does not pull to one side or swerve when braking.

You can check if the brakes are applied evenly by driving straight at a low speed (30–40 km/h) before pressing firmly on the brake while keeping a loose grip on the steering wheel.

The car should keep going straight. Uneven brake force distribution is usually caused by uneven wear on the brakes, and can be fixed by a mechanic.

Anti-lock braking system (ABS)

All cars today are equipped with so called anti-lock brakes. The anti-lock braking system (ABS) prevents the wheels from locking up when braking hard, meaning that the driver is still able to steer the car while braking. Note, however, that the braking distance is not affected by whether or not the car is equipped with ABS brakes.

If there is a problem with the ABS, the car is still safe to drive, but it is no longer according to regulation.

Warning lights for the brakes

Braking system

This light indicates that there is a fault in the braking system, and you should pull over immediately to check the brakes. On some cars the light may also indicate that the hand brake is applied. If the light stays on after the hand brake is released, there is most likely a problem with the braking system.

If the light turns on while driving, this may indicate a fault with the dual-circuit braking system, low brake fluid levels, a parking brake malfunction or a technical fault of some kind.

ABS

This light indicates that there is a fault in the anti-lock braking system. The brakes are still working, but you no longer have any guarantee that the brakes will not lock up under heavy braking.

Steering

For a car to be safe, the steering system must comply with the following requirements.

The car must be easy and safe to steer

All modern cars are equipped with power steering, which makes it easier to turn the steering wheel. If the power steering stops functioning, the car will be much heavier to steer, but it will still be safe to drive and you may continue driving. If the car is equipped with electromagnetic/electromechanical power steering, however, it is not safe to continue driving, as this may cause further damage to the steering system.

To check the power steering the engine must be turned off. By turning the steering wheel to one side and then starting the engine, you can feel how the power steering makes it easier to turn the wheel further.

To check the self-centering, simply check that the steering wheel returns to the centre position on its own after turning.

The car must be directionally stable

A directionally stable car moves in the direction that you steer it, without pulling to either side. If the car pulls to either side when driving straight, this may be caused by unequal air pressure in the tires or incorrect front wheel alignment. If the steering wheel starts shaking around 80–100 km/h, this indicates the front tyres are out of balance.

You can check that your car is directionally stable by driving straight at low speeds. Keep a loose grip on the steering wheel. If the car is directionally stable, it should go straight.

The steering wheel must be self-centering

This means that if you let go of the steering wheel when turning and the car is in motion, the wheels should automatically return to centre and seek to go straight.

To check the self-centering, simply check that the steering wheel returns to the centre position on its own after turning.

The wheels should start turning before you have turned the steering wheel 3 cm.

Steering wheel play refers to how much you can turn the steering wheel before the wheels start turning. Excessive play makes your steering less precise, and a maximum of 3 cm play is allowed. In other words, the wheels should start turning if you turn the steering wheel more than 3 cm to either side.

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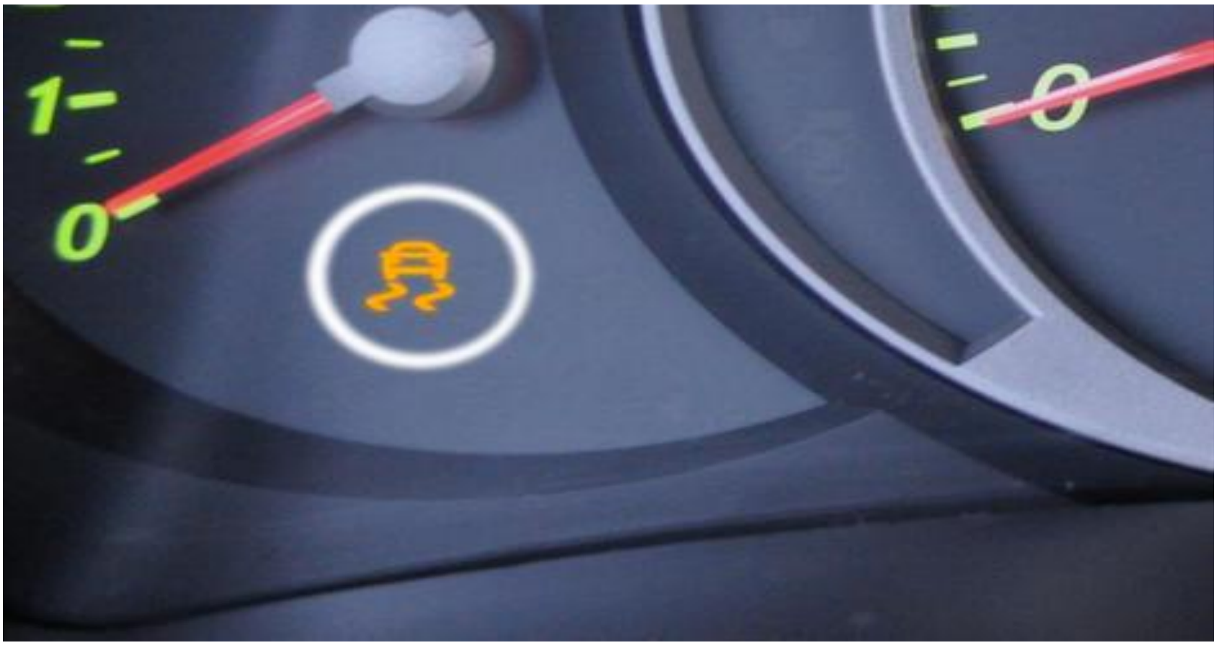
You can check the steering wheel play by turning the steering wheel when the car is standing still. You should not be able to turn it more than 3 cm before the wheels start turning.

Warning lights for the steering system

Electronic stability programme (ESP)

If this light flashes while driving, it means the ESP system is working to regain grip. If the light stays on constantly there is a fault with the ESP system, or ESP has been turned off.



ESP warning light.

Tyres

As a driver, you are responsible for ensuring the vehicle has sufficient grip under the current driving conditions

The tyres greatly affect your car's grip, and should be chosen to ensure the car is safe to drive under the current driving conditions. This may include the use of winter tyres, studded tyres and chains, but also means that you may use summer tyres in the winter, as long as conditions allow it. Conversely, if you are driving in winter conditions, on snowy or icy roads, your car must be equipped with winter tyres even if it is the middle of summer. All-season tyres are not recommended for Nordic driving conditions, and are not the same as studless winter tyres.

Tyre wear and damage

Check your tyres and rims for wear and damage. Typical damage includes cuts, bulges or lumps on the side of the tyre and buckled rims.

Uneven wear is another common problem. Excess wear in the middle or on both sides of the tread indicates incorrect air pressure. One-sided wear may be caused by wheel misalignment.

Tyre dimensions

By "tyre dimensions" we are referring to the size of the tyres. The vehicle registration certificate contains information about correct tyre dimensions and load capacity. Minimum load capacity is listed under "Min. Li", followed by a number.

To make sure your tyres are approved for your car, compare the dimensions and load capacity stated on the tyre with that stated in the vehicle registration certificate.

What do the different numbers mean?

Look at the tyre markings in the picture. The first part (225/45 R 17) indicates the tyre dimensions.

225 — the width (in millimetres) of the tyre at correct air pressure

45 — the height of the tyre as a percentage of the tyre width

R — radial (the alternative is D for diagonal)

17 — the diameter of the wheel rim in inches

The last part (94 T) indicates the load and speed rating of the tyre.

Air pressure

The air pressure in your tyres affects the grip. If the air pressure is too low or too high, less of the tyre's surface area is in contact with the road surface, and the risk of losing grip increases significantly.

Correct air pressure also contributes to an increase in fuel efficiency and more eco-friendly driving.

How do you check that the tyres have the correct air pressure?

You can find information regarding correct air pressure in the owner's manual of your car, or sometimes on a sticker inside the door jamb or the fuel door. Newer cars are equipped with an automatic tyre-pressure monitoring system, which alerts you if anything is wrong. Otherwise you can check the air pressure with a pressure gauge, or by checking for irregular tread wear:

Excessive wear on both sides of the tread indicates that the air pressure is too low.

Excessive wear in the middle of the tread indicates that the air pressure is too high.

Tread depth

Your tyres' tread depth affects your grip, and different tyres are better suited to certain surfaces. Take a closer look at your tyres to check the tread depth. You can also use a tread depth gauge or check the tyre's tread wear indicator.

If tread depth is not sufficient, grip will be poor and there is a higher risk of aquaplaning and increased braking distances. It is therefore not safe to drive with tyres without sufficient tread depth.

There are minimum tread depth requirements for tyres:

Summer tyres must have a tread depth of at least 1.6 mm. If summer tyres are used in the winter season (when studded tyres are permitted), tread depth must be at least 3 mm.

Winter tyres must have a tread depth of at least 3 mm.

Studded tyres

Because the negative environmental impact of studded tyres is larger than that of non-studded tyres, they are only permitted in the winter season. They may, however, be used outside this season, if necessary under the driving conditions.

When are studded tyres permitted?

Northern Norway: From 16 October to 30 April.

The rest of the country: From 1 November to the first Sunday after Easter Monday.

Note!

If your car is equipped with studded tyres, all tyres must be studded.

Lights

Headlights and fog lights

In Norway running lights are mandatory at all times, also during the day. Some cars are equipped with separate running lights, but you may also use low beam (dipped) headlights or fog lights (if conditions require it) as running lights.

Low beam (dipped) headlights

Low beam lights must be white or yellow, and illuminate the road at least 40 metres ahead. The right-hand side light should reach slightly farther ahead than the left. If your low beams are not adjusted properly, you risk blinding oncoming traffic and visibility will be reduced in the dark.

You can check that your low beam lights are working and adjusted properly by turning them on and driving towards a wall.

Main beam/high beam headlights

The main beam lights must be white or yellow, and illuminate the road at least 100 metres ahead. As the main beam lights are strong and may easily blind other road users, you will need to switch to low beam lights when approaching another vehicle.

Curve/fog lights

Curve/fog lights must be white or yellow. Just like with main beam lights, incorrect use of fog lights may blind oncoming traffic. It is prohibited to use fog lights in conjunction with low beam lights, but fog lights may be used as running lights if required by the driving conditions.

To check the cuve/fog lights, turn them on and perform a visual inspection of the front and rear of the car.

Turn signals/hazard warning lights

Turn signals (direction indicator lights)

The car must be equipped with orange direction indicator lights on all four corners.

Hazard warning lights

When the hazard warning lights are activated, this means all direction indicator lights are flashing at the same time. Hazard warning lights should only be used in case of an emergency stop to alert other road users of your presence, or if the traffic situation requires extra attention (i.e. if a queue suddenly forms).

If using hazard warning lights when stopped, remember to also turn on your parking lights to be seen without blinding other road users.

To check the hazard warning lights, activate the lights and perform a visual inspection of all lamps.

Brake lights and tail lights.

Brake lights (stop lamps).

Brake lights must be red.

To check the brake lights, perform a visual inspection of both left and right hand side lamps, as well as high mount brake lights. Ask someone to check for you while you depress the brake pedal.

It is not safe to drive if your brake lights are not working.

Tail lights (rear position lamps).

Tail lights must be red.

To check the tail lights, start the car and perform a visual inspection. Tail lights should turn on automatically when starting the car.

It is not safe to drive in the dark if your tail lights are not working.

Parking lights and registration plate lights

Parking lights

The parking lights must be white and are always lit when headlights, tail lights, registration plate lights or fog lights are lit.

It is important that parking lights are functioning properly when stopped in the dark (low beam lights would blind other road users and make it more difficult to see what is happening).

To check the parking lights, start the car and perform a visual inspection. Parking lights should turn on automatically when starting the car.

Registration plate lights

The registration plate light must be white, and automatically illuminates the licence registration plates when the engine is running.

To check the registration plate lights, start the car and perform a visual inspection. Registration plate lights should turn on automatically when starting the car.

Reversing lights

Reversing lights are common on newer cars, but are not a requirement. If the car is equipped with reversing lights, they must be either white or yellow.

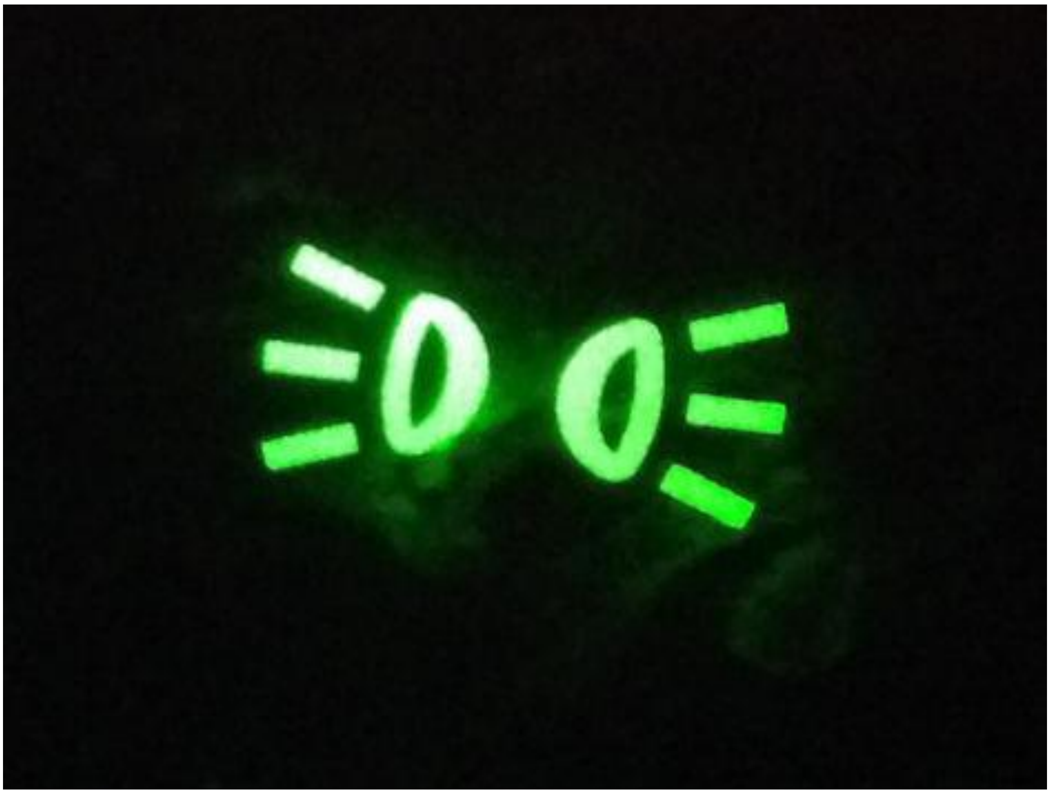
Warning lights

Main beam lights

This light indicates that the main beam lights are on.

Parking lights

This light indicates that the parking lights are on.



Visibility

As a driver, you must always maintain sufficient visibility in all directions. Snow, ice, dew, condensation, stickers, etc. must not restrict visibility or in any way interfere with driving.

Windscreen/windshield wipers

Check your windscreen wipers by turning them on and ensuring they work as intended at different speed settings. Also check the wipers for wear and tear, hardened or loose rubber, deformed wiper blades, lines or bands on the windscreen, and that the blades are pressing properly on the screen.

Windscreen/windshield washers

Turn on the windscreen washers to check that they are working properly, and that there is enough washer fluid. Locate the windscreen washer reservoir under the hood to check the washer fluid level.

Ice and dew

Brush any snow off and scrape all windows clear of ice before driving. To clear ice and dew from the inside of the windscreen, turn the heating and defroster all the way up. To ensure visibility in all directions, turn on the defroster for the rear and side windows, as well as side-view mirrors, if the car is equipped with this.

Engine

The engine powers the car by utilizing energy from fuel or electricity.

Faults and damages in the engine are often expensive to repair, so it is important that you are familiar with the relevant warning lights and what these indicate.

Even though low oil levels are indicated by a warning light on the dashboard, it is still a good idea to regularly check the levels yourself. Open the hood, locate the dipstick, and check if oil levels are between the MIN and MAX markings on the dipstick. Some cars are equipped with systems that check oil levels automatically, and notify the driver if levels become too low.

Battery

The car battery supplies electricity to the electrical equipment in the car, such as lights, windows, heating, air conditioning, windscreen wipers, etc.

The battery also supplies electricity to the engine, and especially the ignition system.

While driving, the dynamo or alternator charges the battery so it doesn't run down. If the battery is flat, you will not be able to start the car without assistance and booster/jumper cables. To avoid damage, follow the instructions for your booster cables carefully.

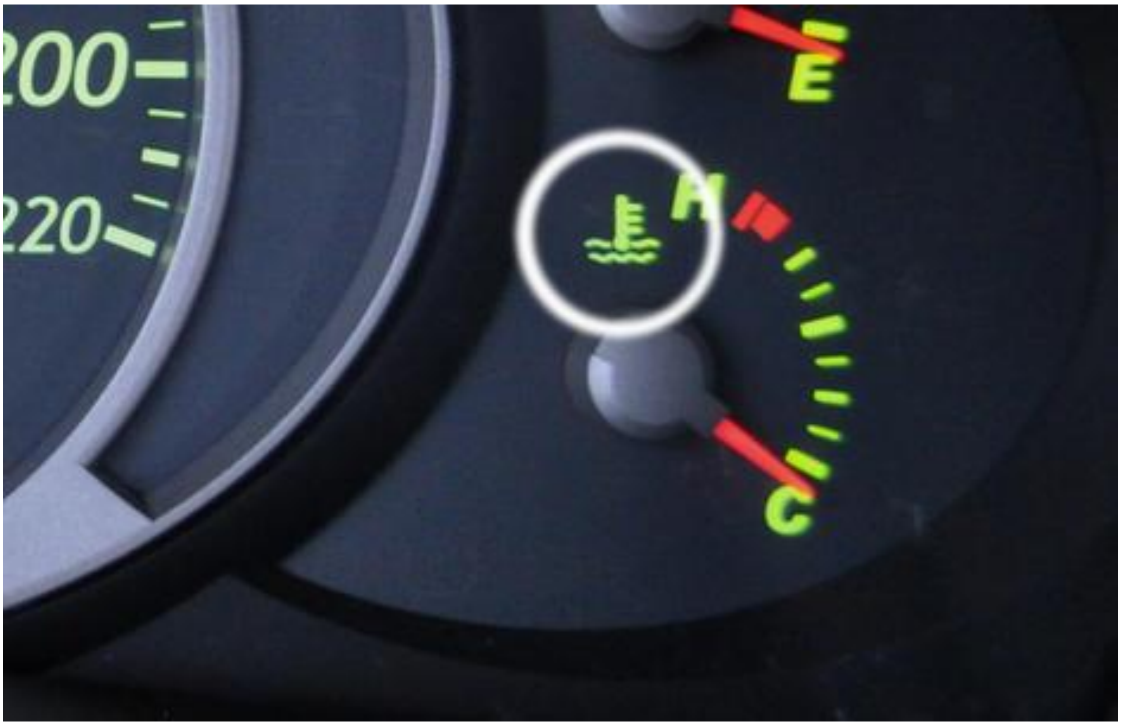
Checking the battery

To check that the battery is securely fastened, open the hood and try moving it. If the battery stays put, it is fastened properly. If the battery is not firmly secured, this may lead to leaks and short-circuits.

Warning lights for engine and battery

Engine temperature

The engine temperature gauge, or indicator warning light, shows you if the engine temperature becomes too high. When the light turns on, or the gauge shows high temperatures, you must immediately stop the car and check the coolant/antifreeze level. If you continue driving an engine that is overheating, it can cause serious damage to your car.



Engine temperature gauge.

Oil pressure

Turn off the engine immediately if this light turns on while driving. If the cause is low oil levels, top up the oil and continue driving. If it is not caused by low oil levels, the car must be taken to a mechanic right away.

Oil pressure warning light.

Battery charge

This warning light indicates that the battery is not charging properly. When the battery runs down, you will no longer be able to start the car without assistance.

Safety

Active and passive safety

When talking about driving safety, we often distinguish between active and passive safety systems.

Active safety

Active safety systems are safety features that actively help avoid accidents. ABS brakes and ESP (electronic stability programme) are examples of active safety systems common in cars today.

Passive safety

Passive safety systems are safety features designed to reduce the effects of an accident that is to protect the vehicle, driver and passengers in the case of an accident. Airbags and seat belts are examples of passive safety systems common in cars today.

Safety equipment

Mandatory safety equipment

All cars are required to be equipped with at least one approved reflective vest and hazard warning triangle.

Reflective vest

The reflective vest should be stored so that it is easily accessible from the driver's seat, i.e. under the seat, in the glove compartment, or in the side door.

In case of an emergency stop, put the vest on before exiting the vehicle. This also applies to any other passengers exiting the car, and it is therefore advisable to keep more than one vest in the car.

Hazard warning triangle

When you need to use the warning triangle, follow these instructions:

Assemble the warning triangle before leaving the car. Remember to put on a reflective vest before exiting the vehicle.

Walk with the reflective side of the warning triangle towards oncoming traffic, so they spot you easily.

Place the warning triangle so that it is clearly visible to other road users. Consider your surroundings to decide the suitable distance for placing the warning triangle when in a densely populated area. Outside of densely populated areas the warning triangle should be placed 100-200 meters behind the car.

Seat belts

It is mandatory for all passengers to wear seat belts if the car is equipped with them, and both driver and passengers may be penalized for not using seat belts. Newer cars also have a warning light and reminder alarm to ensure everyone remembers to fasten their seat belts.

Check the seat belts for wear, tear and damages, and make sure the seat belt tensioner is functioning properly (or the belt will be too loose, and not tighten properly in case of an accident).

The airbag warning light turns on if there is a problem with one of the airbags.

Trailer, weight and load

Driving licence limitations

With a category B licence, you are in general allowed to drive cars with a maximum authorized mass not exceeding 3 500 kg. In addition to this, you are permitted to tow a trailer following the rules outline below.

You are permitted to tow

- a trailer with a maximum authorised mass not exceeding 750 kg — i.e. so that the maximum authorised mass of the car and trailer combined does not exceed 4 250 kg
- a trailer with a maximum authorised mass exceeding 750 kg, as long as the maximum authorised mass of the car and trailer combined does not exceed 3 500 kg

Heavier trailers

Category B code 96: To pull a trailer with a maximum authorised mass exceeding 750 kg, where the maximum authorised mass of the car and trailer combined is between 3 500 kg and 4 250 kg, you need to complete an obligatory course. The course is 7 hours and you will receive a code 96 on your licence upon completion. There is no test to complete the course.

Category BE: To drive a vehicle-trailer combination where the maximum authorised mass of the car and trailer combined is between 4 250 kg and 7 500 kg, you need a category BE licence.

Car and trailer limitations

When driving with a fully loaded vehicle or towing a trailer, you first need to check if the car and trailer can carry the weight. This is stated in the vehicle registration certificate. Be sure to check the vehicle registration certificate of both car and trailer, and if they differ the strictest limitation (lowest weight) applies.

Even if the trailer does not exceed the limit for maximum authorised mass, you still need to check that the actual total weight of the trailer does not exceed what the car is registered to tow. This information is stated in the vehicle registration certificate, under permissible gross trailer weight ('tillatt hengervekt') and permissible gross train weight ('tillatt vogntogvekt'). You will need to know whether the trailer is equipped with brakes or not.

The vehicle registration certificate contains information regarding

- how many people the car is registered to carry (as the driver, it is your responsibility that the car is not used with more people in it than it is registered to carry)
- maximum permissible payload ('nyttelast')
- maximum authorised mass ('tillatt totalvekt')
- permissible gross trailer weight ('tillatt hengervekt')
- permissible gross train weight ('tillatt vogntogvekt')
- maximum tongue weight ('koblingslast')

When towing a trailer

- you may not drive faster than 80 km/h, even if the speed limit is higher
- you may not drive faster than 60 km/h if the trailer is not equipped with brakes and the actual total weight is 300 kg or more

Trailer requirements

In addition to the weight limitations, the trailer must also meet certain technical requirements.

Trailer requirements

- The trailer must be fitted with at least two red, triangular reflectors on the back
- All lights on the trailer must be in working condition
- It is permitted to tow a trailer without brakes, but the permissible trailer weight may be lower, and if the actual total weight of the trailer exceeds 300 kg you may not drive faster than 60 km/h

Load securing

As the driver, it is your responsibility that any load is properly secured and placed in a manner that will not impair safety.

- The load must be secured in such a manner that it does not cause any damage, danger or unnecessary noise, is not dragged along the road or in danger of falling off. The same applies to chains, ropes, etc. used to secure the load
- The load must be placed in such a manner that you maintain sufficient visibility and that it does not prevent you from manoeuvring the car safely
- Neither mandatory lights nor licence plates may be obscured by the load
- Loads overhanging the sides of the vehicle, must not project more than 15 cm
- Loads overhanging the front of the vehicle, must not project more than 1 meter
- The load must be secured in such a way that it will not shift or fall off during transport. The load securing must be able to withstand a force equivalent to the entire weight of the load in the forward direction, and half the weight of the load to the side and rear

Overhanging loads

In general it is Loads overhanging the front of the car or projecting more than 1 m from the rear of the car must be clearly marked at the outer ends.

The marking must

- be alternating red and white in colour, and the red material must be light-reflecting
- have a visible surface area of at least 250 cm² from all directions
- be placed so that it is clearly visible to other road users

If it is dark or if visibility is otherwise restricted, loads projecting more than 1 metre from the rear of the car must also be marked with red lights visible from behind and from the sides. The lights should be clearly visible from a distance of at least 150 meters, but not so bright as to be blinding for other road users.

Occasional transport

If you regularly (e.g. for work) drive with overhanging loads, the requirements are stricter than for someone who on occasion transports timber from the home improvement store. If you are driving with overhanging loads and do not have the proper equipment, you can use other ways of marking the load, such as a coloured plastic bag or any other clearly visible item.

Transporting bicycles

If you are transporting a bicycle mounted sideways on a car with a width of 180 cm or less, the bicycle must be mounted in such a way that the total width of the vehicle (including the bicycle) is still no more than 180 cm. If the bicycle projects out from the sides of the car, it should be mounted approximately in the middle of the car so that it projects equally on both sides.

If the bicycle (rack) obscures the car's lights or licence plate, you must use a lighting board marked with the car's vehicle registration number. The vehicle registration number should be written in black, be of about the same size as on the licence plate, on a white or aluminium-coloured background.

The lighting board must be equipped with tail lights, brake lights, turn signals and reflectors.

Loose objects

If you need to perform an emergency stop at high speeds, an unfortunately placed soda can easily become a deadly projectile. For example, according to research conducted by NAF (the Norwegian Automobile Federation), driving at 50 km/h a thermos could increase in weight from 1.5 kg to 45 kg in a collision! You should therefore be very careful to safely secure all loose objects in the car.

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If it is dark or if visibility is otherwise restricted, loads projecting more than 1 metre from the rear of the car must also be marked with red lights visible from behind and from the sides. The lights should be clearly visible from a distance of at least 150 metres, but not so bright as to be blinding for other road users.

Occasional transport

If you regularly (e.g. for work) drive with overhanging loads, the requirements are stricter than for someone who on occasion transports timber from the home improvement store. If you are driving with overhanging loads and do not have the proper equipment, you can use other ways of marking the load, such as a coloured plastic bag or any other clearly visible item.

Transporting bicycles

If you are transporting a bicycle mounted sideways on a car with a width of 180 cm or less, the bicycle must be mounted in such a way that the total width of the vehicle (including the bicycle) is still no more than 180 cm. If the bicycle projects out from the sides of the car, it should be mounted approximately in the middle of the car so that it projects equally on both sides.

If the bicycle (rack) obscures the car's lights or licence plate, you must use a lighting board marked with the car's vehicle registration number. The vehicle registration number should be written in black, be of about the same size as on the licence plate, on a white or aluminium-coloured background.

The lighting board must be equipped with tail lights, brake lights, turn signals and reflectors.

Overload

Driving with a trailer, or a heavily loaded car, affects the car's handling. If you load your car or trailer beyond the limits stated in the vehicle registration certificate, the car may become dangerous to drive.

What are the consequences of overloading the car?

- The car is more difficult to handle
- Braking distance increases
- The headlights will point further up, which may blind other road users.

What are the consequences of overloading the trailer?

- The front wheels will have fewer grips
- Braking distance increases
- Directional stability decreases
- The headlights will point further up, which may blind other road users

Summary

Highlights from chapter 2

- A car that is in accordance with regulations, complies with all vehicle regulations and standards
- A car that is safe to drive should not pose any danger to the driver, passengers or other road users. If the car is not safe to drive it may not be used, and you will need to pull over and stop the car if you are driving
- You are permitted to tow a trailer with a maximum authorised mass not exceeding 750 kg (as long as the maximum authorised mass of the car and trailer combined does not exceed 4 250 kg)
- As long as the maximum authorised mass of the car and trailer combined does not exceed 3 500 kg, you may also tow trailers with maximum authorised mass above 750 kg
- The car and trailer must also be approved for the weight. Make sure the trailer's actual total weight does not exceed the permissible gross trailer weight stated in the car's vehicle registration certificate. You will need to know whether the trailer is equipped with brakes or not
- As the driver, you are responsible for ensuring all loads are securely fastened and placed in a manner that will not impair safety
- As you may get questions about different technical requirements on your theory test, you will need to learn them. On your practical driving test you will also be asked to demonstrate some of the safety checks

Chapter 3- Human

Introduction

Our physical and mental capabilities and our limitations dictate how we drive. In this chapter we will take a closer look at the challenges we face in everyday traffic and how our senses and thought patterns influence our ability to drive in a safe and sound manner.

Traffic situations and the driving process

A traffic situation is an important concept in driver training. A traffic situation consists of you and your vehicle, other road users, the road, the road rules and road signs, visibility and driving conditions, and the interaction between these factors.

Traffic situations vary from simple ones to highly complex ones. In a complex traffic situation a lot happens which you need to relate to and your brain has to process large amounts of information.

We call this process for the driving process. The driving process describes what you as a driver do and think in the traffic, from a situation occurs to your reaction and response to this situation.

Sensing – Your eyes receive information, but your brain hasn't processed it yet.

Perceiving – Your brain recognizes what your eyes are sensing and you understand what is happening.

Deciding – You evaluate different actions and decide on what course of action you should take.

Acting – You act.

Driving experience = a lower accident risk

One of the main reasons for why experienced drivers are involved in fewer accidents than younger drivers is because they have experienced many different traffic situations over the years. This makes it easier to recognize dangerous situations and it is easier to choose a safer course of action. Experienced drivers have a better set of rehearsed and automated skills which allows them to better concentrate on observation.

The driving process

Sensing

When you are driving you have to focus your attention to what goes on around you in order to spot potential dangers early and plan you driving accordingly. You constantly receive sensory input from your surroundings and your sense of sight is your primary sense, constituting almost 90 % of the input.

Central vision

- directed forwards, with a field of view between 3 and 5 degrees
- perceives details
- works best in well-lit conditions

Peripheral vision

- directed sideward, with a field of view of about 180 degrees
- perceives movement and changing light conditions
- perceives surroundings, shadows and silhouettes

Fixation (to focus on a single location)

- Fixating on a single location takes 2-3 tenths of a second
- When fixating on things not related to your driving, you remove focus from the driving situation. When this happens you will move several meters before you regain your attention to the traffic situation

Perceiving

Perceiving means interpreting and understanding what you are sensing in a traffic situation

In order to interpret and understand you need prior knowledge and acquiring experience. Experience makes it easier for you to recognize certain situations and helps you predict the outcome faster and what actions you have to take.

Deciding

When you have perceived the traffic situation your brain can start deciding on a course of action.

You judge the situation based on prior experience and have to judge different options and what you think the outcome might be.

You decide on what course of action you want to take.

Deciding on a course of action your prior experience will help you make simpler and more efficient choices.

Acting

When you have decided on what actions you will take, the next step is executing your actions.

Sometimes the execution will be simple and other times it will be complex. If something unexpected happens when you're driving at high speeds or you're driving in bad weather conditions it will be difficult to act appropriately in time.

In order to become an accomplished driver you need to practice automating your skills. When you repeat the same action over time, for example changing gears or stopping the car, you will eventually learn to do this without thinking about it. You will then need less time and concentration on the actions themselves and free up capacity to pay attention to the traffic and avoid dangerous situations.

Be aware that cars often handle differently from model to model. If your car has ABS brakes, you will be able to steer the car even if you apply full pressure to the brakes. If your car on the other hand is not fitted with ABS brakes, you will not be able to steer the car when braking hard.

Weight, engine power and steering system will also affect how your car responds to your actions. A heavy car will have a longer braking distance than a light car. You need to consider factors like these when you're driving.

Assessing speed and distance

Research has shown that we have a tendency to judge the distance to an oncoming vehicle as greater than it actually is. This is something you have to take into consideration when you are planning on overtaking traffic or taking a turn to the left in an intersection.

We also have a tendency to perceive the speed of oncoming traffic as slower than it actually is. On distances of 300 meters or more we find it difficult to tell a slow car from a car driving at higher speeds.

Remember!

We think oncoming traffic is farther away than it actually is.

We think oncoming traffic is driving slower than it actually is.

It is especially difficult to judge the speed and distance of an oncoming motorcycle (because motorcycles are smaller).

Objects in the rear view mirror are actually closer than they appear to be.

Your own speed

Speed blindness is what you are experiencing when you're driving faster than you think you are. Misjudging your own speed will give you considerably less time to observe and react if something unforeseen should happen.

What causes speed blindness?

Maintaining the same speed for longer periods will trick your brain into perceiving your actual speed as lower than it is. If you in addition are driving in a featureless environment or the roads are straight and wide this tends to warp your perception of speed further.

Loud music or a generally noisy environment will also mask your sense of speed.

Ways to prevent speed blindness

- Check actual speed by observing your speedometer
- Brake firmly to slow down when the speed limit changes
- Gear down
- Observe your speed by checking your side quickly

Tunnel vision

The faster you drive, the smaller your field of view becomes. This phenomenon is known as tunnel vision.

Your condition

Your ability to interpret the traffic situation correctly and to act in time is severely affected if you are feeling tired or under the influence of drugs or other pharmaceuticals.

Affected conditions can be grouped in two categories

- Alcohol, drugs and medicinal drugs affecting driving performance
- Tiredness, illness, stress and mood

Road traffic act § 21

No one must drive or attempt to drive a vehicle when he is in such a state that he cannot be deemed to be capable of driving safely, irrespective of whether this is due to the fact that he is under the influence of alcohol or any other intoxicating or narcotic agent, or to the fact that he is ill, weak, worn out or tired, or to other circumstances.

Alcohol, drugs and medicinal drugs affecting driving performance

Alcohol limit

The alcohol limit in Norway is 0.02 %.

Blood alcohol concentration decreases by 0.01-0.015 % per hour. There is no way to speed up this process.

The risk of getting killed in a car accident increases by about a thousand times if you blood alcohol concentration is 0.15 % as opposed to 0.05 %.

Drugs and prescription drugs

It should be obvious that driving under the influence of prescription drugs is dangerous, but this is still a common problem. Prescription drugs that affect the central nervous system are usually labeled with a red warning triangle on the packaging.

In 2012 driving under the influence was a contributing factor to over 24 % of all fatalities on Norwegian roads.

Feeling tired and illness

Feeling tired

A lack of sleep can lead to a decrease in your driving skills. Staying awake for more than 20 hours will have the same effect as a blood alcohol level of 0.08 %. Sleep is the only remedy if you're feeling tired. The only remedy is taking a power nap.

Illness

The common cold, flu, headaches, nausea and other ailments can affect you to such extent that you should park your car.

Stress and mood

If you're feeling stressed out or your mood affects your attention and judgement this can lead to dangerous situations in the traffic.

Distractions

Distractions affect you when you are driving and leads to a loss of focus when you're driving.

Typical distractions

- Talking on the phone, even if you're using hands-free
- Checking your phone and texting while driving
- Using the car radio while driving
- Eating or drinking while driving
- Hyperactive kids
- Fidgeting with the GPS while driving

Reaction time

Reaction time is defined as the time it takes from the moment you notice a hazard until you start acting on it. It consists of the three first steps from the driving process - **sensing, perceiving and deciding.**

Your reaction time is normally around one second, but the following factors can increase this

- You're in a complex traffic situation and there are a lot of factors to keep track of.
- Visibility is limited (darkness, fog, and blizzard).
- You have a lack of experience or you have never encountered a situation like this before.
- You're distracted.
- You're under the influence or not feeling well.

Inattention time

Inattention time is the time from a potential hazard develops until you first notice the developing hazard.

In a simple traffic situation the inattention time is usually short, but if you are distracted — or the traffic situation is complex, the inattention time can become much longer.

Example

You're driving 80 km/h and suddenly you notice a moose in the road 100 meters ahead of you. If your inattention time is 1,5 seconds and your reaction time is 1,5 seconds, you will travel for 3 seconds, about 66 meters, even before you decide how you're going to act. You will probably have enough time to steer clear of the moose.

If you had been distracted by talking on the phone in this situation your inattention time would probably have been longer, and you wouldn't have had enough time to avoid the moose.

Summary

Highlights from chapter 3

- A traffic situation consists of you and your vehicle, other motorists, the road, the road rules and traffic signs, visibility and driving conditions, and the interaction between these factors.
- The driving process describes what you as a driver do in the traffic in four steps: sensing, perceiving, deciding and acting.
- According to the Road Traffic Act, you must not attempt to drive if you are not fit to drive.
- It is illegal to drive with a blood alcohol concentration greater than 0.02 %.
- Staying awake for more than 20 hours has the same effect as a blood alcohol concentration of 0.08 %.
- **Reaction time** is defined as the time it takes from the moment you notice a hazard until you start acting on it, while **inattention time** is the time from a potential hazard develops until you first notice the hazard.
- Alcohol, illness, stress or lack of sleep can increase your inattention time.
- Distractions like texting or turning on the radio seriously slows down your reaction and inattention time.

Chapter 4- Responsibilities

Introduction

With driving comes many responsibilities and you are required to know your obligations both as a driver and owner of a vehicle. You need to be familiar with relevant legislation, regulations and rules, and the consequences of violating them.

Vehicle condition

Before driving

You are required by law to ascertain that the vehicle is in accordance with regulations and safe to drive, before driving.

While driving

You are also responsible for ensuring the car is safe to drive while driving it. If a problem occurs while driving, you may keep driving only if the car is still safe to drive.

When lending your car to someone else...

- It is your responsibility to make sure the car is safe to drive when handing it over.
- It is also your responsibility to make sure the person borrowing your car has a valid license and is fit to drive.

EU inspections

The owner, or whomever the owner has granted access to the motor vehicle, is responsible for making sure the vehicle undergoes periodic roadworthiness tests (EU inspections).

Vehicles with a permissible maximum weight of 3500 kg or less must undergo an EU inspection (periodic road worthiness test) in the fourth year after first-time registration. EU inspections then need to be performed every other year. The time of registration is stated in the vehicle registration certificate.

Inspection month

The month in which your car is to be presented for inspection is decided by the registration plate number of the car.

An easy to remember rule is that the last digit in the vehicle's registration number correlates to the month of inspection, i.e. cars with registration numbers ending in 1 must be inspected by the end of January, cars with registration numbers ending in 2 must be inspected by the end of February, etc.

There is an exception in the case of numbers ending in 7 (which undergo the inspection in November) and 0 (which undergo inspection in October). No cars are inspected in July or December.

Example

You car's registration number is DF 47895. The inspection month is therefore May, since the last digit is 5 (cf. the main rule).

Child passenger safety

The driver is responsible for ensuring that all children under the age of 15 are properly secured. The proper child safety equipment depends on the child's age, weight and height:

- Children below 135 cm in height, or weighing less than 36 kg, must always use approved child restraints that are correct for their weight.
- It is recommended that children are transported in rear-facing car seats until the age of four.
- Children below 140 cm in height must not be transported in the front seat unless the airbag is deactivated.
- If approved child restraints are not available, children between 135 and 150 cm in height may use standard seat belts.

Responsibilities when towing a car:

If you ever have to tow a car, there are certain rules you need to be aware of:

- Only one motor vehicle may be towed at a time.
- The car being towed must be equipped with functional brakes.
- The rope, pole or bar used for towing must be clearly marked.

The rules for use of car lights are no different when towing, so do not use hazard warning lights.

Be aware that some cars should not be towed, as this may damage the vehicle. If you are towing a car with automatic transmission, check the owner's manual for information on how long and at what speeds it is safe to tow.

Motor vehicle registration
In general, all cars and trailers must be registered before they can be used.

To register a vehicle in Norway for the first time, you will need to contact a Driver and Vehicle Licensing Office ('trafikkstasjon'). You will need to show them that you have taken out motor vehicle liability insurance, and that all fees and taxes have been paid.

Registered vehicles are given a vehicle registration number (the identifying code on the registration plate on your car) and a vehicle registration certificate.

The vehicle registration certificate consists of two parts:

- Part 1, which you **must always carry when driving.**
- Part 2, which you must not keep in the car, as it can be used to register a change of ownership.

Change of ownership

When buying or selling a car, you will need to notify the Norwegian Public Roads Administration ('Statens vegvesen') of the change of ownership. The following rules apply to a change of ownership:

- The NPRA must be notified within 3 days after the change of ownership.
- The previous owner is responsible for sending in the notification of sale form. All claims and taxes will be the responsibility of the previous owner until the NPRA has been notified.
- The notification of sale can be filled in online, mailed or delivered in person to a Driver and Vehicle Licensing Office. If you send it in online, both previous and new owner must use the online form.

- If you do not use the online form, the NPRA recommends using part 2 of the vehicle registration certificate as the notification of sale.
- Both former and new owner must sign the same notification of sale. If the online form is used, both must approve the form online.
- When the notification of sale is registered, the new owner will receive a letter with further instructions.
- The new owner may not use the car until he or she has received the new vehicle registration certificate.

Liability insurance (third party insurance)

In general, drivers are financially responsible for damages caused by the car they are driving, regardless of whether the driver was careless or just unlucky.

To make sure that anyone who sustains an injury in traffic accident is entitled to compensation from an insurance company, all registered motor vehicles must be covered by motor vehicle liability insurance (traffic insurance/third party insurance).

The minimum compulsory motor vehicle liability insurance covers:

- injuries sustained by another person
- damages to other vehicles (if the driver of that vehicle cannot be blamed for the damages)
- damages to another person's property (e.g. if you drive into a house or a fence)
- injuries sustained by the driver himself/herself (i.e. you, if you are driving)

Liability insurance does not cover damages to your own vehicle. To cover damages to your own vehicle, comprehensive vehicle damage insurance (in Norway often called 'kasko') is necessary.

Insurance and recourse

If the insurance company has to pay for damages that you are to blame for, they may demand to have all or part of the amount reimbursed by you. This is called recourse.

The insurance company may invoke their right of recourse if you

- caused the damage or injury on purpose, or through gross negligence
- were under the influence of alcohol (or other intoxicants)
- had stolen the car or were using it for illegal purposes
- did not hold a valid driving licence
- were practice driving illegally

Passengers in the car may also receive less, or lose the right to compensation altogether, for instance if they have got in a car they know to be stolen.

If the accident was caused by negligent driving, the insurance company may demand you reimburse them for any damages paid out as a result of the accident.

Sanctions

Traffic violations and penalties
As a driver you must be familiar with — and obey – all relevant legislation and regulations.

If you violate any of these regulations, the authorities (police, parking authorities, NPRA ('Statens vegvesen'), etc.) may give you

- verbal warnings
- fees
- on-the-spot fines, regular fines and imprisonment
- penalty points and licence suspension
- vehicle-related penalties, e.g. your vehicle or registration plates may be confiscated, you may be prohibited from using it or a lien (security for a debt) may be placed on the vehicle

Drunk-driving

As mentioned in chapter 3, in Norway it is illegal to drive with a blood alcohol concentration above 0.2 ‰ (0.02 %). The Road Traffic Act contains penal provisions for driving under the influence of alcohol or other intoxicants.

If you are convicted of drink-driving, you may get

- a fine, if your blood alcohol concentration is below 0.5 ‰
- a fine and probation or imprisonment, and your driving licence will be suspended for at least one year, if your blood alcohol concentration is above 0.5 ‰, but below 1.2 ‰
- a fine and imprisonment, and your driving license will be suspended for at least one year, if your blood alcohol concentration is above 1.2 ‰

Note!

You can be convicted for driving under the influence even if your blood alcohol level is below 0.2 ‰, if your driving has been affected by the alcohol.

Other provisions concerning driving and intoxication

If you have been involved in a traffic incident that might be cause for a police investigation, you must not consume alcohol (or other intoxicants) in the first 6 hours following the incident. This no longer applies after you have either provided a blood sample, or been told by the police that they do not intend to take one.

Professional drivers must abstain from consuming alcohol (and other intoxicants or sedatives) during working hours, as well as the 8-hour period immediately preceding. This stipulation only applies to those who in their occupation transport people or goods.

Penalty points

A penalty point system is in place in Norway, meaning that for certain traffic offences you will receive penalty points.

If you receive 8 points or more within 3 years, your licence is suspended for at least 6 months.

During the probation period — the first 2 years after you get your licence — you will receive twice as many points for traffic offences.

Example

Running a red light normally gives 3 penalty points, but during the probation period the driver will instead get 6 penalty points.

Summary

Highlights from chapter 4

- As the owner of a motor vehicle, you are responsible for ensuring the car is safe and according to regulations before driving. As the driver you are also responsible for ensuring the car is safe to drive while driving it.
- The owner is responsible for making sure the vehicle undergoes periodic roadworthiness tests (EU inspections) in the fourth year after first-time registration, and then every other year thereafter. The month in which your car is to be presented for inspection is decided by the registration plate number of the car.
- As the driver, you are responsible for ensuring that all children under the age of 15 are properly secured. Children below 135 cm in height, or weighing less than 36 kg, must always use approved child restraints that are correct for their weight.
- All cars and trailers must be registered before they can be used.
- All registered motor vehicles must be covered by motor vehicle liability insurance (traffic insurance).
- Drunk-driving is punishable by fines, probation and imprisonment, depending on your blood alcohol level. Blood alcohol levels above 1.2‰ are punishable by fines and imprisonment.

In the next chapter we will go through all the elements of the traffic system, and the rules regulating behaviour in traffic.

Chapter 5- The Traffic System

Introduction

The traffic system consists of four parts: traffic management, roads, vehicles and road users. Your behavior in traffic and the interaction between the parts of the traffic system are largely governed by the rules and regulations in the Road Traffic Act and the traffic rules, together with road marking, signs and traffic lights.

What governs traffic?

Without a set of rules and guidelines for everybody to follow, traffic would be chaotic and dangerous. Laws, rules and regulations regulate how the elements in the traffic system should interact with each other, and give us guidelines for how we should conduct ourselves in traffic.

Through legislation the government has given us both general and specific rules for how to conduct ourselves in traffic, vehicle requirements, construction of roads and how traffic is governed.

Important laws and regulations

Right of way, stopping and parking, speed limits, changing lanes, using lights and signals, etc. are all governed through different laws and regulations.

The Road Traffic Act

The Road Traffic act consists of many regulations governing our behavior in traffic. Section 3 — the **basic rules of traffic** — is the main provision concerning our behavior in the traffic.

The Road Traffic Act applies to all motor traffic in Norway, but also applies to cyclists and other road users, as long as they are travelling on a road or in an area where other vehicles are travelling as well.

The Road Traffic Act also includes important provisions regarding:

- speed
- driving under the influence
- traffic accident obligations

- practice driving
- transgressions and punishment

The traffic rules

«Forskrift om kjørende og gående trafikk» – better known as the traffic rules – contains regulations about behaviour in the traffic.

The traffic rules apply to public and private roads, streets or places open for general traffic.

In the traffic rules you will find regulations regarding

- use of the carriageway, vehicle placement on the carriageway and changing of lanes
- right of way
- pedestrians
- reversing and turning
- special speed regulations
- stopping and parking

Basic rule of traffic

The third article in the Road Traffic Act is the basis for the code of conduct in traffic, and is may be better known as the basic rule of traffic. You should always drive with the basic rule of traffic in mind. When somebody fails to follow this rule dangerous situations can arise, and violating this article can lead to the confiscation of your driving licence.

The basic rule of traffic states:

«A driver should show consideration and be alert and cautious so that he does not cause damage or risk, and so that other traffic is not unnecessarily obstructed or inconvenienced.»

When you are driving it is your fundamental duty to be:

- **Considerate** – Take into consideration the needs of your fellow drivers.
- **Alert** – Concentrate and be alert when you are driving.
- **Cautious:** – Drive in a manner that reduces the risk for accidents and injuries. Reduce speed according to the conditions and show the others drivers by driving in a clear and timely fashion what you are planning to do.

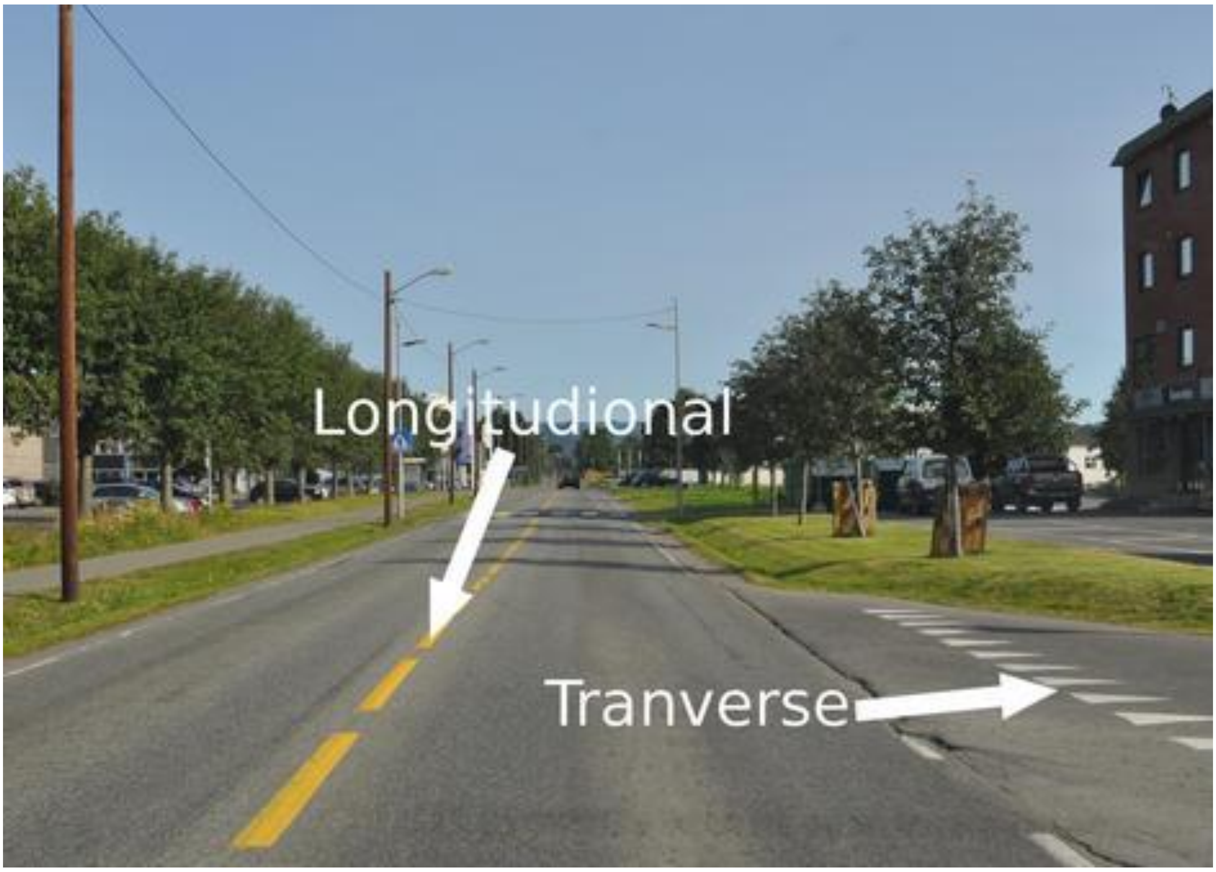
The Road

The road is an integral part of the traffic system.

In the traffic rules a road is defined as "any public or private road, street or open square (including lay-bys, car parks, stopping places, bridges, winter roads except marked tracks for snow scooters, ferry quays or other quays immediately adjoining a road) that is open to general traffic".

Road markings

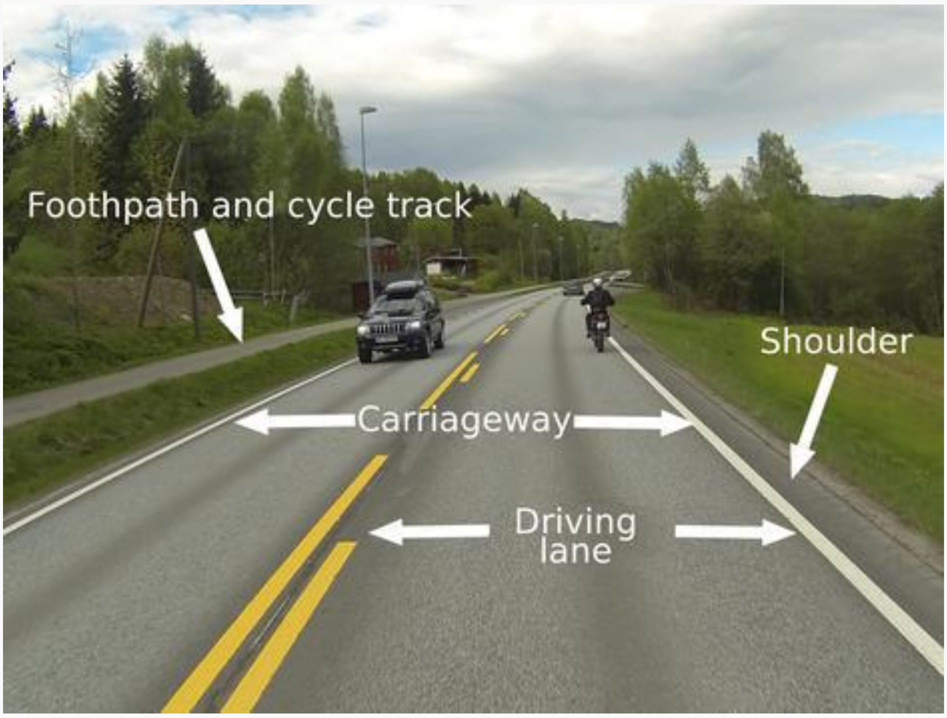
- Longitudional markings (langsgående) are used to divide the carriageway into lanes and to define the borders of the carriageway
- Transverse markings (tverrgående) are used to indicate stop line, right of way lines and pedestrian crossings.
- Yellow markings divide between traffic in opposite directions
- White markings divide between traffic in the same direction.



Parts of the road

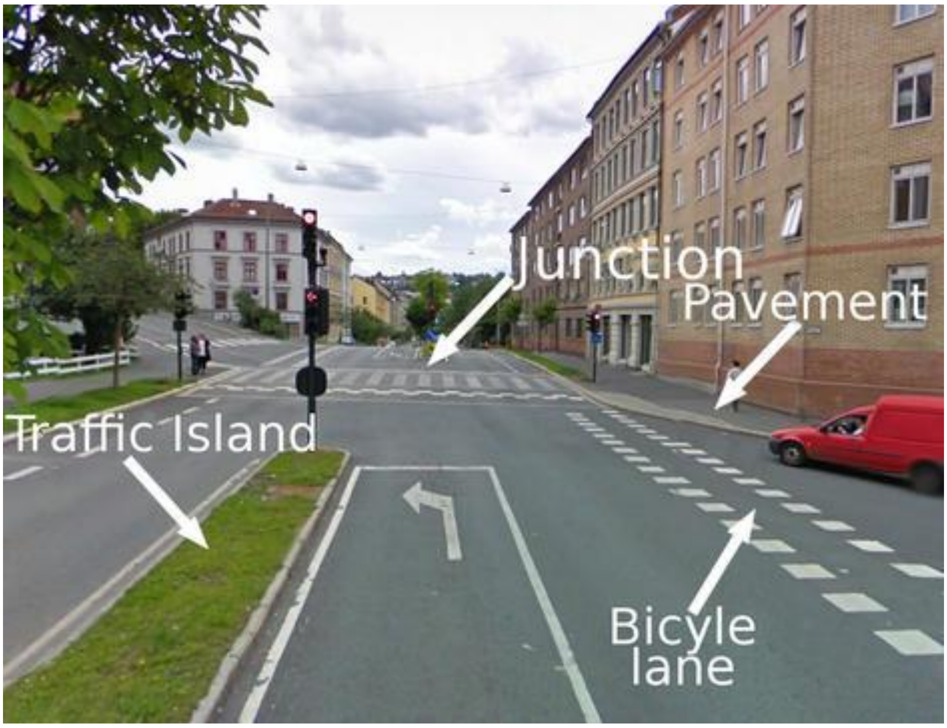
The traffic rules contain a lot of definitions for the parts of the road.

- **Carriageway (kjørebane)** – The part of the road used for normal traffic.
- **Driving lane (kjørefelt)** – Each of the longitudinal parallel lanes into which the carriageway is divided by road markings, or which is wide enough for a stream of traffic
- **Footpath and cycle track (gangvei og sykkelvei) -** Any path or track designated by an official traffic sign for the use of pedestrians or cyclists or a combination of pedestrian and bicycle traffic. Such path or track is separated from other roads by a strip of grass, a ditch, fence, curbstones or by other means
- **Shoulder (veiskulder)** – The part of a road that lies outside its marked edge.



- **Junction (veikryss)** – Any point at which a road crosses or merges with another road.
- **Bicycle lane (sykkelfelt)** – A lane designated for the use of cyclists by an official traffic sign and road markings

- **Traffic island (midtdeler)** – A solid or painted object in a road that channelizes traffic.
- **Pavement (fortau)** – Any paved strip designated for the use of pedestrians and separated from the carriageway by curbstones



Parts of the Road

The road and road markings

Road markings in the middle of the carriageway

Lane line

A lane line may be crossed if the traffic rules allow it, as the lane line tells you that you will normally be able to get a clear view of the road ahead in this area. Lane lines may be yellow or white. The line is a broken line consisting of short lines with large gaps in between. When the speed limit is 50 km/h or slower the line will be 1 meter and the gap will be 3 meters. If the speed limit is 60 km/h or higher the length increases to 3 meters for the lane and 9 meters for the gap.

Lane line

Warning line

A yellow warning line indicates that visibility ahead is too restricted for normal overtaking. However, this does not necessarily mean that overtaking is prohibited. Note that overtaking in general is prohibited if it is dangerous or may cause an accident, or if it inconveniences other road user. The line is a broken line consisting of long lines with short gaps in between. When the speed limit is 50 km/h or slower the line will be 3 meters and the gap will be 1 meter. If the speed limit is 60 km/h or higher the length increases to 9 meters for the lane and 3 meters for the gap.

Warning line

Combined Lines

Combined lines are used if visibility differs depending on the direction you are driving. You should observe the line closest to you. In the picture oncoming traffic is following the lane line while traffic in your direction should follow the warning line.

Prohibitory line

A prohibitory line in an unbroken line and crossing it is prohibited. In the picture you see a double prohibitory line.

Double prohibitory line

Hatched area

Crossing a hatched area is not allowed. Hatched areas are often used in junctions or to separate between traffic on semi-motorway.

Arrows

Arrows indicate which directions the lane is intended for. Since these arrows can be hard to spot in busy traffic or in snowy conditions you will often find them repeated on information signs above the road in larger junctions.

If you are planning on going left in this junction, you should select the left lane.

Road markings on the side of the carriageway

Dividing line

A dividing line consists of equally long lines and gaps and is used to separate a lane from a bicycle lane or a bus bay. Dividing lines are also used to separate lanes intended for public transportation and taxi from the other lanes.

A bicycle lane

Edge line - unbroken

An unbroken edge line is white and follows the edge of the carriageway. It is usually not allowed to cross this line. Unbroken edge lines are used on both one-way and two-way carriageways.

Unbroken edge lines on both sides of the carriageway.

Edge Line - broken

On roads narrower than 6 meters the carriageways is not divided into lanes and there are no center line markings. Broken edge lines are used to mark the edge of narrow roads like this.

Broken edge line

Road markings across the carriageway

Give way line

A give way line is a transverse road marking consisting of small triangles, and marks where you should stop when giving way in a junction. Give way lines are also used to mark the entrance points in a roundabout.

Give way line.

Stop line

A stop line is a transverse unbroken line that marks where you should stop before a traffic light or a stop sign.

The lanes end in stop lines in front of the junction.

Traffic Signs

Signs

As a road traffic user you have to be able to recognize the different groups of signs, and understand what they mean for your driving.

In this chapter we will take a look at the different groups of signs, but for a complete review of every sign please check chapter 10, "List of signs."

Important!

On the official theory test you will be thoroughly tested in your knowledge about signs. Some questions test if you know what a certain sign means, while other questions will test if you know how to use this knowledge in different traffic situations.

You need to know about every sign and what that means.

Warning signs

A warning sign contains information about potential hazards ahead. The hazard can be road irregularities like sharp turns, slippery road or a bumpy surface. Warning signs can also be used to warn about certain kinds of road users or certain groups of animals which you should be aware of.

Warning signs are usually triangular with white background and red border. Warning signs used temporarily because of road works may have a yellow background rather than the normal white.

Warning signs can have supplementary signs indicating distance to the hazard, the size of the hazard or other information about the hazard. The supplementary sign can also contain arrows, symbols or additional text about the upcoming hazard.

The placement of warning signs varies according to the speed limit.

- If the speed limit is 60 km/h or lower signs are placed between 50-150 meters in front of the hazard.
- If the speed limit is 70 km/h or higher signs are placed between 150-250 meters in front of the hazard.
- If the speed limit is 90 km/h or higher signs are often placed 400 meters in front of the hazard.

Children

Give way signs

Give way signs regulate who should give way at a junction or on a stretch of road. The shape and color of these signs varies according to their purpose. You can read more about give way signs in chapter 7.1 which deals with the regulations regarding giving way.

Give way signs can have supplementary signs indicating distance to the junction or illustrating the course of a priority road.

Give way.

Prohibitory signs

Prohibitory signs prohibit certain actions on a stretch of road or a place. The signs usually are valid in the direction you are driving from the placement of the sign to the first junction (some signs need to be relieved by another sign. The sign indicating speed limit is also a prohibitory sign.

Prohibitory signs are usually round, with a red edge on a white background.

Prohibitory signs can have supplementary signs indicating where or when the prohibition is valid.

No motor vehicles.

Mandatory signs

Mandatory signs usually concern which lane you should use or in what direction you can leave a junction

Mandatory signs are round with white symbols on a blue background

The arrows indicate in which direction you should leave the junction.

Informative signs

Informative signs specify that certain regulations either apply or no longer apply. Informative signs are usually square with white or black symbols on a blue background. Signs showing how the lanes proceed and have a white background color.

Informative signs may have supplementary signs.

Pedestrian crossing

Service information signs

Service information signs inform about road service, tourist attractions, landmarks etc. Two or more symbols can appear on the same sign.

Service information signs regarding road service are rectangular with black symbols on a white background with blue edges, while signs regarding tourist attractions and landmarks are rectangular with brown symbols on white background with brown edges.

Emergency telephone and fire extinguisher

Direction signs

Direction signs inform about place names, establishments, routes, lane choice, roads and distance to destination. Unlike the other groups of signs, direction signs don't have supplementary signs. Instead they can contain text, number and symbols.

Color coding on direction signs

- Geographical destination motorway – blue background and white text
- Geographical destination other roads – yellow background and black text
- Other destinations – white background and black text
- Temporary direction – orange background and black text
- Tourist destination – brown background and white text

The large yellow sign on the right hand shows an exit guide.

Supplementary signs

Supplementary signs are only valid when they are together with the main sign and they clarify, expand or limit the validity of the main sign.

If the supplementary sign is used in conjunction with a temporary sign it can have a yellow background.

The first supplementary sign indicates that the main sign only is valid if you are taking right in the upcoming junction. The second supplementary sign defines when parking is allowed. The last supplementary sign informs where the road is uneven.

Marker signs

Marker signs inform about obstacles in or near the road, or about the course of the road. Traffic cones and other boundary markers are also considered to be marker signs

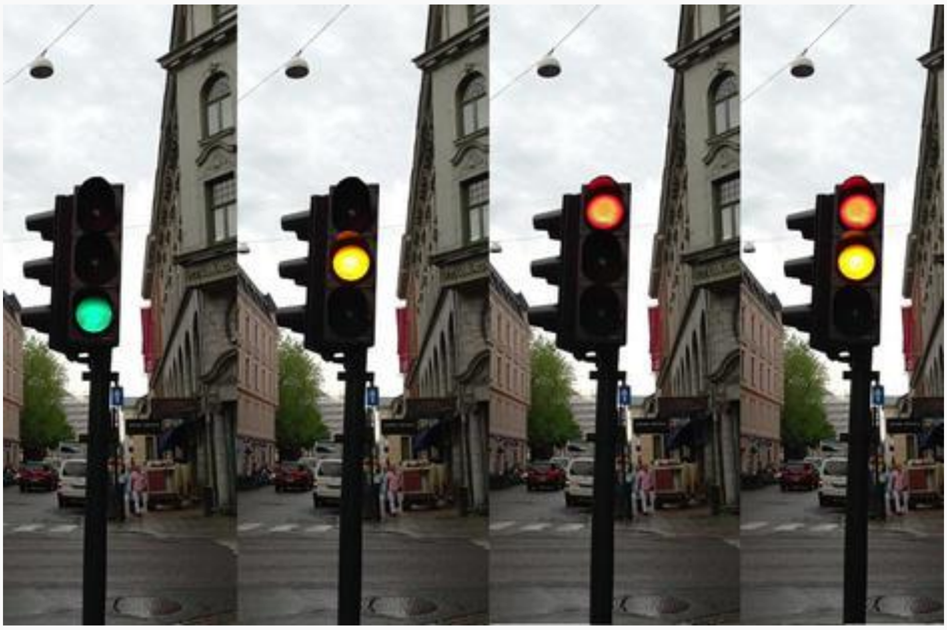
Temporary marker signs have a red background with white or yellow symbols.

Directional marking

Traffic lights

Junctions with heavy traffic are often regulated with traffic light. In recent years many junctions with traffic lights have been replaced with roundabouts, but we still find many traffic lights in city centers.

- **Green light** – go if the way is clear.
- **Yellow light** – the traffic light is about to turn red and you have to stop. If braking in time is difficult due to your speed, it is better to drive through the junction.
- **Red light** – stop.
- **Red and yellow light** – the traffic light is about to turn green.
- **Blinking yellow light or no light** – the traffic light is out of order and signs and give way regulations apply.



The four phases of the traffic light

Pedestrian crossing lights

- A red man means that you have to wait.
- A green man means that you can cross the road.

- A blinking green man indicates that the sign soon will change to red signal.

Pedestrian crossing lights

Arrow signals

- The signal is only valid for you if you are going in the direction of the arrow.
- Arrow signals can have one, two or three openings. If the arrow signal is unlit you should follow the main traffic lights.

Arrow signal

Other traffic lights

Lane control lights

- A red cross prohibits using the lane.
- A green arrow allows using the lane.
- A yellow arrow indicates that the lane is closing further ahead and you have to change to the lane in the arrow's direction.

Lane control lights

Grade crossing signal

- Red blinking signal - Stop by the stopping line or in safe distance to the level crossing.
- White blinking signal - Crossing is safe.

Grade crossing signal

Blinking arrow

- The arrow shows you on which side you should pass.
- Blinking light arrows are usually used in road work.

Blinking light arrow

Police

Sometimes regulations, signs and traffic lights are not enough to ensure a safe and efficient traffic flow. Typically when there has been an accident or the road is partially blocked because of road work, the traffic needs to be controlled manually.

Statens vegvesen and the police sometimes conduct traffic controls to check if vehicles are in proper condition or that people are not driving under the influence.

People with traffic control authority

- **Police** – Directs traffic past traffic accidents or helps controlling traffic jams. The police also conducts speed controls, blood alcohol concentrations controls etc.
- **Customs** – Customs officers can conduct controls at the borders checking for smuggled goods. They can also confiscate car registration plates if your vehicle registration certificate is missing.

- **Statens vegvesen** – Vegvesenet conducts many of the technical controls also conducted by the police.
- **Road work** – Road work will sometimes require additional control of the traffic flow and construction workers in orange or luminescent clothes and traffic lights can control the traffic.

Traffic control duties

You are obliged to stop when being signaled for a control. Be prepared to show the necessary documents and follow the instructions given to you.

Both Police and vegvesenet have the authority to control traffic.

Be alert in situations like this and follow the directions you are given.

The Pyramid of Authority

From time to time you will encounter situations where traffic regulations, road markings, signs and traffic lights all come in conflict with one another. When this happens you can use the *the pyramid of authority* to decide what or who takes precedence.

On top of the pyramid you will find **police**. You should always follow the instructions given by a police officer, even if it goes against traffic signs, road markings or traffic lights.

Right below police we find **traffic lights.** When you approach a junction with both traffic lights and give way signs, you should ignore the signs unless the traffic lights are blinking yellow.

Under traffic lights you find **signs and road markings**. You normally have to give free way to traffic coming from a road on your right hand side. If a sign indicates that you are driving on a priority road or that traffic from the right has a give way sign, then this precedes the traffic regulations

At the bottom of the pyramid, we find **traffic regulations.**

Note!

On your theory test you don't typically get questions about the pyramid of authority, but you can get questions where you have to judge traffic situations based on the pyramid of authority.

Traffic users

The road traffic act and the traffic rules require that you have to pay attention to the other traffic users. As a driver you have both moral and legal liability and the requirements imposed on you are strict - Much stricter than the requirements imposed on the vulnerable road users

The traffic rules contains provisions that demand that you are extra careful around pedestrians, children, elderly and handicapped people.

In order to contribute to a safe and efficient traffic flow you should also show extra consideration for heavy vehicles and two-wheelers.

Vulnerable road users

Cyclists, pedestrians and non-motorized road users are much more prone to injury in an accident. In the traffic rules there is a regulation that states:

Drivers are obliged in particular to maintain sufficiently low speed so that they can, if necessary, stop when passing

- children on or near the road
- school crossing patrols
- persons using a white stick or guide dog for the blind
- other persons who can be assumed to suffer from a disability or illness which hinders them in traffic
- trams or buses that have stopped or are in the process of stopping at a stop

Cyclists

Bicycle paths are becoming quite commonplace, but you still meet a lot of cyclists along the main road. Sharing the road with vulnerable road users necessitates that drivers take extra precautions.

The interaction between drivers and cyclists

- Keep sufficient distance. When you're driving past a cyclist you should at least have a safety margin of about 1,5 -2 meters.
- Bicycle paths are for cyclists only.
- If you are turning right in a junction remember that you have to give way to cyclists who are going straight ahead in the junction. They might hiding in your blind spot.
- Even though cyclists are following the same rules and regulations, you should be aware that they don't always follow these regulations.

Show respect!

Aggressive driving or recklessly attempting to overtake a cyclist is not only dangerous, it is illegal!

Pedestrians

The traffic rules contain regulations which require you to show extra caution towards pedestrians.

You should always:

- allow pedestrians sufficient space on the road.
- give way to pedestrians when crossing a footpath, or pavement. The same applies to driving in pedestrian streets or pedestrianized residential streets.
- give way at pedestrian crossings where traffic is not controlled by the police or by traffic lights,
- give free passage to disembarking passengers if you are intending to overtake a tram on the right hand side at a stop without a traffic island.
- avoid stopping on a pedestrian crossing.

Note!

Pedestrians and cyclists are especially vulnerable in dark driving conditions because it can be difficult to see them. Use the correct lighting, adjust your speed and chose a position that gives you the best view

Children

You are obliged to maintain sufficiently low speed so that you can, if necessary, stop when passing children on or near the road, Why do we have to take extra precaution when passing children?

- Children are small and therefore often difficult to notice.
- Children are impulsive and unpredictable.
- Children's vision and hearing are not fully developed (children do not see the same as adults; typically the field of vision is more narrow than that of adults).
- Children have difficulties correctly assessing speed and distance.
- Children do not necessarily understand the traffic rules and the dangers of traffic.

Take extra precaution when you are driving in an area where you know children might be playing. They are easily hidden behind parked cars!

Motorcycles and Mopeds

Statistics show that accidents involving a two wheeler and a car usually are due to the driver of the car failing to observe the two-wheeler.

Because motorcycles and mopeds are relatively small there are some situations where accidents might happen. Pay special attention to two wheelers coming from the right at a junction, and remember to check for two-wheelers coming up on your side when you are planning to make a turn in a junction. Two-wheelers also tend to disappear behind oncoming traffic when you are planning on taking a left in a junction.

To detect two-wheelers in traffic more efficiently you should:

- Be particularly mindful of them.
- Chose a position that gives you a better overview.
- Use your mirrors often.
- Check your blind spots before turning or changing lanes.

Large vehicles

Due to their size and mass large vehicles have some limitations when it comes to maneuvering in the traffic. This is something drivers of smaller cars should take into consideration when driving.

Differences between large vehicles and smaller vehicles:

- Speed. Larger vehicles have considerably slower acceleration.
- Weight. Increased weight causes longer braking distances
- Wheelbase. Increased wheelbase leads to a larger turning radius.
- Visibility. It can be difficult for the driver of a large vehicle to observe what is happening in front, behind or on the sides of the vehicle.

How can you contribute to a better traffic flow?

- Make space for large vehicles when they are turning in a junction
- Since large vehicles need considerably more time to decelerate and accelerate, you should in some situations waive your right of way.

Note!

Always keep in mind that potential dangers might be hidden from view by large vehicles.

Trams

When you a driving in an area where you know you can expect trams, you should take the following into consideration. A tram is both big and heavy and need even more time than a lorry to come to a complete stop. A tram also lacks any maneuverability beyond its tracks. In the traffic rules this is taken into account and in most situations you have to give way to a tram. The exception is give way signs which applies all road users.

Remember the following:

- You should always give free passage to a tram
- It is allowed to pass a tram on the right hand side, but beware of the limited visibility.
- When passing a tram at a tram stop, lower your speed. If you're passing the tram on the right hand side remember to give passengers coming of the tram free passage.
- A tram takes longer to brake and needs more space when turning, so keep a good distance.

Summary

Highlights from chapter 5

- The traffic system consists of four parts: **traffic management, roads, vehicles and road users.**
- In the **road traffic act** we find **the basic rule of traffic** which states: **A driver should show consideration and be alert and cautious so that he does not cause damge or risk, and so that other traffic is not unnecessarily obstructed or inconvenienced.**
- The traffic rules contain regulations about positioning on a carriageway, right of way, stopping and parking, etc and applies to public and private roads, streets or places open for general traffic.
- According to the road traffic act ««a driver shall be alert to the official road signs, signals and road markings, and must comply with prohibitions and instructions so given»
- The traffic rules contains provisions that demand that you are extra careful around walking, children, elderly and handicapped people.
- The pyramid of authority governs the relationship between traffic rules, road markings and signs, traffic lights and the police

Chapter 6- Driving

Introduction

Before driving in traffic for the first time, you should practice basic observation techniques and think about how best to make your intentions clear to other road users. You should also have knowledge of how a car behaves in relation to speed, road grip and braking.

The rules of observation

When driving, you must constantly gather information about the traffic situation. As much as 90 % of this information is taken in through sight. You interpret other road users' behaviour and adjust your own driving accordingly. But for other road users to correctly interpret your behaviour and be able to adjust their driving accordingly, it is important that you clearly show your intentions and act in good time. By following the rules of observation, you can more easily anticipate what other road users will do, while also making it easier for them to anticipate your behaviour.

The rules of observation

1. Look far ahead.
2. Keep your eyes moving.
3. Get the big picture.
4. Always look for a way out.
5. Be seen and understood.

1. Look far ahead

Looking far ahead helps you notice potential hazards as early as possible, so you can adjust your driving accordingly.

It is also easier to stay on course and position the car correctly on the roadway when paying attention to the traffic situation further ahead.

How far ahead should you look?

- In densely populated areas you should try to see as far ahead as the end of the street.
- On country roads the speed is higher, and you should therefore try to see at least 400 meters ahead to avoid hazards and drive safely.
- In the dark you should try to see further ahead than just the stretch illuminated by your own headlights. Even main beam lights only illuminate about 100 meters ahead, but you will be able to spot a pedestrian wearing a reflector at 300 meters distance.

2. Keep your eyes moving

To get a complete overview of the traffic situation, especially when travelling at higher speeds, you will need to keep your eyes moving. As mentioned in chapter 3, the central vision is only a small part of your field of vision. When the speed increases, it becomes more difficult to notice things going on outside of the central vision field, and it is easy to develop tunnel vision. To compensate for this you must move your eyes more actively to see what is going on side roads further ahead, in traffic behind you and on your sides.

Forwards and to the sides

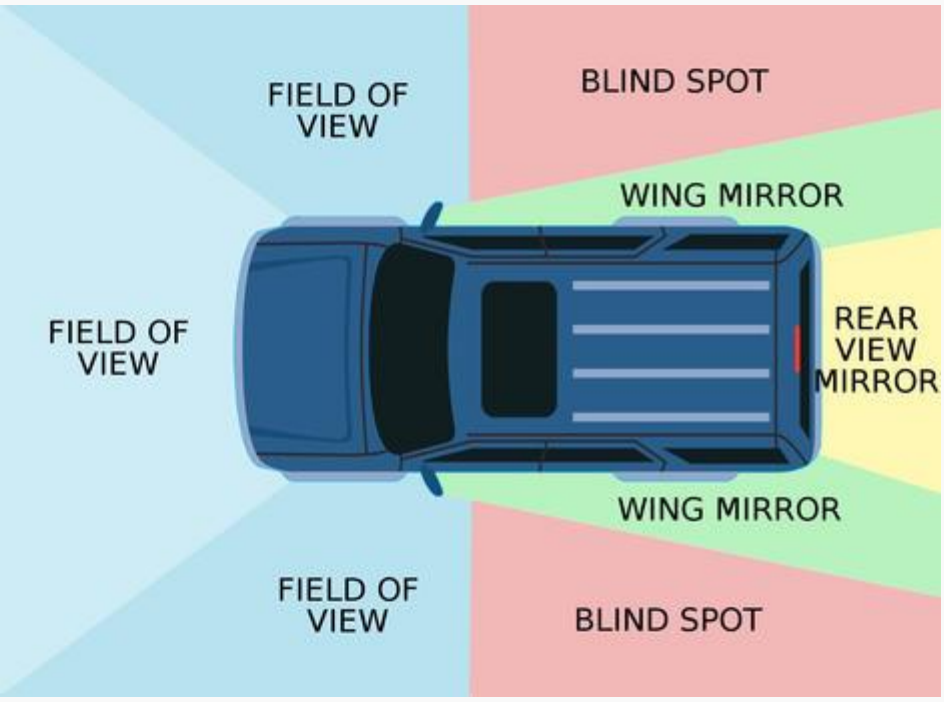
- Keep your eyes moving
- Look to the sides several times to make sure you take in all relevant information.

Behind you and to the sides

- Check your mirrors frequently.
- When braking, you need to check your mirrors to make sure traffic behind you is also able to stop in time.
- Check your mirrors and blind spot before turning in a junction, changing lanes, overtaking or performing any other lateral change in position.

Blind spots

The blind spot is the area on the sides and behind your vehicle that you are unable to see in your side-view mirrors and rear-view mirror. Before changing lanes or turning in a junction, you therefore need to check your blind spot to make sure it's clear.



3. Get the big picture

Get an overview of the situation and try to anticipate how it will develop. In general, you should be able to anticipate what might happen for the next 10 seconds in traffic.

When your speed is 50 km/h, you are travelling approximately 14 m/s. Anticipating how the traffic situation will develop for the next 10 seconds, means you should be able to anticipate what will happen the next 140 meters ahead.

If you are driving at 100 km/h, you will in 10 seconds have travelled about 280 meters.

4. Always look for a way out

Unforeseen events and dangerous situations may always arise, no matter how observant and alert you are. Always look for a way out, so you are prepared — even for the unexpected.

By keeping a good distance to the car in front of you, your visibility increases and it is easier to look for possible ways out. Prepare to brake when you notice a potential hazard, so you are ready in case anything happens.

5. Be seen and understood

As a driver, being seen and understood by other road users is obviously important. You should therefore always drive in a manner that makes your intentions clear, so that other road users are able to anticipate your actions and adjust to them.

In the next part we will look more closely at how you can show other road users what you intend to do.

Tips for being seen and understood

- Position your vehicle correctly in the lane, well in advance. E.g. if you are on a multi-lane carriageway, you should position yourself in the left-hand lane early.
- Adjust your speed in good time. E.g. when approaching a pedestrian crossing, you should start braking early to indicate to the pedestrians that you intend to uphold your duty to yield.
- Use your direction indicators and brake lights to signal in good time.
- Seek eye contact with other road users whenever possible, to make sure they have seen and understood you (motorcyclists, especially, are taught to do this).
- Signal by flashing your headlights or using the horn to warn other road users of hazards (the traffic rules prohibit any unnecessary or inconsiderate use of sound or light signals).

Cooperation

In traffic, being able to correctly interpret other road users' actions, as well as making sure they understand what you are doing, is crucial.

If you assume that most road users by and large abide by the traffic rules and regulations, there are several assessments you can make based on their signaling, speed and positioning on the road. You will also be able to determine whether another road user has spotted you, and act accordingly. This forms the basis of cooperation in traffic.

The car's indicator and brake lights are our main way of signaling our intentions to other road users, but we'll first take a look at how your choice of speed and positioning affects cooperation in traffic

Traffic flow

Cooperation facilitates a safe and efficient movement of traffic.

Choose a speed that makes you understood

Your choice of speed says a lot about your intentions. For instance, when slowing down before a junction, other road users will assume that you are going to make a turn or give way for crossing traffic.

When approaching a roundabout (where you have a duty to yield), you are required to slow down well before reaching the roundabout, to show that you will uphold your duty to yield.

When approaching a pedestrian crossing, you are also required to lower your speed in good time to show the pedestrian that you intend to stop to let them past.

Position the car to make yourself understood

Positioning your car correctly in your lane, will also help other road users understand your intentions.

If you for instance position yourself to the right before a junction, close to the road shoulder or pavement, you are telling other road users that you intend to turn right in the upcoming junction.

If you, however, position yourself to the left in your lane before a junction, other road users will assume that you are planning to turn left in the junction.

It is just as important to keep a neutral position, i.e. staying in the middle of your lane, if you are going straight in the junction, so as not to confuse other road users.

Give clear signs

What are signs?

You give signs by using your direction indicator lights when turning or moving sideways. When braking, your brake lights are a sign to the road users behind you that you are going to apply brakes.

When do you give signs?

It is stated in the traffic rules that you must give signs when turning or moving sideways (e.g. when changing lanes or moving out to pass a parked car).

For other road users to be able to understand your intentions — and adjust their own driving accordingly — you need to give clear signs in good time.

Clear signs means that the signs you give should not be easily misunderstood. E.g. if you are turning right in a junction, you should inform other road users by using both brake lights (push lightly on the brake pedal a few times) and indicator lights.

If possible, you should turn on your direction indicator lights 5-8 seconds before turning or moving sideways. But keep in mind that turning on your indicators too early may lead to misunderstandings. E.g. when changing lanes in a roundabout, it does not make sense to signal 5 seconds in advance.

Use signals to warn of hazards

What are signals?

Signals refer to using the car's horn or flashing the headlights.

When do you use signals?

Signals are used to warn other road users of a hazardous situation you have just passed, so that they may reduce their speed. Many people also flash their headlights to alert other road users that their headlights are off or not functioning properly.

You should always be careful not to use signals unnecessarily. It is stated in the traffic rules that unnecessary or inconsiderate use of sound or light signals is prohibited.

Speed limits

The main rules regarding speed are stated in the Road Traffic Act and the traffic rules.

In Norway we have two general speed limits that apply if there are no signs indicating otherwise. When speed limits are cancelled, the general speed limits will apply.

General speed limits

- In densely populated areas: 50 km/h.
- Outside densely populated areas: 80 km/h.

The legislation does not define the term 'densely populated', but when driving into a densely populated area, there will usually be a speed limit sign indicating 50 km/h.

Adjust your speed to the conditions

The most important rule regarding speed is that you yourself need to determine how fast it is safe to drive under the conditions.

The driver is required to adjust the speed to the area, road and driving conditions, as well as the traffic situation, and so that other road users are not unnecessarily obstructed or inconvenienced. The driver must always be in full control of the vehicle.

The driver must be able to stop within the distance which he can see to be clear, and before reaching any foreseeable obstruction.

The rules entail that you as a driver always need to decide for yourself if it is safe to drive the speed limit, or if you need to keep a lower speed.

Speed limits and traffic signs

The general speed limits apply if there are no signs indicating otherwise.

Speed limit signs

Speed limit signs apply from where they are posted until a new sign indicates a new speed limit. When the speed limit changes, signs will be posted on both sides of the road. If the sign is only posted on the right-hand side of the road, this is meant as a reminder of the current speed limit.

The sign *End of speed limit* indicates that the new speed limit is 80 km/h.

Signs posted on both sides of the road inform you that a new speed limit applies.

Speed limit zone signs

In residential areas, another type of speed limit sign is often used — speed limit zone signs.

The speed limit indicated by the zone sign applies on all roads until it is cancelled. If there are no signs indicating otherwise, the general speed limit of 50 km/h applies after passing the sign *End of speed limit zone*.

Zone signs apply on all roads within the zone, until it is cancelled.

Warning signs and recommended speeds

Warning signs warn of potential hazards along the road, and you are normally required to adjust your speed to avoid any dangerous situations.

Many places the warning sign will be combined with a supplementary sign indicating a recommended speed. Typically these signs are posted in sharp bends outside of densely populated areas.

On certain stretches you should keep a lower speed to avoid any dangerous situations.

Special provisions regarding speed

There are several provisions regarding speed that apply in certain situations:

Drivers are obliged to maintain a sufficiently low speed and, if necessary, stop, when passing:

- children on or near the road
- school crossing patrols
- persons using a white stick or guide dog for the blind
- other persons who can be assumed to suffer from a disability or illness which hinders them in traffic
- trams or buses that have stopped or are in the process of stopping at a stop

Other provisions regarding speed

- In a pedestrian precinct or residential street, you may not drive faster than walking speed (about 5-10 km/h).

- If you are towing a trailer (this also applies to road trains/lorries, etc.), you may not drive faster than 80 km/h, even if the speed limit in the area is higher.
- If you are towing a trailer without brakes and actual total weight exceeding 300 kg, you may not drive faster than 60 km/h.

If you are driving with a trailer, you also need to check what speeds the trailer is approved for. E.g. if the trailer is approved for speeds up to 50 km/h, then you may not drive faster than 50 km/h.

Road grip

By grip, hold or traction, we refer to the vehicle's ability to stay on the surface of the road and move in the direction the driver wishes (i.e. not skid). Good grip is essential to be in full control of the vehicle at all times. There are two main factors that affect road grip:

- **The car's kinetic energy** — This is the energy a car has because of its weight and speed. The more kinetic energy a vehicle has, the weaker grip there is for braking and turning. A heavy car travelling at high speeds has weaker grip than a smaller car travelling at lower speeds.
- **Gripping ability between tyres and road surface** — The quality of your tyres, as well as the surface you are driving on, is the other main factor affecting grip. If you are driving on icy roads in zero degree weather with worn summer tyres, grip will be very poor, and there is a high risk that your car will slide in a different direction than you are driving.

To a large extent you are therefore able to affect road grip by selecting suitable tyres and driving at speeds that are safe for the road and driving conditions.

Calculating kinetic energy

When your speed doubles, the kinetic energy is quadrupled. This means that you need four times the energy to stop a car going 60 km/h, than a car going 30 km/h. Increasing the speed to 90 km/h will increase the kinetic energy by nine times, compared with what it was when the car was only travelling at 30 km/h.

Note!

Kinetic energy changes with the square of the change in speed.

When braking hard you risk losing grip.

Finding the square of a number

To find the square of a number, you need to multiply the number with itself. E.g. the square of 2 is (2 x 2 =) 4, the square of 3 is (3 x 3 =) 9, and so forth.

Grip and handling

Grip is used when accelerating, braking and turning. To maintain grip when going through a bend, you therefore need to reduce your speed *before* turning. Otherwise there is a risk the car's kinetic energy will push the car forward, instead of through the bend.

When you lose grip, the wheels start spinning and you lose control of the vehicle. When the wheels lock up, you also lose control. The wheels lock up when braking hard in a car without ABS brakes.

Grip may be reduced to as much as 1/4 when driving in winter conditions. Good quality winter tyres provide better grip, but you will still need to drive more carefully than you would on dry summer roads. It is often a good idea to test the driving conditions carefully before heading into traffic.

Aquaplaning

When driving on wet roads, you should be aware of the risk of aquaplaning. Aquaplaning occurs when there is water between the tyres and the road surface. The tyres lose contact with the road surface, causing you to lose control of the car. The risk of aquaplaning increases with speed, and often occurs at speeds above 60 km/h.

Braking and braking distances

To make sure all learner drivers understand how important it is to focus on driving and maintaining a safe speed, the curriculum requires you to be able to calculate changes in reaction distance/thinking distance and braking distance.

Of course, you are not expected to perform these calculations at all times while driving, but being able to calculate these things makes you more aware of the forces at play when driving, and how much braking distance increases with only a small increase in speed.

Reaction distance, braking distance and stopping distance

You are out driving and notice a hazard further down the road. You quickly decide to brake and stop before reaching the hazard. How much time passes from the hazard first appeared until the vehicle has come to a complete stop? How far does the car travel before it stops?

You might remember the elements of the driving process — sensing, perceiving, deciding and acting. Using these elements as a starting point, we will look more closely at braking.

- **Sensing** — Some time passes from the moment a hazard appears until you notice it. This time is called perception time or inattention time ('uoppmerksomhetstid' in Norwegian), and the distance you travel during this time is called the *inattention distance*. Many factors influence your inattention time, and usually you will therefore not have to calculate this in questions about braking, but keep in mind that a driver who actively uses the rules of observation, is likely to discover hazards more quickly than they otherwise would.
- **Perceiving and deciding** — The time that passes from the moment you notice a hazard until you start reacting to it, is called the reaction time, and the distance you travel during this time is called the *reaction distance*. The reaction distance is affected by factors such as the driver's level of experience, alertness and maturity.
- **Acting** — The distance you travel from the moment you start braking until the car has stopped, is called the *braking distance*. The braking distance is affected by road grip, the car's kinetic energy and how hard you brake.

Reaction distance + braking distance = stopping distance

Stopping distance consists of both reaction distance/thinking distance and braking distance. Many people confuse the terms braking distance and stopping distance, so make sure you understand the difference between them.

Calculating reaction distance

How do you calculate the reaction distance?

To calculate the reaction distance, you need to know the reaction time and the speed. Since reaction time is stated in seconds, it is useful to convert the speed to meters per second (instead of km per hour). To convert from km/h to m/s, divide the speed by 3.6. Multiply the answer with the reaction time to get the reaction distance.

Example

You are driving on a country road at 60 km/h, when you notice a hazard. It takes you 2 seconds to decide on a course of action. What is the reaction distance?

Firstly, convert the speed into metres per second: **60 / 3.6 = 16.67**
Then multiply the speed by 2: **16.67 x 2 = 33.32**
The reaction distance is therefore **approximately 33 meters.**

How do you calculate changes in reaction distance?

To calculate how much the reaction distance changes, when the speed changes, you first need to find the change in speed. You find the change in speed by calculating how many times faster or slower the new speed is in relation to the old speed: **New speed / old speed = change in speed**

The reaction distance changes just as much as the speed, so to calculate the new reaction distance, you just need to multiply the old reaction distance with the change in speed: **Change in speed x old reaction distance = new reaction distance**

Example

Your speed increases from 60 km/h to 90 km/h. That means your reaction distance increases to (90 / 60) x 33.32 = 50 meter.

Calculation tips

- To convert from km/h to m/s, divide the speed by **3.6**.
- To find the change in speed, divide the **new speed** by the **old speed**.

Calculating braking distance

How do you calculate braking distance?

Because there are so many factors that affect braking distance in a situation, calculating the braking distance is much more difficult than calculating the reaction distance.

The braking distance is affected by the speed, road surface, road and driving conditions, weight, how hard you brake, the condition the brakes are in, and many other things. Therefore, you are not expected to calculate the braking distance just by knowing the speed.

Instead there are tables of braking distances at different speeds and road conditions, and you are welcome to memorize these, but if you remember that the braking distance (when emergency braking) on dry asphalt is 2 meters when driving at 20 km/, you can calculate the braking distance for other speeds and road conditions. Let's take a look at how.

Braking distances on different road conditions

- On wet summer asphalt emergency braking distances can double compared to that on dry asphalt.
- In winter conditions emergency braking distances can quadruple compared to that on dry asphalt.

How do you calculate changes in braking distance?

The change in braking distance is mainly affected by the change in kinetic energy. Just like kinetic energy, the braking distance changes with the square of the change in speed.

To calculate the change in braking distance, you therefore need to first find the change in speed. As mentioned before, you find the change in speed by dividing the new speed by the old speed. Because the braking distance changes with the *square* of the change in speed, you also need to multiply the change in speed by itself.

To find out what the braking distance is after the change in speed, multiply the old braking distance with the square of the change in speed that you just calculated.

Example

You perform an emergency brake at 40 km/h. The braking distance is 8 meters. What would the braking distance have been if you were driving at 60 km/h?

The change in speed from 40 km/h to 60 km/h: new speed / old speed = 60 km/h / 40 km/h = 1.5

The square of the change in speed: 1.5 x 1.5 = 2.25

The braking distance after the change in speed = braking distance before change in speed x the square of the change in speed, i.e.: 8 meters x 2.25 = 18 meters

The braking distance at 60 km/h would have been **18 meters**.

Speed	Reaction distance 1 second	Braking distance dry tarmac	Stopping distance
20 km/h	5,6 meters	2 meters	7,6 meters
30 km/h	8,3 meters	4,5 meters	12,8 meters
40 km/h	11,1 meters	8 meters	19,1 meters
50 km/h	13,9 meters	12,5 meters	26,4 meters
60 km/h	16,7 meters	18 meters	34,7 meters
70 km/h	19,4 meters	24,5 meters	43,9 meters
80 km/h	22,2 meters	32 meters	54,2 meters
90 km/h	25 meters	40,5 meters	65,5 meters
100 km/h	27,8 meters	50 meters	77,8 meters

Braking distances when emergency braking on dry roads

Note!

- The braking distance changes **much more** than the speed.
- If the speed is doubled, braking distance is quadrupled. If the speed is halved, braking distance is reduced to one fourth. If you triple the speed, braking distance will be nine times as long!

Calculating stopping distance

How do you calculate the stopping distance?

The stopping distance consists of the reaction distance and braking distance. Stopping distance therefore tells you how far you travel from the moment you first notice a hazard until the car has come to a complete stop. E.g. if the reaction distance is 10 metres, and the braking distance is 20 metres, the stopping distance is 30 metres.

When calculating the change in stopping distance when the speed changes, keep in mind that:

- the reaction distance is multiplied by the change in speed

- the braking distance is multiplied by the square of the change in speed

Example

You are driving at 40 km/h. The reaction distance is 11 meters and the braking distance is 8 meters. What would the stopping distance have been if you were driving at 80 km/h?

The change in speed from 40 km/h to 80 km/h: 80 km/h / 40 km/h = 2

The reaction distance at 80 km/h: 11 meters x 2 = 22 meters

The braking distance at 80 km/h: 8 meters x (2 x 2) = 32 meters

The stopping distance at 80 km/h: 22 meters + 32 meters = 54 meters

The stopping distance increases from 19 meters at 40 km/h, to 54 meters at 80 km/h.

Summary

Highlights from chapter 6

- The rules of observation makes it easier to be both seen and understood: Look far ahead, keep your eyes moving, get the big picture, always look for a way out, and drive so that you are seen and understood.
- Cooperation facilitates a safe and efficient movement of traffic.
- Choose a speed that makes you understood: Slow down in good time to make your intentions clear.
- Position the car to make yourself understood: position the vehicle to the side if you are turning, keep to the middle of your lane if you are going straight.
- Give clear signs: Use indicator lights when turning or moving sideways. When braking your brake lights will notify road users behind you that you are braking.
- Unnecessary or inconsiderate use of sound or light signals is prohibited! Use signals (using the horn or flashing the headlights) to warn other road users of hazardous situations.
- If there are no signs indicating otherwise, the speed limit in densely populated areas is 50 km/h, and 80 km/h outside of densely populated areas.
- Adjust your speed to the area, road and driving conditions, as well as the traffic situation: You must be able to stop within the distance which you can see to be clear, and before reaching any foreseeable obstruction.
- By grip, hold or traction, we refer to the vehicle's ability to stay on the surface of the road and move in the direction the driver wishes (i.e. not skid). By selecting suitable tyres and driving at speeds that are

safe for the road and driving conditions, you can affect road grip significantly.

- *Inattention distance:* The distance you travel from the moment a hazard appears until you notice it.
- *Reaction distance:* The distance you travel from the moment you notice a hazard until you start reacting to it. To calculate the reaction distance, multiply the reaction time by the speed in metres per second.
- *Braking distance:* The distance you travel from the moment you start braking until the car has stopped. Braking distance changes with the square of the change in speed. On wet summer roads braking distances may be twice as long — in winter conditions they are easily four times as long.
- *Stopping distance:* The sum of reaction distance and braking distance.

In the next chapter we will go through positioning and give way rules, as well as proper behaviour in different traffic situations you are likely to encounter when driving in urban areas.

Calculation tips

- To convert from km/h to m/s, divide the speed by 3.6.

- To find the change in speed, divide the new speed by the old speed.
- To find the square of a number, you need to multiply the number with itself. E.g. the square of 2 is (2 x 2 =) 4, the square of 3 is (3 x 3 =) 9, and so forth.

Chapter 7- Driving in City

Introduction

In the city traffic is often heavy; there are many junctions, roundabouts, one way streets, and vehicles changing lanes all the time. In addition you have to be aware of pedestrians and bicyclists.

In this kind of traffic you will need to have a good knowledge of the rules for right of way, and interact with buses, trams and vulnerable road users. You also have to pay attention to positioning and the distance to the other road users.

What is obligation to give way?

Obligation to give way is letting other drivers drive before you. We will take a closer look at what this entails and who should give way to whom in different situations. Let us start by looking at the definition of obligation to give way in the Road traffic act.

Road users for whom one has an obligation to give way must not be obstructed or distracted.

Having an obligation to give way implies not being in somebody's way, to not be a hindrance. If there isn't any road users close by, there is no one to give way to and you can drive

A road user who has an obligation to give way shall demonstrate this clearly.

In good time reduce your speed so that the other road users realize that you intend to give way. If drivers on the priority road have to reduce their speed because they are uncertain about what you are planning, you have violated your obligation to give way.

Priority-to-the-right rule

The priority-to-the-right rule is the general rule of obligation to give way.

This means:

- Unless signs or other traffic rules takes precedence, you always have to give way to traffic coming from the right at a junction.
- The same applies when you intend to turn left at a junction and approaching vehicles will end up on your right hand side.

Everybody has to give way!

From time to time you will run into traffic situations where everybody has an obligation to give way. Who should take the initiative to go first?

This is a typical situation where cooperation is needed: Initially, the driver arriving first should go first, but it is crucial that the driver communicates these intentions and that the other drivers show sign of giving way.

The grey car is taking a left turn and has to give way to the approaching car.

Obligation to give way - other regulations

In addition to the priority-to-the-right rule there are other obligations to give way rules addressing certain places or situations.

Obligation to give way when you are coming from a parking lot, bus stop etc.

When you are pulling out from a parking spot, private road, courtyard or similar places you always have to give way to the traffic on the road. The same applies if you are coming from the curb, road shoulder or when you want to cross a cycle path or the pavement.

Obligation to give way for the bus

On roads with speed limit 60 km/h or lower you have to give way for the buss if the driver is signaling.

Giving way on narrow roads

If the road is too narrow to let two cars pass each other safely, both parts have to slow down and try to make sufficient space for passing. On longer stretches of narrow road you will sometimes find meeting points where you can stop and let the oncoming traffic pass.

If part of the road is obstructed on your side, you have to give way for oncoming traffic and vice versa.

Give way when changing lane

When you are about to change lanes you will have to give way to vehicles in the lane you are changing to. This also applies if you are coming from the curb, shoulder or are making other lateral movements. If the lanes in one direction are merging into one lane, traffic in both lanes needs to take care to accommodate each other in turns, so that traffic merges like a zipper.

Changing lanes

- Use your indicator lights to inform the other drivers and check your mirrors and blind spots before you change the lane. Remember that you also have to give way to the traffic coming from behind before changing lanes.
- Make the turn early to better observe and make sure the other drivers understand your intentions. The regulations require you to show your

intentions to give way, so be absolutely sure that you can change lanes without obstructing the other traffic.

Give way to pedestrians

Pedestrians are protected by regulations in the traffic rules and the police enforce these regulations with vigor.

Giving way to pedestrians and cyclists:

- At pedestrian crossings where traffic is not controlled by the police or by traffic lights, drivers shall give way to pedestrians who are already on the pedestrian crossing or who are about to enter it.
- Drivers intending to turn shall give way to pedestrians and cyclists going straight ahead.
- Drivers intending to turn shall give way to pedestrians and cyclists on the road being entered.
- Drivers intending to cross the sidewalk shall give way to pedestrians and cyclists.

Provide free passage

In order to ease the navigability for certain road users, the traffic rules state that you must take appropriate action to provide free passage. Providing free passage means letting them pass and reducing the risk for accidents by pulling over to the side of the road.

You should always provide free passage to

- emergency vehicles with flashing blue lights (ambulance, police car, fire truck
- trams and trains

You should also give free passage to

- Pedestrians in groups under the supervision of a leader
- Processions and funeral processions
- Convoys of military vehicles or convoys of civil defence vehicles

Note!

Emergency vehicles are exempted from observing speed limits and other rules of the road providing they use due regard for the safety of others.

Give way and priority signs

In some cases give way and priority signs are used to improve traffic flow.

Give way

- This sign tells you that you have to give way to traffic from both directions on the crossing road.

Give way

Stop

- You have to give way to traffic on the crossing road and it is mandatory to stop completely as close to the stop line or the crossing road as possible before continuing
- The sign is used in hazardous and complex crossroads where parts of the crossing road are obstructed from view.

Stop

Priority road and priority crossroads

It can be beneficial for the general traffic flow if drivers on the larger roads where traffic is heavy don't have to give way for traffic coming from the right. These roads are called priority roads.

Priority road

- This sign informs that you have priority at all following crossroads up to an end of priority road, give way or stop sign.

The supplementary sign informs you that you are driving on national road 165.

Priority Crossroads

- The sign warns of a dangerous junction where road users on the crossing road have to give way. Note, however, that the sign does not indicate whether the road you are driving on is a priority road or not, and it can therefore stand both on priority roads and ordinary roads.

Underskiltet informerer om hvor langt det er til krysset

Oncoming traffic and giving way

On some narrow stretches of road where it is difficult for two cars to safely pass each other, the right of way can be determined by signs.

Give way to oncoming traffic

- While this sign might look like a prohibitory sign, it is actually a type of give-way sign. The sign is called give way for oncoming traffic, and indicates that you are prohibited from driving onto a narrow stretch of road if this will hinder oncoming traffic

You have to stop for oncoming traffic.

Oncoming traffic has to give way

- This sign is a type of give-way sign, and indicates that *oncoming traffic must give way.* This sign is normally posted on narrow roads where there is not enough room for two cars to pass each other.

Oncoming traffic has to give way.

As a rule of thumb: The red arrow represents the one who has to give way.

Supplementary signs

Course of priority road

- The sign shows a diagram of the junction, whereas a thick line indicates a priority road and a thin line indicates roads where Give-way or Stop signs are posted.

Course of priority road

Positioning on the road

Correct positioning on the road sends a clear message to the other road users about your intentions and helps prevent conflicts with them.

Positioning on the road is to some extent regulated in the traffic rules:

- Insofar as circumstances permit, vehicles shall be driven on the right-hand side of the road
- Drivers shall keep well within the lane boundaries (the lane is in this case the area between the yellow and white road markings)
- On carriageways with two or more lanes in the same direction, the right-hand lane shall be used when the traffic rules do not require or permit use of the left-hand lane(s)
- If you are driving on a road with multiple lanes in the same direction, you must position your car before making a turn. If you are going left, chose the left lane, if you are going to the right chose the right lane.
- Bicycles and other vehicles that do not have a motor may be driven on the right-hand road shoulder.

Distance

When you are driving you have to think about your position relative to the other road users.

Why is distance in the traffic important?

Keeping a good distance to other road users decreases the risk for accidents.

A good distance will give you a better overview in the traffic and you will have more time to react if something unexpected happens.

What is a good distance?

In the traffic rules there is a regulation where it is stated:

- The distance to the car in front should be such that there is no risk of collision should the car in front slow down or stop.
- You should keep enough distance for vehicles overtaking to be able to pull in between you and the vehicle in front.

Three second rule

What is enough distance varies from situation to situation. As a rule of thumb, you should at least always have a distance of 3 seconds.

Situations where 3 seconds isn't enough

- The roads are slippery.
- Driving at high speeds (above 80 km/h).
- The car behind you is larger/heavier than your own vehicle (as the braking distance will be longer).

Multiple driving lanes in the same direction

On carriageways with two or more lanes in the same direction, the right-hand lane shall be used when the traffic rules do not require or permit use of the left-hand lane(s)

When signs or road markings allow or order, the middle or left lane can be used.

Remember the rules to give way when changing lanes.

In the traffic rules there are specified instances where the middle or left lane can be used

- The middle or the left-hand lane can be used for overtaking.
- The middle or the left-hand lane can be used for when traffic is heavy and this will help with the flow of traffic.
- Left-hand lane is to be used if you are planning to take a turn to the left in the upcoming junction.

Public transport lane

You are not allowed to use the right-hand lane if this is a public transport lane unless you are driving an electric or hybrid car or a two-wheeler.

One-way traffic

In many one-way streets there is enough space for traffic in two columns. You should position yourself like you would on a multi-lane road.

If you are going to the left in the end of the street, you should position the vehicle to the left in the carriageway.

If the road isn't wide enough for traffic in two columns, positioning should be like on a regular road.

If you are driving in the left-hand lane in a one-way street remember that:

- Pedestrians and cyclists might not be aware that the street is one directional and thus not prepared to meet traffic in the left-hand lane.
- At the end of the road traffic coming from the left might not be aware that there is traffic coming from the left-hand lane.

Space enough for cars in two columns

Signs and choosing lanes

Sometimes your position in the road is determined by mandatory signs.

Mandatory driving lane

This sign informs that drivers must pass the sign on the side to which the arrow is pointing.

Mandatory driving lane

Free choice of lane

This sign is typically in use in one way streets with two lanes.

Free choice of lane

Mandatory driving direction

The sign states that drivers must leave the junction in the direction indicated on the sign.

Mandatory driving direction

Mandatory roundabout

Traffic rules regarding roundabouts apply.

Mandatory roundabout

Junctions

A junction is defined in the traffic rules as any point at which a road crosses or merges with another road.

When you are going to make a turn in a junction you should think about your position and show the other road users what you are planning to do. The general rule is that you should keep to the right if you are going to the right, and keep to the left if you are going to the left. If you are going straight ahead you should take a middle position of your lane or in the right lane if there are more than one lane.

Junctions with traffic lights

In heavily trafficked junctions and pedestrian crossings, traffic is often regulated with traffic lights.

If the traffic light is flashing yellow or not working, traffic signs and the give way rule applies. If there are give-way signs in the junction then you have to the give way to traffic from both sides on the crossing road, if there are no signs then the give way rule applies. Rembember that you always have to give way to pedestrians crossing a pedestrian crossing when the traffic lights are flashing yellow.

Junction with arrow signals

The arrow signal only applies to you if you are going in that direction. When the arrow turns green you don't have to give way to anybody (pedestrians crossing lights will be red).

Remember to choose the correct lane based on where you are going in the junction.

When you are turning left in a junction

- Turn on your indicator lights, position yourself towards the middle of the carriageway and reduce your speed early.
- Make sure that you don't overdo your positioning to the left and obstruct oncoming traffic.
- If both you and the oncoming vehicle are going to your respective left, you should pass each other on the left hand side. This helps the flow of the traffic in a junction.
- In a situation like this you must always be alert to (motor) cyclists or small cars that may be hidden behind the foremost oncoming vehicle.

- If oncoming vehicles are going straight through the junction you will have to give way and wait for them to pass before you can make you turn.
- Be aware that there might be crossing pedestrians on the road being entered.

Remember correct use of indicator lights, positioning and reduction of speed.

Turning right in a junction

- Turn on your indicator lights, position yourself towards the shoulder/curb and reduce your speed early.
- By taking this position early you prevent two-wheelers from coming up on your right hand side and you reduce the risk of conflict. If the two-wheeler is going straight ahead in the junction and you are going to the right you might run into each other, especially if the two-wheeler is hidden in you blind spot.

Be aware of crossing pedestrians when turning right.

Roundabouts

The rules for driving through a roundabout are something many drivers struggle with. It might help if you think about a roundabout as an ordinary junction.

And just like in an ordinary junction it is crucial that the other road users get a clear understanding about your intentions and which direction you are planning to drive.

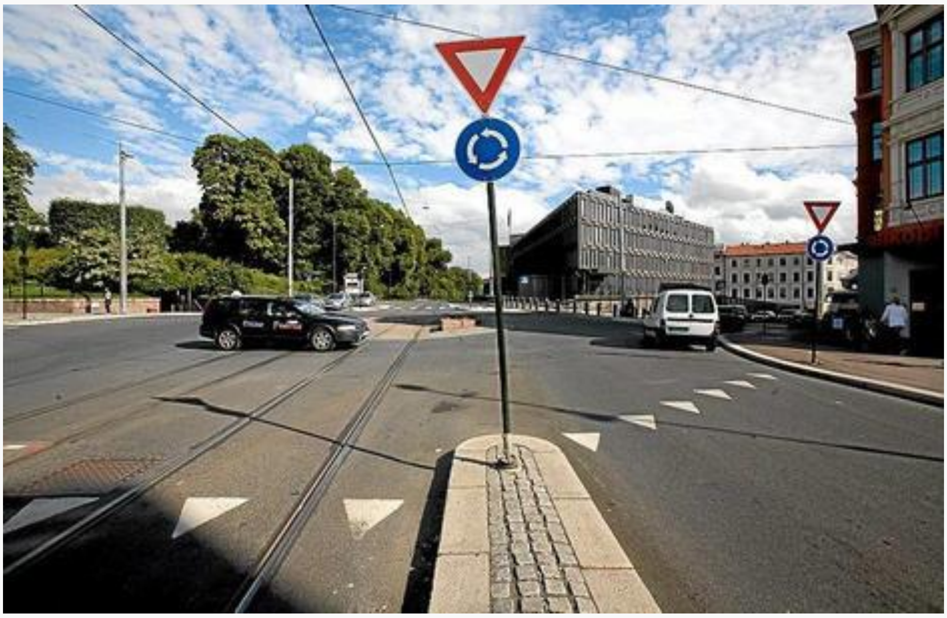
- Give sign with your indicator lights early
- Position your vehicle correctly before entering the roundabout
- Slow down so that you have time to get an overview, and at the same time signaling that you will give way for traffic in the roundabout.

Giving way in a roundabout

In Norway almost every roundabout has give-way signs. This means that you have to give way for traffic already driving in the roundabout.

Make sure that your speed is sufficiently low when you are approaching a roundabout. This makes it easier for you to observe if there are pedestrians right after the roundabout. It also sends a signal to the other road users that you are going to give way.

Remember that you have to give way when changing lanes in a roundabout, so check your mirrors and blind spot before changing lane.



The majority of roundabouts in Norway have give way signs.

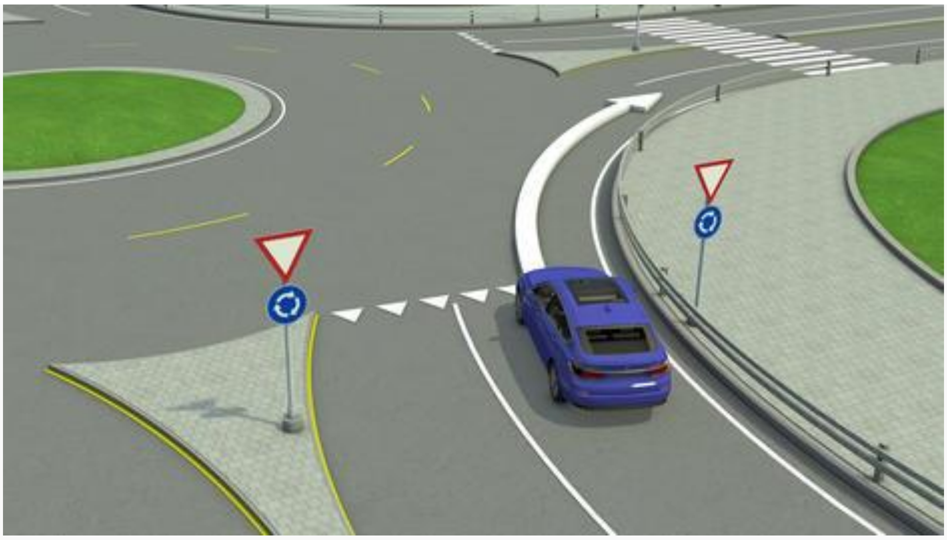
You will always have to give way for traffic coming from your left in a roundabout.

Going to the right in a roundabout

When you are going to the right in a roundabout you should always choose the right lane - Regardless of whether the roundabout has one or two lanes.

This is how you drive:

- Turn on your indicator lights in good time ahead of the roundabout. This will inform the other drivers about where you are heading
- Position your vehicle to the right hand side of your lane and reduce speed to get a better overview. Reducing speed also shows that you will give way if necessary.
- Enter the roundabout when there is a large enough gap and turn out of the roundabout on the right hand side.
- Remember to look for pedestrians as you leave the roundabout.



Turn on your indicator lights early.

Going straight ahead in a roundabout

In a roundabout with one lane

- Position your vehicle to the right hand side of your lane and reduce speed to get a better overview. Reducing speed also shows that you will give way if necessary
- Enter the roundabout when there is a large enough gap.
- Turn on your indicator lights when you drive past the first exit of the roundabout. This signals that you are about to leave the roundabout and oncoming traffic can enter the roundabout.
- Remember to look for pedestrians as you leave the roundabout.

In a roundabout with two lanes

- As a general rule you should choose the right lane before entering the roundabout, unless road markings or signs indicate otherwise.
- If you chose the left lane you will have to make a lateral lane change before leaving the roundabout. Remember that you have to give way when making lateral lane changes.

A roundabout with two lanes

Going to the left in a roundabout

In a roundabout with one lane

- Turn on your indicator lights well ahead of the roundabout. This will inform the other drivers about where you are heading.
- Position your vehicle to the left hand side of your lane and reduce speed to get a better overview. Reducing speed also shows that you will give way if necessary
- Enter the roundabout when there is a large enough gap.
- Turn on your indicator lights when you drive past the next to last exit of the roundabout.
- Remember to look out for pedestrians as you exit the roundabout.

In a roundabout with two lanes

- If you are going to the left in a roundabout with two lanes you should choose the left lane before entering the roundabout.
- When you drive past the next to last exit you must change lanes in order to exit the roundabout from the correct lane.

Going to the left in a roundabout

Reversing

When you reverse or turn around you have to give way to other road users.

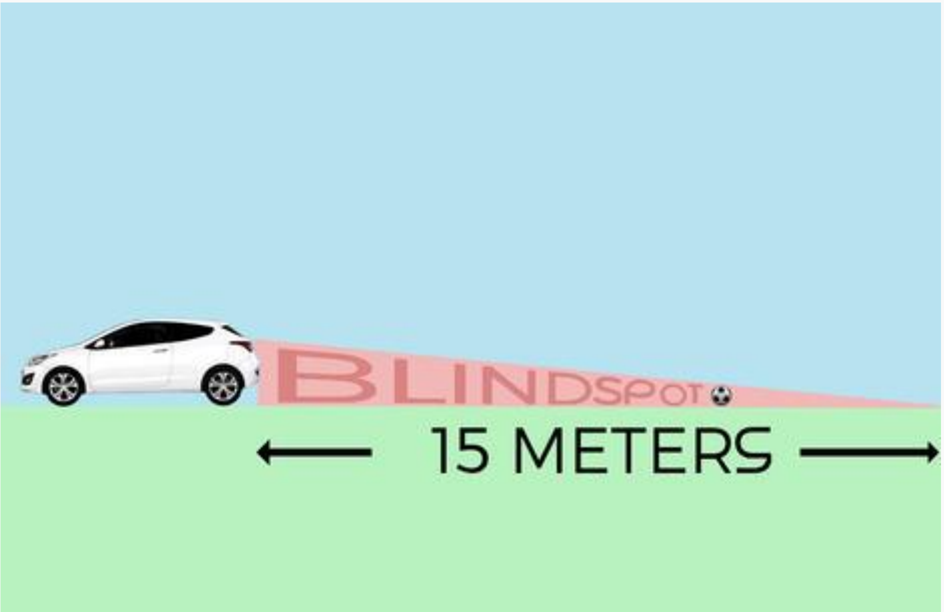
Important regulations regarding reversing and turning:

- If your view is inadequate, you must not reverse or turn around except under the guidance of another person or if you have made sure that danger or injury cannot arise as a result. The person guiding you should be behind the car, preferably your left hand side.
- It is prohibited to reverse or turn around on a motorway and on a 'road for motor vehicles' or in a motorway or 'road for motor vehicles' acceleration or deceleration lanes
- Maintain a low speed when reversing
- Remember to also monitor the front of the car when reversing, as the front will move sideways when you reverse and turn at the same time.

Hidden in the blind spot

Children are unpredictable and can be difficult to spot, especially when they disappear in your blind spot behind the car. Many modern cars have small rear windows with a high orientation and in some cases you can get a blind spot reaching between 7 and 15 meters behind your car.

Never trust that you can see everything from the driver's position. Take an extra check around the car if you're not 100 % certain.



The blind spot behind the car can be considerable.

U-turn

Because of the risks associated with reversing it is recommended that you try to make a U-turn without actually reversing.

The safest strategy is to make the turn on the side of the road or on the shoulder if it is broad enough. If this is not possible you should consider making a U-turn.

You can make a turn/U-turn on all roads except on a motorway, 'road for motor vehicles' or in motorway and 'road for motor vehicles' acceleration or deceleration lanes.

It is not allowed if there is a No U-turn-sign.

No U-turn allowed

Stopping and parking

Stopping and parking on public roads is regulated in the traffic rules and with traffic signs.

What counts as parking?

According to the traffic rules, parking is defined as 'any stationing of a vehicle' - even if the driver does not leave the vehicle - with exception of shortest possible stops for the purpose of embarking or disembarking or loading and unloading

What counts as stopping?

Parking is defined 'any stationing of a vehicle, even if the driver does not leave the vehicle', with the exception of stops as described above.

Example of stopping and parking

- If you're stopping to let a person off, this counts as stopping but not parking. So it is allowed to let passengers off even if there is a sign with 'No parking' present.
- This is not allowed if the sign is a 'No stopping' sign instead.

Remember!

You always have to consider the basic rule of traffic when you are planning on stopping or parking.

This means that you should not stop or park if this in any way can lead to dangerous situations or if this will interfere with the traffic in any way.

You are also obligated to secure your car to prevent unauthorized use of it when you leave it behind. This means that you have to lock your car.

Parking signs often have supplementary signs showing when parking is allowed.

Stopping - the traffic rules

It is prohibited to make a stop at the following places:

- On a bend with restricted visibility, in tunnels, on the crest of a hill or at any other place where visibility is restricted.
- In a junction or less than 5 meters from a junction.
- Partly or wholly on pavements, foot paths or cycle tracks.
- On pedestrian crossings or bicycle crossings or less than five metres before such crossings.
- On motorways and 'roads for motor vehicles' or on motorway and 'road for motor vehicles' acceleration or deceleration lanes
- Less than 5 meters from a level crossing.
- In lanes reserved for public transport, in high-occupancy vehicle lanes or in bicycle lanes, except for buses or trams at stops
- At a widening of the road for a bus, taxi or tram stop or less than 20 meters from an official traffic sign for such stop. Stopping for disembarkation or embarkation purposes is excepted if it does not obstruct buses, taxis or trams

It is prohibited to stop in tunnels.

Parking - The traffic rules

In addition to the places listed in the last slide there are some places where parking is not allowed.

Parking is prohibited in the following places:

- In front of entrances and exits for vehicles
- In passing places, in the full breadth of the road and for the whole length of the widened section of the carriageway
- In pedestrian streets
- In pedestrianized residential streets other than in specially marked spaces
- On the carriageway on priority roads with a speed limit greater than 50 kph

Note that there are no regulations against parking on the left side of the road, but the basic rule of traffic always applies - so parking on the other side of the road is only allowed provided you are not causing any danger or interference with traffic.

You can stop to let of a passenger, but parking is not allowed here.

Parking — the signs

No parking

This sign indicates that it is prohibited to park on the side of the road where the sign is posted. You are allowed to briefly stop to load/unload goods or allow people to get in or out of your car.

The sign applies up until a junction, or a new parking sign (*No stopping*, *No parking*, *Parking zone*, *End of parking zone* or *Parking*) is posted.

No parking

No stopping

This sign indicates that it is prohibited to stop – **not even for letting someone on or off** – on the side of the road where the sign is posted. The sign applies from where it stands and up until a junction, or until another sign (*No stopping*, *No parking*, *Parking zone*, *End of parking zone* or *Parking*) is posted.

No stopping

Parking allowed

This sign indicates that parking is permitted, as long as it is not in violation of any of the traffic rules concerning stopping or parking.

If the sign is accompanied by supplementary sign 'Extent of stopping or parking regulation', it is permitted to park in the direction indicated even though such parking goes against the standard provisions in the traffic regulations concerning stopping and parking.

If the supplementary sign states that parking is reserved for specific vehicles or group of road users, parking is prohibited for others

Parking sign

Supplementary signs - Time

A parking sign will often have a supplementary sign indicating when it is allowed to park there. If this supplementary sign is used in conjunction with a *No parking* or a *No stopping* sign it will indicate when it is illegal to park or stop at the place.

Example:

If this supplementary sign (se picture) is used in conjunction with a *No parking* sign it would mean that parking is prohibited:

- Monday to Friday between 8:00 and 17:00.
- Saturdays between 8:00 and 15:00.
- Sundays and public holidays between 18:00 and 20:00.

Please note:

- Black figures apply to weekdays except Saturdays.
- Black figures in parentheses apply to Saturdays.

- Red figures apply to Sundays and public holidays.

The supplementary sign indicates when the *No parking* sign applies.

Summary

Highlights from chapter 7

- Road users for whom one has an obligation to give way must not be obstructed or distracted. A road user who has an obligation to give way shall demonstrate this clearly in good time by reducing his speed or stopping.
- The priority to the right rule is the general rule of giving way.
- Insofar as circumstances permit, vehicles shall be driven on the right-hand side of the road. On carriageways with two or more lanes in the same direction, the right-hand lane shall be used when the traffic rules do not require or permit use of the left-hand lane(s)
- It is permitted in the traffic rules to use the center or the left-hand lane when overtaking, or if necessary for an efficient traffic handling.
- When you are making a turn, position your vehicle to the corresponding side of the lane. Remember indicator lights, reducing speed and your position.
- Always keep at least a 3 second gap to the vehicle in front of you.
- When you are changing lanes you have to give way to vehicles driving in that lane. The same rule applies when you are coming from the curb or any other lateral positioning.
- The rules for driving in a roundabout and a junction are the same. When you're entering a roundabout you normally have to give way to vehicles already in the roundabout.
- Drivers intending to reverse or turn around shall give way to other road users. If your view is limited you will need someone to guide you.
- Parking is defined in the traffic rules as any stationing of a vehicle, even if the driver does not leave the vehicle. Exceptions are shortest possible stops for the purpose of embarking or disembarking or loading and unloading.

In the next chapter we will take a look at driving in high speed traffic including highways, tunnels and level crossings. We will also take a closer look at overtaking, night driving and driving abroad.

Chapter8- Driving on Country Roads

Introduction

Driving on country roads and motorways differs from driving in a city environment in many ways. Speeds are normally higher, and there are some situations you are more likely to encounter in this type of traffic.

Some situations, such as driving in the dark and overtaking, can be both difficult and even dangerous without the necessary skills and knowledge.

Motorways

Motorways are high-capacity roads designed for safe driving at higher speeds. Statistically, you are less likely to get in an accident on a motorway than on other roads.

Motorways must comply with several requirements:

- Carriageways for opposite directions of travel are separated by a median strip or central traffic barrier (dual carriageway).
- There are at least two lanes in each direction of travel.
- There are no crossings on the same level; other roads, railways and footpaths cross through tunnels or bridges.
- Properties along the road do not have direct access to the motorway.
- All exiting and entering is either at the beginning or end of the road, or through slip roads and designated acceleration and deceleration lanes.

In addition to properties of the road, there are special provisions regarding driving on motorways:

- Motorways are only for vehicles that can drive at speeds of at least 40 km/h.

- Mopeds are not permitted on motorways, because they may be difficult to spot and accidents may occur as a result of turbulence from larger vehicles.
- Pedestrians and cyclists are not allowed on motorways.
- It is prohibited to reverse, turn around, stop or park on motorways.

On motorways, traffic moving in opposite directions is always physically separated.

This sign indicates that the road is classified as a motorway.

Semi-motorways

A semi-motorway, also called a "road for motor vehicles", is also intended for driving at higher speeds, but is usually not of the same standard as a motorway. Semi-motorways are for instance not required to have more than one lane in each direction of travel, and traffic moving in opposite directions is not necessarily physically separated.

Semi-motorways do not have many physical characteristics, but the roads are generally of good quality, and similarly to motorways, properties along the road do not have direct access to the road.

The same traffic rules apply on motorways and semi-motorways:

- Semi-motorways are only for vehicles that can drive at speeds of at least 40 km/h.
- Mopeds are not permitted on semi-motorways.
- Pedestrians and cyclists are not allowed on semi-motorways.
- It is prohibited to reverse, turn around, stop or park on semi-motorways.

Note how there is only one lane in each direction of travel, and that there is no median strip or central traffic barrier on this semi-motorway.

This sign indicates that the road is classified as a semi-motorway.

Speed limits on motorways and semi-motorways

The speed limit will normally be posted. The highest speed limit in Norway today is 110 km/h. If there are no signs indicating otherwise, the general speed limit outside densely populated areas applies — 80 km/h.

As mentioned before in this course, the maximum permitted speed for cars towing a trailer is 80 km/h. This applies everywhere, also on motorways and semi-motorways. Also remember that the maximum permitted speed for towing a trailer that is not equipped with brakes and with an actual total weight exceeding 300 kg, is 60 km/h. This also applies everywhere.

Acceleration lanes

On motorways, all entry is through dedicated acceleration lanes (except where the motorway begins). Acceleration lanes may be found on other types of roads as well.

On acceleration lanes, the rules for merging (the zipper rule) apply:

It is stated in the traffic rules that drivers "in acceleration lanes shall adjust their driving speed to the speed of the traffic in the lane to be entered."

It is also stated that drivers on the motorway are obliged to "facilitate the merging of traffic from the acceleration lane" by

- adjusting their speed
- keeping enough distance to the vehicle in front that it is safe to pull into the gap between them
- changing over to left-hand side lanes, if necessary

Use the acceleration lane to adjust your speed to the traffic on the motorway.

Deceleration lanes

On motorways, all exiting is through dedicated deceleration lanes (except where the motorway ends). Deceleration lanes may be found on other types of roads as well.

Using the deceleration lane

It is stated in the traffic rules, that "on carriageways with two or more lanes in the direction of traffic a driver intending to make a right turn shall in good time move into the lane furthest to the right." This means that

- you must move over to the right-hand lane in good time (before reaching the deceleration lane)
- you must quickly move over to the deceleration lane when reaching it. Remember to check your blind spots before changing lanes!
- unless the traffic situation demands it, you must maintain your speed until you have completely entered the deceleration lane (so as not to obstruct traffic behind you)

In the deceleration lane you can reduce your speed safely, without obstructing traffic behind you.

Lane reduction

Merging lanes

The normal rules for changing lanes do not apply when the number of lanes is reduced by two lanes merging into one.

In this case, drivers in both lanes need to take care to accommodate each other in turns, so that traffic merges like a zipper.

But be sure to signal clearly and in good time, and check your mirrors and blind spot before changing lanes.

End of lane

Sometimes the number of lanes is reduced by one of the lanes ending.

When a land ends, road users in that lane must yield as they normally would when changing lanes.

Tip!

An easy-to-remember rule is that the red arrow on the sign indicates the person who has a duty to yield. If both arrows are red, then neither has the right-of-way, and the rules for merging apply.

When the number of lanes is reduced, the rules for merging usually apply.

Priority roads

To ensure efficient traffic flow on main roads with high traffic volume, these roads are often classified as priority roads. The sign *Priority road* indicates that the road is a priority road, and that road users on side roads and crossing roads have a duty to yield as indicated by *Give way* or *Stop* signs on these roads.

The sign is usually repeated after every junction, to inform anyone joining the road, and is cancelled by the sign *End of priority road*, or a *Give way* or *Stop* sign.

When joining the priority road from a side road, you must give way to traffic already on the priority road. This means that you must not obstruct or inconvenience road users on the priority road. You should therefore check the rear-view mirror and quickly increase the speed when driving onto a priority road, so that road users on the priority road do not have to slow down to accommodate you.

There are special provisions that apply on priority roads:

- It is prohibited to park on the carriageway of priority roads with a speed limit greater than 50 km/h.
- Overtaking right before or in a junction is permitted on priority roads.

Improving traffic flow

During rush hour there will often be a continuous flow of traffic on priority roads, making it difficult for drivers coming from a side road to enter the road without obstructing or inconveniencing road users on the priority road. When driving on a priority road, you should show consideration and sometimes let drivers from side roads in ahead of you to facilitate better traffic flow.

The sign indicates that the priority road ends, and normal give-way rules apply.

Overtaking

As a driver, you will often find yourself in situations where you may have to overtake someone.

It is therefore important that you understand the risks involved, and establish routines for assessing if and how to overtake someone.

In general, overtaking is only permitted on the left-hand side, but in some situations it is permitted to pass on the right-hand side. There are also situations in which overtaking is prohibited.

Because visibility is restricted by the upcoming bend, the road is marked with a warning line.

Overtaking on the right

The vehicle in front is turning left

If the vehicle in front is turning left in the upcoming junction, and is waiting for the road to clear, you may pass on the right-hand side. There has to be sufficient space to pass safely. Be aware that if the vehicle you are passing is large, it may be difficult for oncoming traffic to notice you.

In this situation the car in front is turning left, and you are therefore permitted to pass on the right.

Heavy traffic in both lanes

If there is more than one lane in the direction of travel, and there is heavy traffic in both lanes of traffic, overtaking on the right is permitted when following the flow of traffic. This typically occurs during rush hour, on the main roads into and out of larger cities, or right before major holidays.

When there is heavy traffic in both lanes, you are permitted to pass on both sides.

Trams

When overtaking a tram, it is permitted to pass on both sides, unless doing so violates any other traffic rules, signs or road markings. Look out for any passengers getting off the tram when passing.

It is permitted to pass trams on the right-hand side, but look out for passengers getting off the tram.

Situations where overtaking is prohibited

Overtaking is prohibited

- when visibility is in any way restricted, e.g. by a bend or a hilltop (unless overtaking in a lane that may not be used by oncoming traffic)
- when approaching a pedestrian crossing, if visibility of the crossing is obstructed
- when signs prohibit it or warn of situations where overtaking is especially dangerous

- when road markings prohibit it
- right before or in a junction

Overtaking is nonetheless permitted in or right before a junction when

- crossing traffic has a duty to yield
- the carriageway has two or more lanes in the direction of travel
- the vehicle in front turns left or is clearly preparing to make a left turn, and you are overtaking on the right
- traffic in the junction is controlled by traffic lights or by the police

Overtaking two-wheel mopeds and motorcycles keeping to the right is nonetheless permitted

- right before or in a junction
- when visibility is not obstructed

The sign indicates that overtaking is prohibited. The prohibition applies to motor vehicles with more than two wheels. The prohibition does not apply to overtaking on the right, when this is otherwise permitted.

Before overtaking

Before overtaking another vehicle, you must ensure that

- overtaking is permitted
- the road is clear of obstructions and oncoming traffic sufficiently far ahead, so that you are able to safely overtake the other vehicle
- the vehicle in front has not indicated that it wishes to change lanes or overtake someone
- no vehicles behind have started overtaking
- it is clearly possible to quickly re-enter the stream of traffic, should this be necessary

The road markings indicate whether overtaking may be safe:

- A lane line indicates that it is safe to overtake, as long as there is no oncoming traffic.
- A warning line indicates that visibility is too restricted for overtaking to be safe.
- A prohibition line indicates that overtaking is prohibited. Crossing a prohibition line is grounds for license suspension.

Overtaking and being overtaken

When you are overtaking another vehicle, you must

1. use your direction indicator lights to signal other road users
2. check your mirrors and blind spot before pulling out
3. keep a safe distance when passing the other vehicle
4. signal to indicate you are pulling back into the lane
5. pull back into the lane only when you can see the front of the other vehicle in your rear-view mirror

When you are being overtaken, you are required to facilitate a safe overtaking:

- Stay as far to the right as possible.
- Do not increase your speed, making the overtaking distance longer for the vehicle overtaking you.
- Slow down or pull over, if necessary. Standing your ground in a situation like this could have very dangerous or even fatal consequences!
- Use your mirrors actively, and remember to check your blind spots.

Overtaking distance

The overtaking distance is the distance it takes to pass another vehicle, from pulling out of your lane until you pull back in ahead of the vehicle you have passed.

It is difficult to calculate exactly how many meters you need to overtake someone, as there are many factors that affect the situation, such as how close you are to the vehicle in front when you pull out, how fast you are going when pulling out, how quickly you pull out, etc.

You are therefore not expected to be able to calculate this, but you are expected to understand how the difference in speed between your vehicle and the vehicle you are overtaking affects the overtaking distance.

This difference in speed is called the **relative speed.** If the relative speed doubles, the overtaking distance is halved.

Example

You are overtaking a car moving at 70 km/h. You are driving at 80 km/h. The relative speed is therefore 10 km/h, and the overtaking distance is approximately 500 meters.

If the vehicle you are overtaking reduces his speed to 60 km/h, while you still travel at 80 km/h, the relative speed will be 20 km/h — i.e. double what it was.

Since the overtaking distance is halved when the relative speed is doubled, the overtaking distance would be 250 meters if the relative speed increases to 20 km/h.

Clear stretch of road needed for overtaking

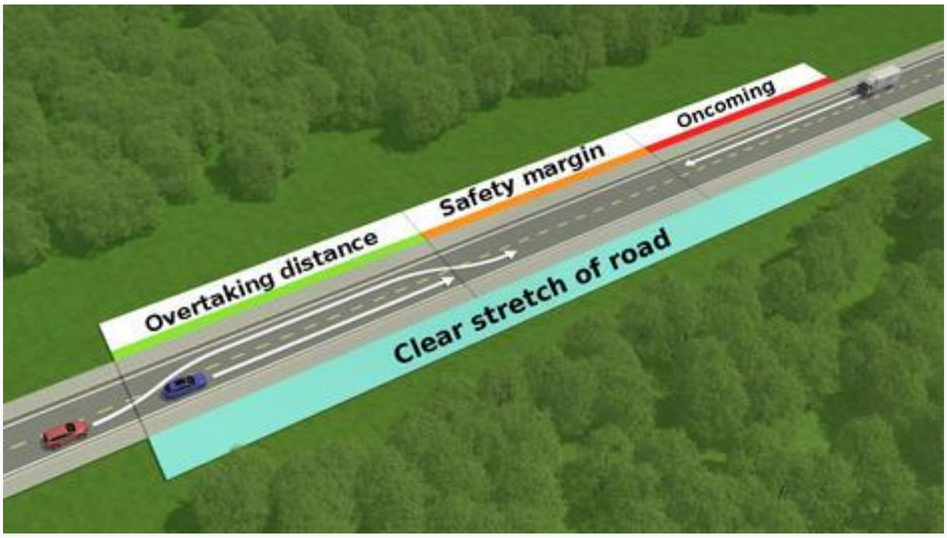
One of the most important and most difficult assessments you have to make before overtaking someone, is calculating how long it will take you and if there is a sufficiently clear stretch of road to safely overtake another vehicle.

The clear stretch of road needed for overtaking consists of three parts:

- the overtaking distance
- a safety margin equal to the overtaking distance
- the distance travelled by oncoming traffic

If we assume that oncoming traffic is travelling at about the same speed as you are, we can simplify this as: **overtaking distance x 3 = clear stretch needed for overtaking**

This means that if the overtaking itself takes 250 meters, you will need a clear stretch of 750 meters to safely overtake someone.



The clear stretch needed for overtaking consists of the overtaking distance, a safety margin equal to the overtaking distance, and the distance travelled by oncoming traffic while you are overtaking.

Driving in the dark

As you know, as much as 90 % of the information you take in while driving is through sight. Obviously then, a lot of information is lost when driving in the dark, and accident statistics show that the risk of accidents is much higher in the dark, than in daylight.

The risk of running into pedestrians, especially, is significantly higher in the dark, than when driving in daylight.

What you see in the dark depends on how much light your car emits, and how much of this is reflected back into your eyes:

- Dark clothing reflects very little of the light from the car's headlights (approx. 5-10 %).
- Light clothing reflects a lot more, up to 80 % of the light.
- A decent reflector will reflect almost all the light emitted by the car.

The correct speed = your responsibility!

Both the Road Traffic Act and the traffic rules require you to adjust your speed to the conditions. When driving in the dark, this means you often have to reduce your speed to drive safely.

Limitations of the eye

The human eye functions well in the dark, but only after gradual adaptation. It takes about 30 minutes for your eyes to get used to the dark after having been in out in daylight. If you are exposed to bright lights in the dark, your night vision worsens, and it takes another 30 minutes for your eyes to accustom themselves to the dark again.

Because it takes time for your eyes to adapt to the dark, be aware that your vision will be impaired when

- driving from daylight into a tunnel or dimly lit car park
- coming out of a well-lit place, such as a service station or convenience store
- you have been blinded by the lights of an oncoming vehicle

Blinding

Problems with glare and blinding are common when driving in low sun, or when driving in the dark and your eyes are exposed to bright light from the headlights of oncoming cars.

Tips for handling glare and blinding

- Slow down immediately if you've been blinded.
- Avoid staring into the headlights of oncoming cars when driving in the dark. Fixing your eyes to the right-hand side of the road helps to

avoid blinding, and makes it easier to notice pedestrians, animals or obstructions along the road.

Stopping and parking in the dark

There is a special provision in the traffic rules stating that parking lights "shall be switched on while a vehicle is standing still or parked on a road when the light or visibility conditions make it necessary in order to make the vehicle visible to other road users."

It is prohibited to use dipped or main beam lights when the vehicle is stopped or parked in the dark, as these lights may blind oncoming traffic and reduce visibility.

Use of lights when driving in the dark

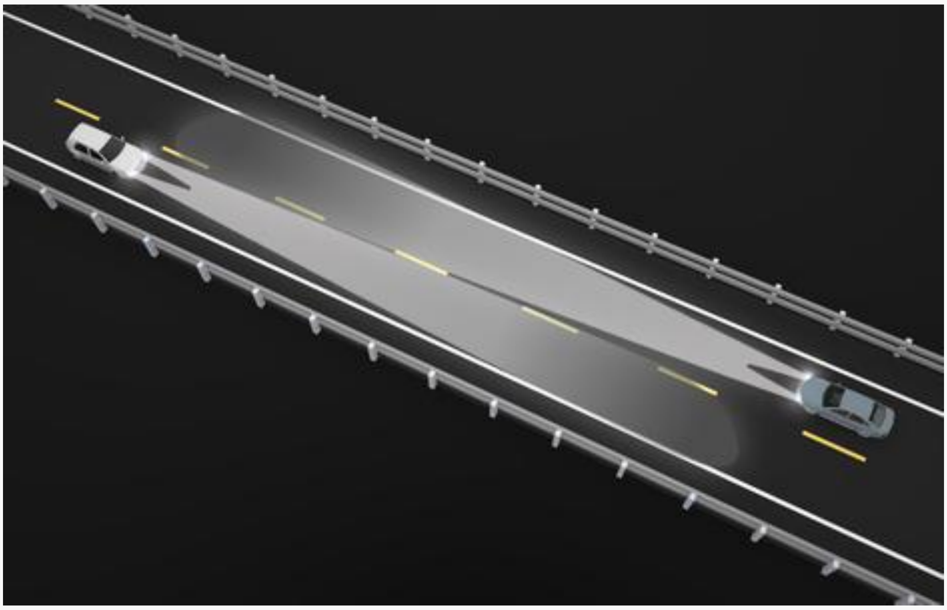
When driving in the dark, it is essential that you use the car lights properly. As you learned in chapter 2, the main beam lights illuminate the road at least 100 meters ahead, while the low beam (dipped) lights only illuminate the road at least 40 meters ahead. When driving in the dark it is therefore preferable to use the main beam lights, whenever possible.

Because the main beam lights are brighter and angled higher than low beam lights, you will still need to switch to low beam lights when you risk blinding vehicles in front or oncoming traffic with your headlights.

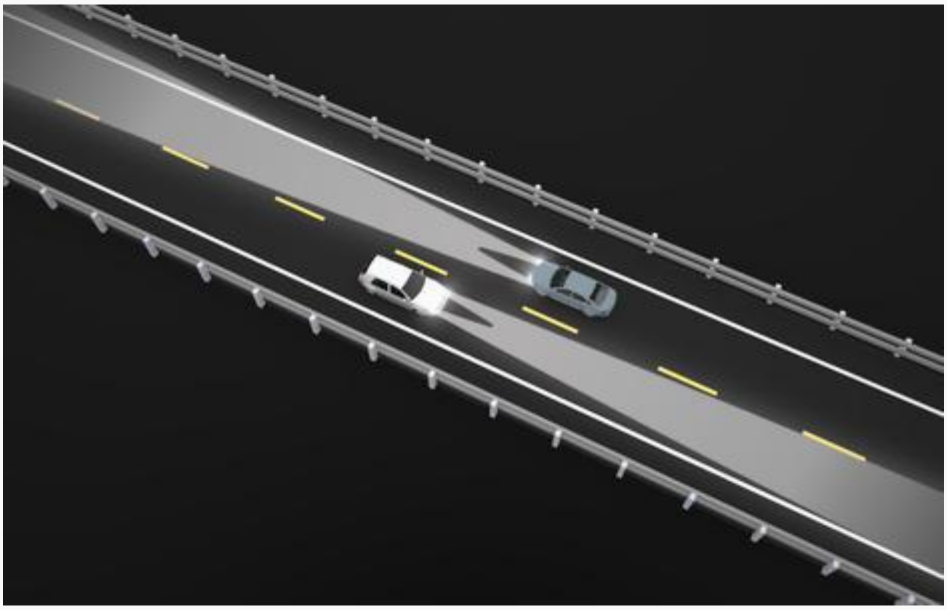
When two cars meet in the dark, both drivers shall

- switch from main beam lights to low beam lights when the distance between them is 200-300 meters
- reduce their speed (because visibility decreases when you see the lights of the oncoming vehicle)
- fix their eyes on the right-hand side of the road (to avoid blinding, and to look out for pedestrians and animals along the road)
- switch back to main beam lights approximately two car lengths before meeting (as main beams lights are directed straight ahead, you won't blind each other)

When closing in on another vehicle in the dark, you should switch to low beam lights when your headlights start illuminating the vehicle in front. This usually occurs when the distance between you two is about 100-200 meters (depending on how far your main beam lights reach).



When the cars are about 200-300 meters apart, both drives must switch to low beam lights.



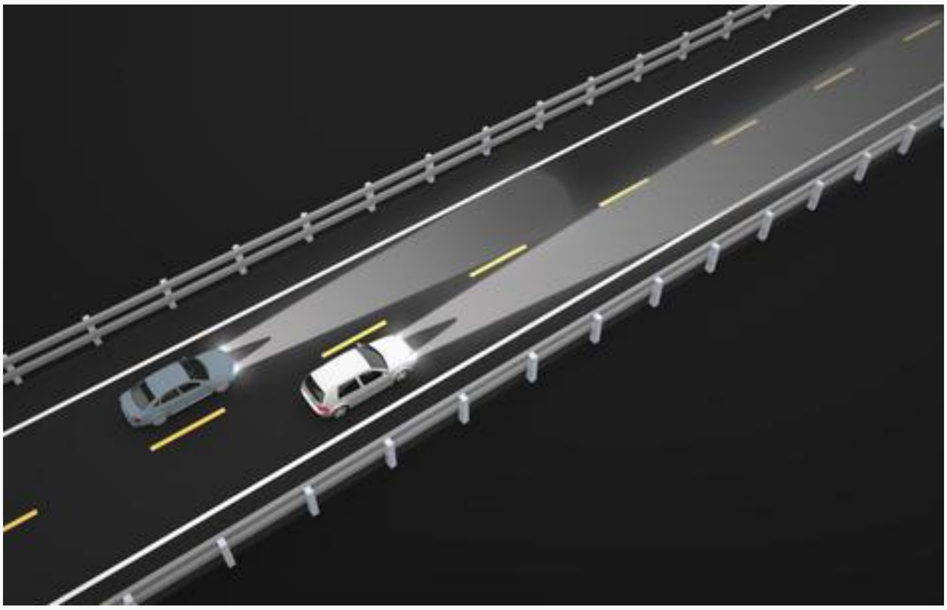
When the cars are about two car lengths apart, it is safe to switch back to main beam lights.

Overtaking in the dark

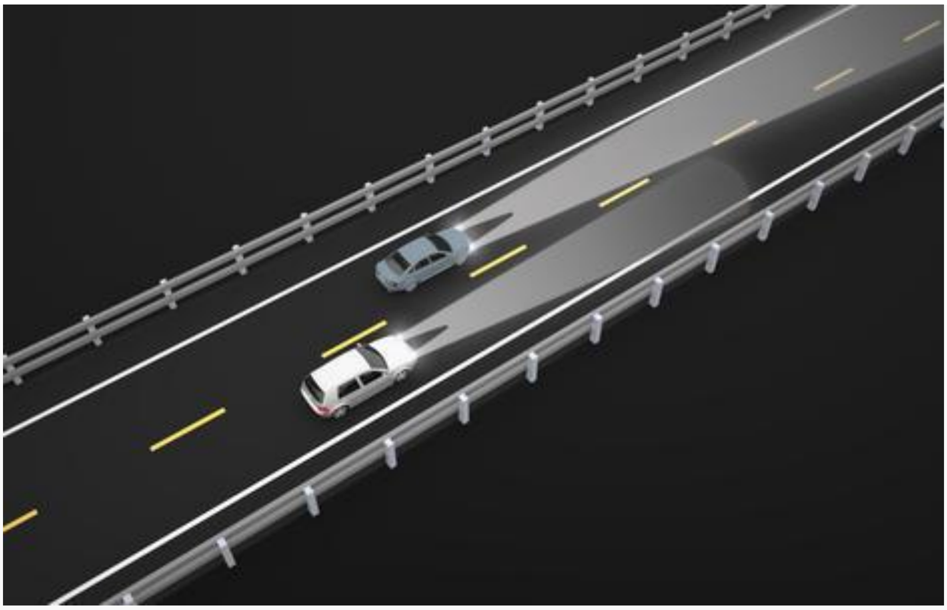
Overtaking in the dark can be very risky. Look out for pedestrians or animals along both sides of the road.

Overtaking in the dark

- When overtaking someone in the dark, you must switch to low beam lights when your headlights start illuminating the vehicle in front.
- Switch back to main beam lights when you have pulled out and the front of your car is aligned with the rear of the car you are passing.
- If another vehicle overtakes you in the dark, keep your main beam lights on until the other vehicle passes you.



When the blue car is just behind the white car, the blue car should switch to main beam lights.



The white car keeps the main beam lights on until the blue car passes.

Tunnels

Before driving into a tunnel, there are several potential hazards you should be aware of, so you can drive through as safely as possible.

Risks and hazards in tunnels

- Visibility may be poor.
- The road may be slippery in the tunnel.
- In some areas there may be animals (sheep, goats, cattle) in or near the tunnel entrance, seeking shelter from the sun or rain.

Safe driving in tunnels

- Slow down when approaching a tunnel.
- Be prepared to defog your windows. Heating up your windshield when it's cold outside will help remove condensation on the inside of the windshield, while the windshield wipers may help with the outside.
- Avoid overtaking in tunnels, unless there are multiple lanes in the same direction of travel.

- Use main beam lights whenever possible, but make sure they don't blind other road users.

Mandatory training in tunnel driving

As of 2016, tunnel driving is a mandatory part of driver training, as well as both the practical test and the theory test. You also need to know how to act in the case of an accident in a tunnel.

Tunnel safety

Traffic accidents in tunnels are not common, but when they occur, the consequences can be disastrous. This is why the authorities make large investments in tunnel safety, and it is important that you as a driver know how to act in the case of an accident.

- Do not drive into a tunnel that is closed with a barrier or red light.
- Pay attention to signs along the road, indicating where the nearest emergency exits, phones and fire extinguishers are.
- Use the emergency phone, if possible — the operator will automatically see where you are calling from, and can more easily assist you.
- If your car breaks down, you get a flat tyre or you for some reason are unable to continue driving, try maneuvering the car into the nearest emergency lay-by and exit the vehicle.
- In case of a fire, use the fire extinguishers in the tunnel, as these will set off the fire alarm at the Road Traffic Central.

Keep your distance!

If you notice a queue is forming ahead of you in the tunnel, make sure you stop far enough away from the vehicle in front that you are able to turn around if necessary. If there has been an accident and you need to leave your vehicle behind, leave the keys in the ignition so that emergency personnel may easily move it.

Level crossings

A level crossing is a place where the railway line crosses the road at the same level.

There are rarely accidents on level crossings, but the few that do occur are often very serious. Most level crossings are therefore safeguarded in different ways, and all of them are clearly marked and forewarned by traffic signs.

Level crossings are usually safeguarded with

- barriers/gates
- special light signals
- sound signals

This level crossing is safeguarded by barriers and light signals. The signs indicate a single-track railway.

Traffic signs for level crossings

There is always a certain level of risk involved when driving across a level crossing. In addition to signs posted at the crossing, signs also forewarn the crossing well before reaching it.

The signs tell you a lot about the crossing itself. First you will see a warning sign, which indicates whether the crossing is safeguarded with barriers or gates.

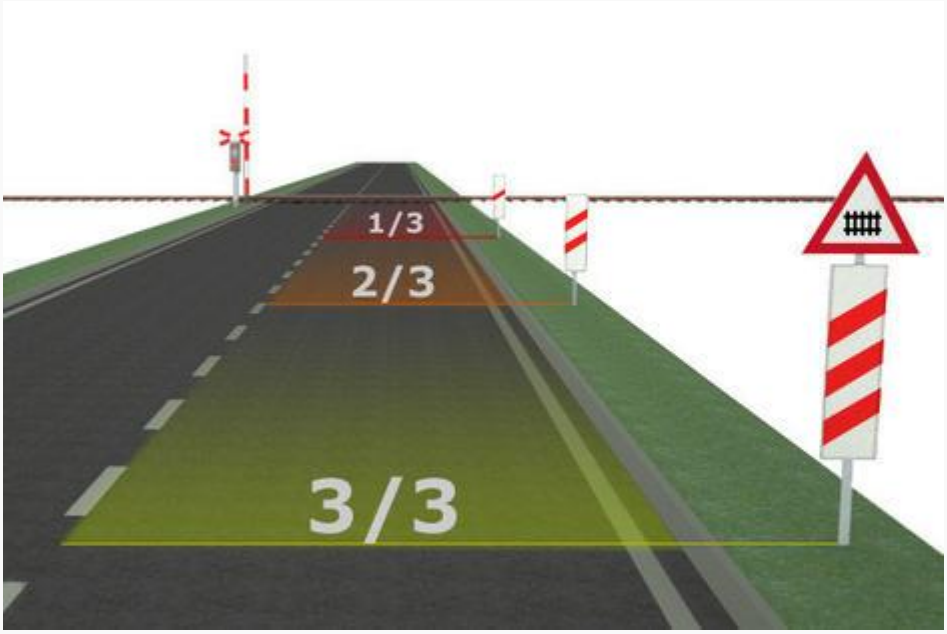
Keep in mind that there may be more than one train on the way, so never cross over until you are absolutely certain it is safe to do so — even if a train has already passed.

This sign is posted before level crossings that are not safeguarded by gates or barriers.

Distance markers

Before many level crossings, distance markers will be posted to indicate how far away the crossing is.

The distance markers count down to the level crossing. The distance is divided into 3 parts, with the number of diagonal lines on the sign indicating how far away you are from the crossing. The first marker, which has three lines, is posted together with the warning sign, indicating that 3/3 (the whole) of the distance from the sign to the crossing is left. Two lines indicate 2/3 of the distance is left, and one line indicates 1/3 of the distance remains.



Distance markers indicate how far away the crossing is, in this case a level crossing safeguarded with barriers.

Note! The distance from the warning sign to the level crossing depends on the speed limit:

- 60 km/t or lower: 50-150 meters
- 70-80 km/t: 150-250 meters
- 90 km/t or higher: about 400 meters

At the level crossing, there are signs indicating if it is a single- or multiple-track railway

Light and sound signals at level crossings

Light signals

Light signals at level crossings are fitted with two lights — a red, flashing light on top, and a white, flashing light below it. As the crossing signals could be faulty or not working, you must always make sure that it is safe to cross before driving onto the crossing.

- A red, flashing light indicates that you must stop before the stop line, or at a safe distance from the signal.
- A white, flashing light indicates that you may drive across. White, flashing lights also indicate that the light signals are working normally.

Sound signals

Some level crossings are also safeguarded with sound signals. The alarm sounds when the light is flashing red.

Stop before the stop line or at a safe distance from the signal, when the light is flashing red.

Driving across a level crossing

Remember the following when approaching a level crossing:

- Pay attention to warning signs and distance markers, and prepare for the level crossing.
- Reduce your speed so that you are able to stop if a train or tram is approaching.
- Even if the barriers are up and the white light is flashing, you must always check to make sure it is safe to cross before driving across. You must give free passage to trams or trains passing the level crossing.

Note!

It is prohibited to stop (and thereby also to park) less than 5 meters from a level crossing. The reason is that parked cars reduce visibility for other road users.

Driving abroad

When driving a Norwegian car abroad, the vehicle must show the international license plate country code near the rear-end registration plate. If it is not already integrated into the registration plate, you will need to get a nationality sticker. Before driving abroad you should also check that your insurance is valid outside of Norway.

Be aware that the risk of accidents if often much higher than in Norway.

The traffic rules are fairly similar in most countries, but there are some notable exceptions:

- Many countries use white road markings to separate traffic travelling in opposite directions (e.g. Sweden, Denmark and Finland).
- The speed limits vary from country to country.

- Be careful when changing lanes on high-speed roads, as traffic is moving very quickly.
- Give way rules may vary, so do not assume other road users will give way because you are coming from the right.
- Daytime running lights are prohibited in some countries.
- Rules regarding drink-driving vary from country to country.

Left-hand traffic

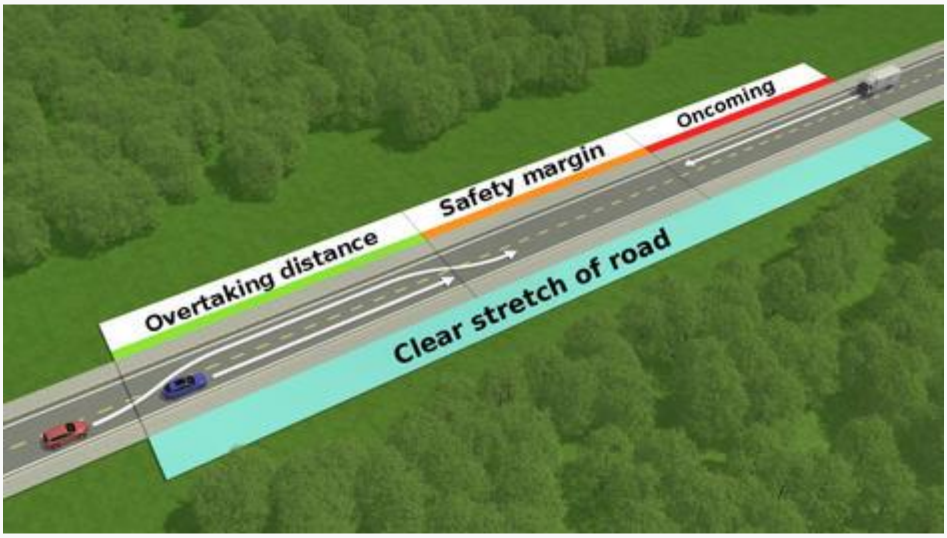
Several countries have left-hand traffic, meaning you drive on the left-hand side of the road. The most notable ones are the United Kingdom, Australia, Japan, Thailand, Pakistan, Indonesia and India. In these countries the steering wheel is on the right-hand side. You are still permitted to drive a car with the steering wheel on the left, but visibility of oncoming traffic will be reduced.

Summary

Highlights from chapter 8

- Motorways and semi-motorways are intended for driving at higher speeds. To prevent danger and problems with traffic handling, the quality of the road is high, and properties along the road do not have direct access to the road. Motorways (unlike semi-motorways) must comply with several requirements, e.g. traffic in opposite directions must be separated by a median strip or central traffic barrier.
- Motorways and semi-motorways are only for vehicles that can drive at speeds of at least 40 km/h — mopeds, cyclists and pedestrians are not allowed on the road. It is also prohibited to reverse, turn around, stop or park on motorways and semi-motorways.
- If you are overtaking someone, you need to assess if and how you can do so safely and legally. In general, you overtake on the left.
- *The overtaking distance* is the distance you travel when overtaking someone, and depends on the relative speed between you and the vehicle you are overtaking. If the relative speed doubles, the overtaking distance is halved.
- *The clear stretch of road needed for overtaking* refers to the length of road ahead that must be clear for the overtaking to be considered safe. The clear stretch of road needed for overtaking consists of the overtaking distance; the distance travelled by oncoming vehicles while you are overtaking, and a safety margin, and is calculated as approximately three times the overtaking distance.
- It takes about 30 minutes for your eyes to get used to the dark after having been in out in daylight.
- When driving in the dark it is preferable to use main beam lights whenever possible. To avoid blinding other road users, you will still need to switch to low beam lights when the distance to the car in front is 100-200 meters, or when the distance to oncoming vehicles is 200-300 meters. When overtaking in the dark, switch back to main beam lights when the front of your car is aligned with the rear of the car you are passing.

- In tunnels visibility may be poor, so reduce your speed when approaching a tunnel. Be aware that the road may be slippery inside the tunnel, and that there may be animals seeking shelter in the tunnel.
- Level crossings are clearly marked and forewarned by traffic signs, and most are safeguarded with barriers/gates, and special light and sound signals. Even if the barriers are up and the light is flashing white, you must always check to make sure that it is safe to cross before driving onto the level crossing. Remember that you must give free passage to trams or trains passing the level crossing.



The clear stretch of road needed for overtaking = 3 x the overtaking distance

Chapter 9- Traffic Accidents and the Environment

Introduction

If you were to come across a road accident, would you know what to do or how to help? This chapter deals with your obligations in the event of an traffic accident and how to perform basic first aid.

The chapter ends with a look at the how traffic affects the environment, and what you can do to drive in a more eco-friendly manner.

Accidents and your obligations

According to Norwegian law you are required to assist in case of an accident, even though you are not directly involved.

Keeping your cool can be difficult and there will often be chaos and confusion at the scene of the accident. It is your responsibility to ensure that the correct measures are being taken, and that they are done in the right order.

- Start by securing the area. This will help to prevent new accidents. Place warning triangles at strategic places and park your car with hazard lights and/or headlights on, ideally facing approaching traffic. You should also try to move any vehicles that may be endangering or obstructing traffic. For you own safety remember to put on your reflective west before you exit your car.
- Get an overview of the scene of the accident. Who are injured and who needs to be tended to first?
- If you have a phone call **113**.
- Administer first aid.

To not help is illegal!

If you ignore an injured person on purpose you can be prosecuted and risk going to jail and lose your rights to drive for life.

Accidents - additional obligations

Most accidents are of the less serious kind, with only minor injuries and little damage to property. Chances are you might be involved in a minor accident one day, and knowing what you should do is important.

- All parties involved have a mutual duty to provide name and address (and if the vehicle is borrowed, also the name and address of the owner).
- Alert the police of the accident if anyone has died or sustained serious injuries.
- Alert the owner or the police if you have caused any material damages and there is no one present to represent the affected party.

In order to decide the question of guilt, it is important to secure proper documentation. Write down what happened, try to illustrate the course of action, locate witnesses and get their names and addresses.

Remember!

You are expected to provide the police your name and address.

First aid

After securing the area at the scene of an accident, you should start administering first aid. Start by calling 113. They will assist you and tell you how to proceed.

In case your phone is unavailable or there is no signal, this is how you should proceed.

Before administering first aid you need to find the answer to three questions:

1. In what kind of state is the injured person?

Is the injured person conscious, are the airways free and how is the breathing? Look for visible injuries and check skin color and temperature.

2. What does the injured person say?

Is the injured person in any pain and if yes: what kind of pain?

3. What can you do?

Based on this information you need to decide on what kind of first aid you should administer.

What can you do?

The person is conscious

If the person is conscious, you should ask about things that can help you form a picture of the person's condition. Try to be calm and confident as this can help giving the person a sense of security.

The person is unconscious

Try to gently shake the person awake. If this is unsuccessful, check that the airways are clear.

An unconscious person cannot secure their own airways if something suddenly should block it, so tilt the person's head backwards and put the person in recovery position.

If the person is breathing with gasping, short breaths you must immediately start CPR.

Unconscious? - The helmet must come off!

If a motorcyclist is unconscious the guidelines are clear: The helmet must be removed in order to secure free airways. Worst case scenario, it is better to risk a neck injury than the motorcyclist dying from suffocation.

CPR

Administering CPR the correct way

- The injured person should ideally lay with a flat back on a hard surface.
- Start with 30 chest compressions, then 2 breaths.
- Place the heel of one hand towards the end of their breastbone, in the center of their chest. Lean over, with your arms straight, pressing down vertically on the breastbone, and press the chest down by 5-6cm at a rate of twice a second.
- Breaths. Ensure that the injured person’s airway is open and pinch the nose firmly closed. Take a deep breath, seal your lips around the person's mouth and blow into the mouth until the chest rises. Remove your mouth and allow the chest to fall. After 2 breaths, go back to the compressions.
- Continue until medics arrive or the person gets better.

Bleeding

First aid - Mild or moderate bleeding

- Clean the wound and bandage it.

First aid - Arterial Bleeding

- If blood is gushing out in pulses, and it looks bright and frothy this might indicate an arterial bleeding.
- Put direct pressure on the wound with your fingers, using a sterile dressing if possible, to stop blood escaping and raise the part of the body that's injured so that it's above the heart. This will reduce the flow of blood to the wound and help stop the bleeding. Lay the injured person down with the head low and the legs raised and supported.

First aid - Internal bleeding

- Internal bleeding can be difficult to detect. Typical symptoms are cold, clammy and pale skin, and a clouded consciousness.
- Secure the airways and put the person in recovery position.
- Don't give the injured person anything to drink if you have suspicions of internal bleeding.

Circulatory failure/Shock

Circulatory failure

- Cold and clammy skin, fast breathing, low blood pressure and abnormally low/high pulse are typical symptoms.

Treating circulatory failure

- There isn't really much you can do to help a person with circulatory failure. Put the person in recovery position and should a cardiac arrest occur, perform CPR immediately.

Hypothermia

Keep the injured person warm and dry to prevent hypothermia.

If the person is still in the car you should consider not moving the person unless this is absolutely necessary. This will prevent further aggravating any potential back or neck damage.

'Nullvisjonen'

A great number of people is injured and killed in traffic every year. There has been a positive development in the recent years and in 2015 there was an all-time low of 125 casualties.

In order to minimize the number of serious accidents the government has set a goal for the road safety work: 'Zero killed and zero permanently injured' - the so called ‘**nullvisjonen’** (the zero vision).

In 1970 the number of people killed in the traffic was 560, but in 2015 the number had shrunk down to 125. The current goal is to reduce this number further with a third by the year 2020. It is worth noting that traffic has increased enormously in this period so when you measure the number of casualties against kilometers travelled the development has been very positive.

Accident statistics

Statistics for the last ten years shows that many of the causalities are relatively inexperienced drivers, primarily young males.

Why are young males overrepresented in the accident statistics?

- The area of the brain that holds our capability for judgment and risk assessment is not fully developed until the age of 25. As a consequence, many male drivers act immature in traffic and come off as impulsive and erratic.
- Girls usually mature long before boys and this is reflected in the accident statistics which is dominated by young male drivers.
- Peer pressure is also a major factor. Many young male drivers want to show off their skills and are also very susceptible to the whims and wishes of the other passengers.

Environmentally friendly choices

As a driver there are a lot of things you can do that is beneficial for the environment. Best of all would be to let the car stay behind, but this is an unrealistic choice for many people. Here are some other tips.

- Avoid using studded tires if possible. One studded tire can tear up as much as 5 - 20 grams of asphalt dust for each kilometer driven.
- Use a preheater. Cold starting an engine can cause large emissions of toxic fumes.
- Maintenance of the car. By taking your car regularly to a car service center you reduce both emissions and pollution.
- Drive an electric car. No fuel=no emissions.
- Avoid unnecessary short trips. All cars today are equipped with catalytic converters to reduce emissions from exhaust gas, but they are most effective when the engine has reached running temperatures (after having driven about 6 km). A car therefore pollutes most during the first couple of kilometers driven. Instead, try using public transportation, a bike or why not just walk?

Eco driving

How you drive will also have a significant impact on the environment.

- Skipping gears. Change straight from 2nd gear to 4th gear.
- Use the highest gear possible. This will minimize the fuel consumption.
- Accelerate firmly. Be sure to avoid exceeding 3000 rpm. Acceleration consumes fuel, so try to keep a steady speed once you have reached your cruising speed.
- Exploit the terrain. You can save fuel by letting go of the gas pedal early in long downhills and just ahead of hilltops.
- Plan your route. By avoiding detours and traffic jams you can save a lot of fuel.

Bonus

Eco driving makes your driving safer and saves you money.

Summary

Highlights from chapter 9

- According to Norwegian law, you are required to assist in case of an accident, even though you are not directly involved
- When you arrive at a scene of an accident you should first secure the area and get an overview. Then call 113 and start administering first aid to those who need it
- All parties involved in an accident have a mutual duty to provide name and address (and if the vehicle is borrowed, also the name and address of the owner)
- Alert the police of the accident if anyone has sustained serious injuries
- You can contribute to a better environment by making eco-friendly choices. Use your legs!

Chapter 10- Traffic Signs

Warning signs

Dangerous curve

Dangerous curves

- 

  Steep hill — upwards

- 

  Steep hill — downwards

- 

Road narrows — on both sides

- 

Road narrows — on the right side

- 

Uneven road

- 

Road hump

- 

  Road work

- 

  Loose chippings

- 

  Landslide

- 

  Landslide

-

- 

Slippery road

- 

Dangerous shoulder

- 

Opening or swing bridge

- 

Quayside or river bank

- 

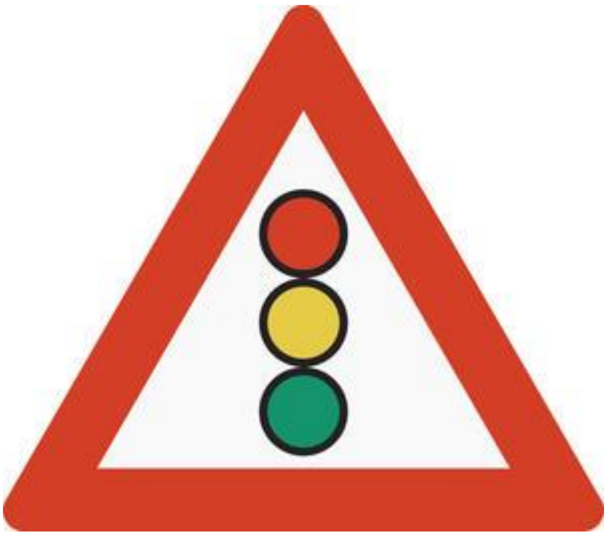
Tunnel

- 

Dangerous junction

- 

  Roundabout

- 

  Traffic signals

- 

Level crossing with a gate or barrier

- 

Level crossing without a gate or barrier

- 

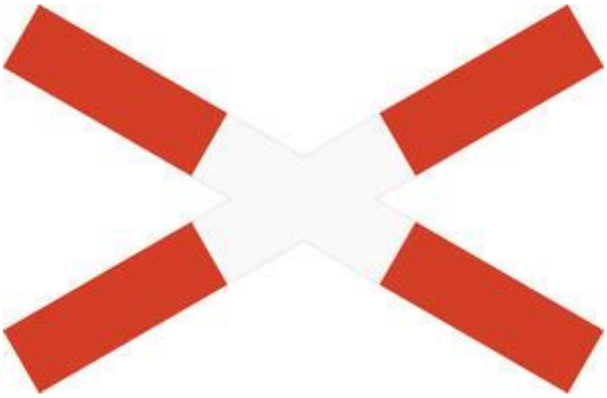
  Countdown marker

- 

  Countdown marker

- 

Countdown marker

- 

Level crossing — single track

Level crossing — multiple tracks

- 

Tramway

Distance to pedestrian crossing

Children

Cyclist

Moose

•

Reindeer

Deer

Cattle

Sheep

Two-way traffic

Traffic queues likely

Low-flying aircraft

Crosswinds

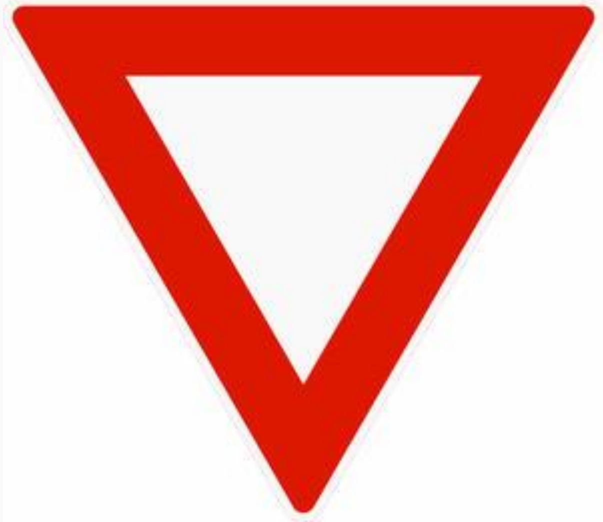
Traffic accident

Skiers

Riders

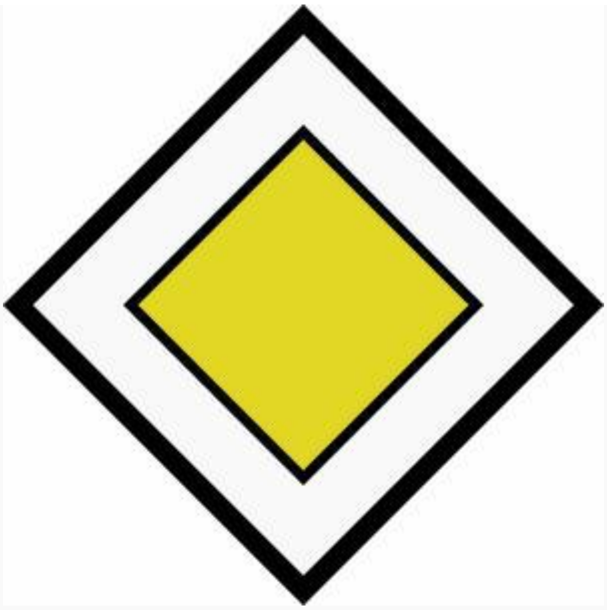
Other danger

Give way and priority signs



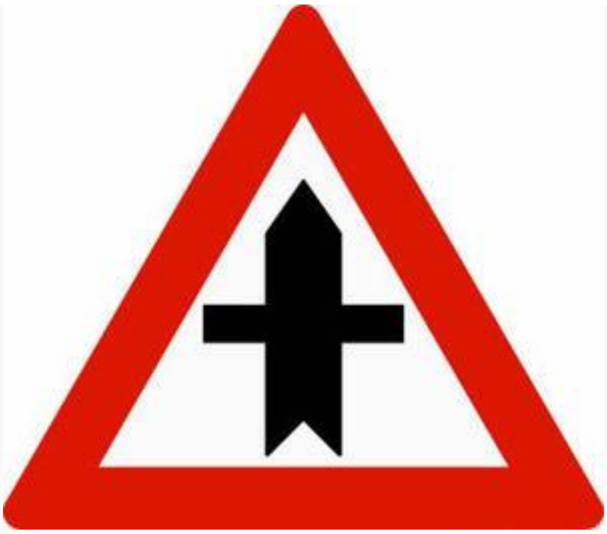
Give way

Stop

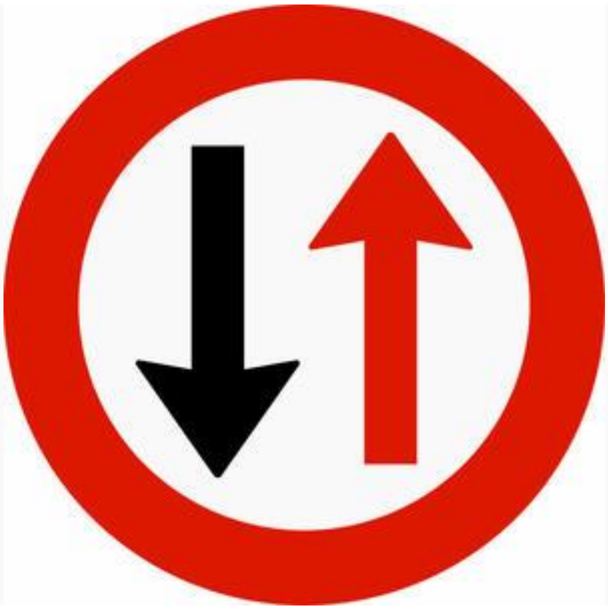


Priority road

End of priority road



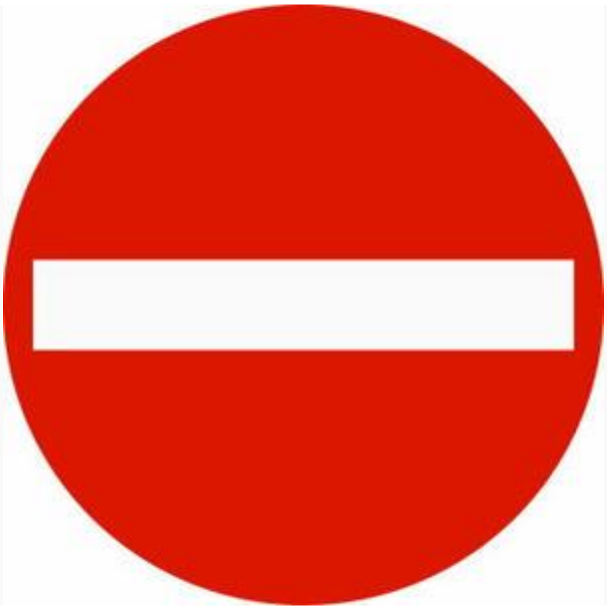
Priority junction



Give way for oncoming traffic

Priority over oncoming traffic

Prohibitory signs



No entry

No vehicles

No motor vehicles

No tractors or motor vehicles slower than 40 km/h

No motorcycles or mopeds

No Lorries

No cyclists

No pedestrians

No pedestrians or cyclists

No riders

No transport of dangerous goods

No motor vehicles with more than two wheels and total weight above the stated weight

Width limit

Height limit

Length limit

Weight limit for vehicles

Weight limit for Lorries

Axle weight limit

Bogie weight limit

Stop for stated purpose (control)

Stop for customs

No right turn

No left turn

No U-turn

No passing

No passing for lorries

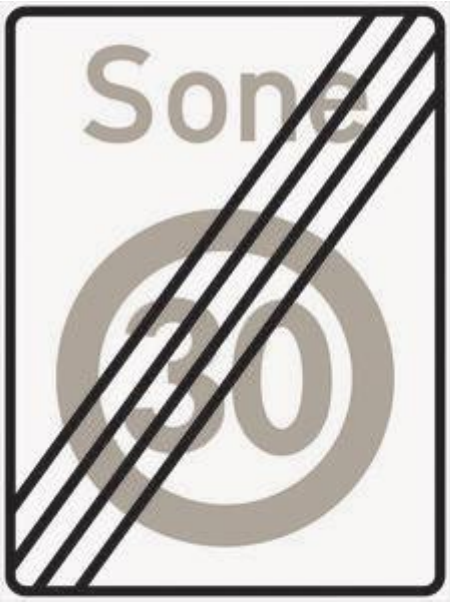
End of no passing

End of no passing for lorries

Speed limit

End of speed limit

Speed limit zone



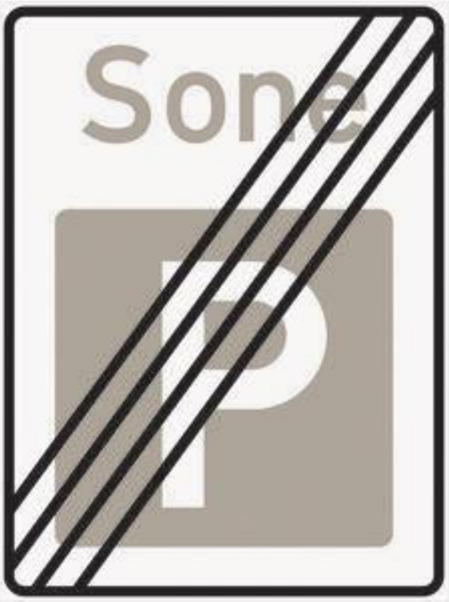
End of speed limit zone

No stopping

No parking

Parking zone

Parking zone



End of parking zone

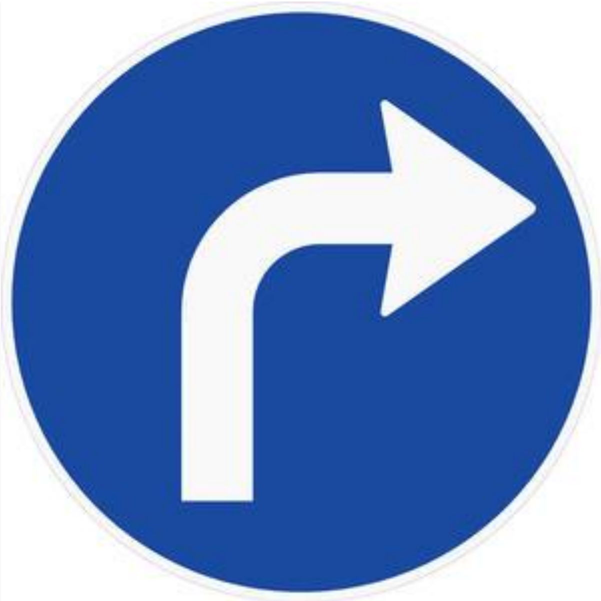
End of parking zone

Mandatory signs

Mandatory driving direction — proceed right

Mandatory driving direction — proceed left

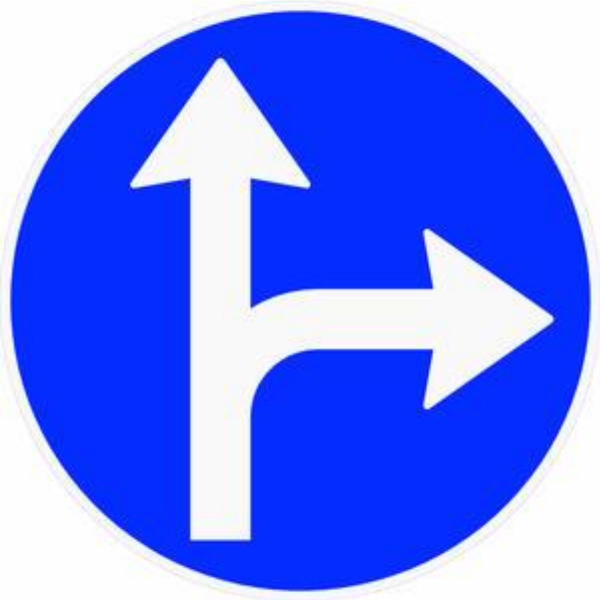
Mandatory driving direction — proceed straight

- 

  Mandatory driving direction — turn right

- 

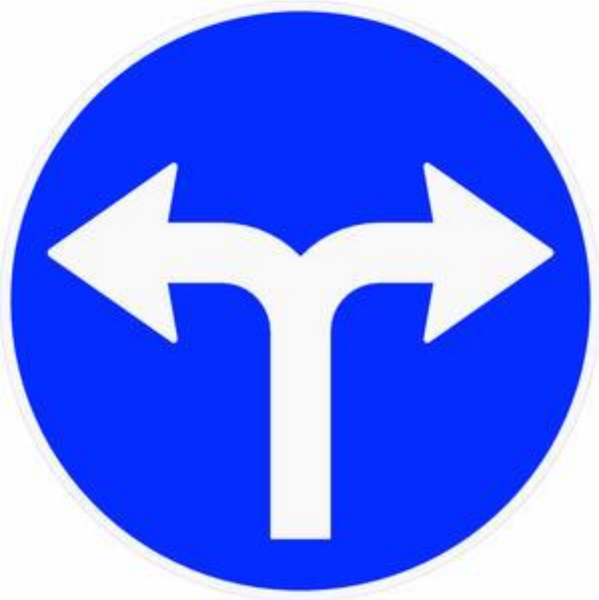
  Mandatory driving direction — turn left

- 

Mandatory driving direction — drive straight or turn right

- 

Mandatory driving direction — drive straight or turn left

- 

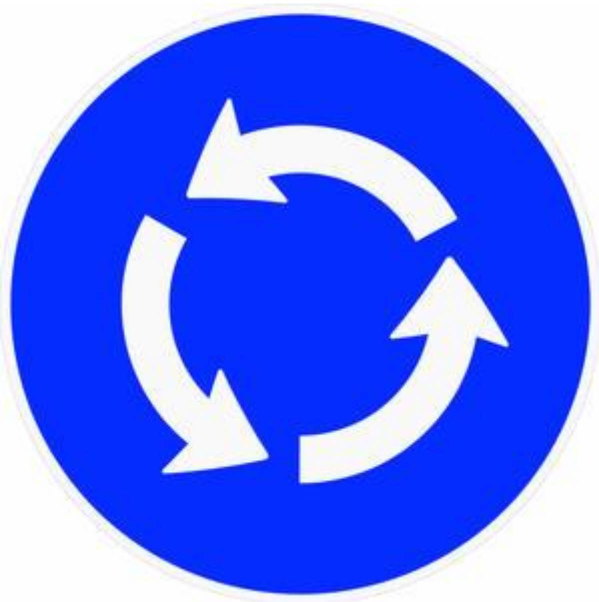
Mandatory driving direction — turn right or left

- 

Mandatory lane — pass on the right

- 

Mandatory lane — pass on the left

- 

Mandatory roundabout

Informative signs

- 

Motorway

- 

Semi-motorway / Road for motor vehicles

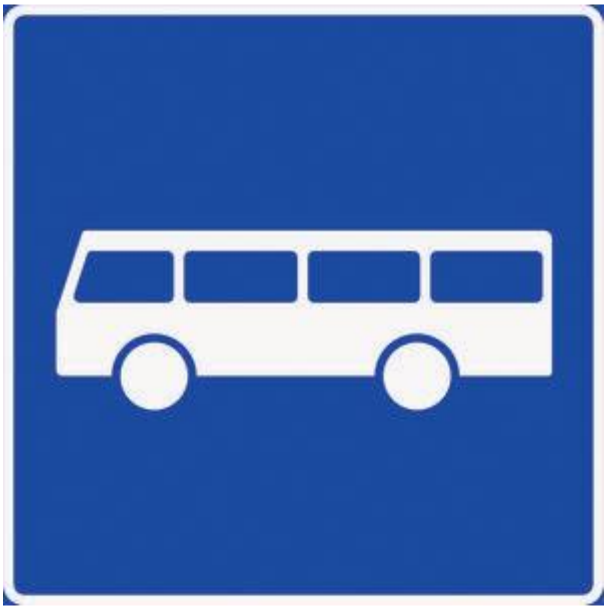
- 

End of motorway

- 

End of semi-motorway / End of road for motor vehicles

-

- 

Public transport lane — bus lane

- 

Public transport lane — bus and taxi lane

- 

  High-occupancy vehicle lane

- 

  End of public transport lane

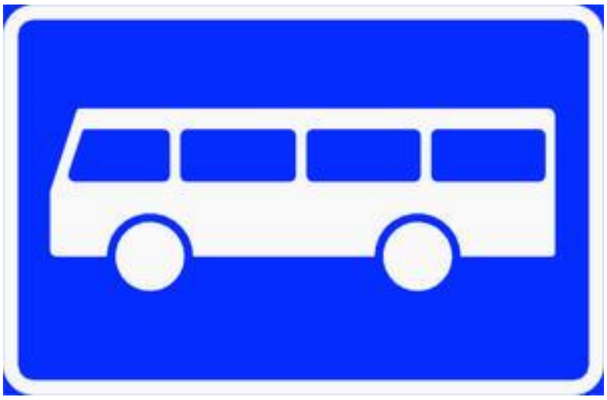
-

- 

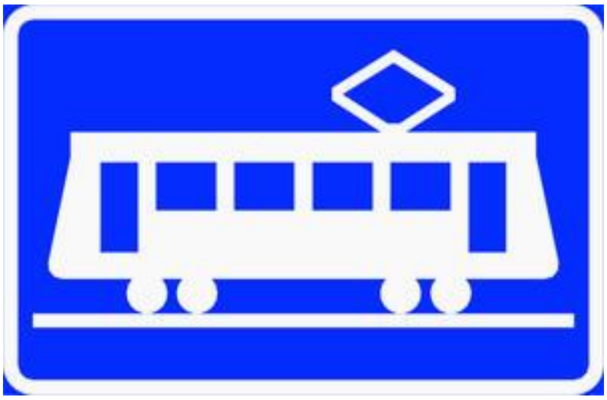
End of public transport lane

- 

End of high-occupancy vehicle lane

- 

  Bus stop

- 

  Tram stop

- 

  Taxi stop

- 

Pedestrian crossing

- 

Footpath

- 

  Cycle track

- 

  Cycle lane

-

- 

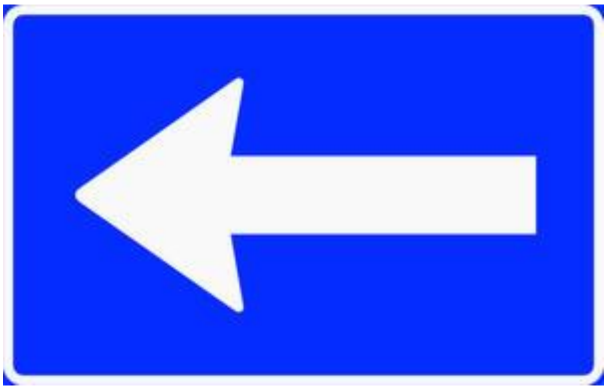
Footpath and cycle track

- 

Meeting/passing place

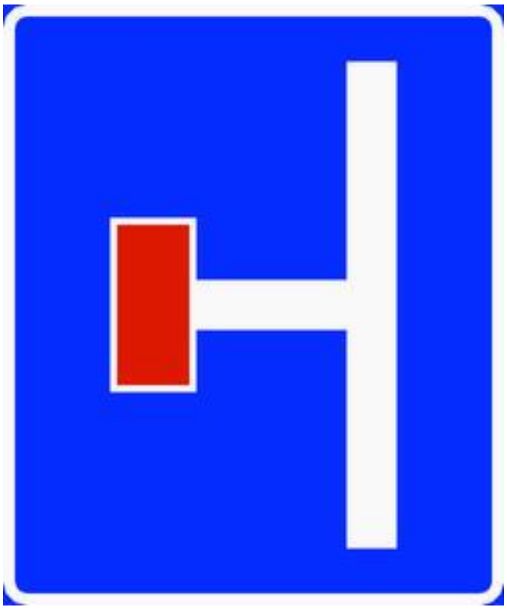
- 

  One-way traffic

- 

  One-way traffic

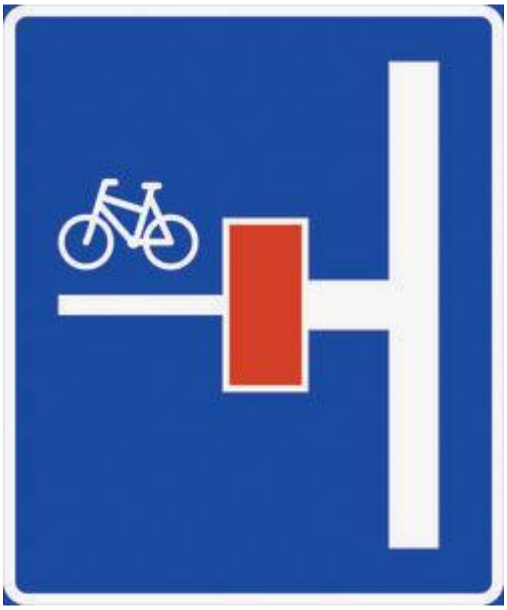
-

- 

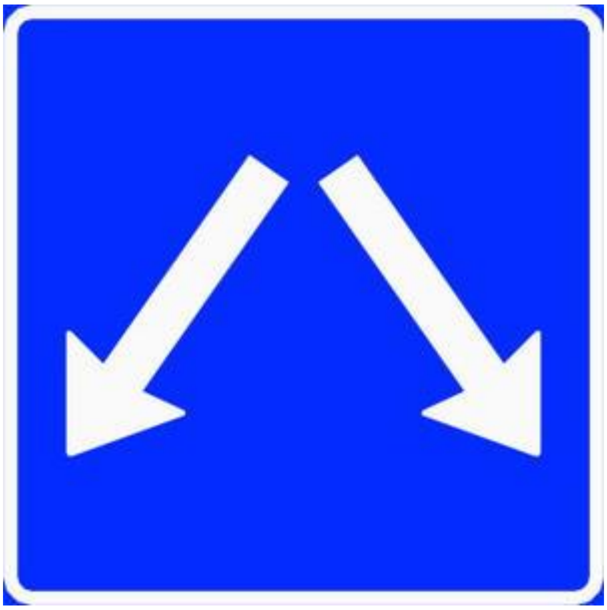
No through road / Dead end

- 

No through road / Dead end

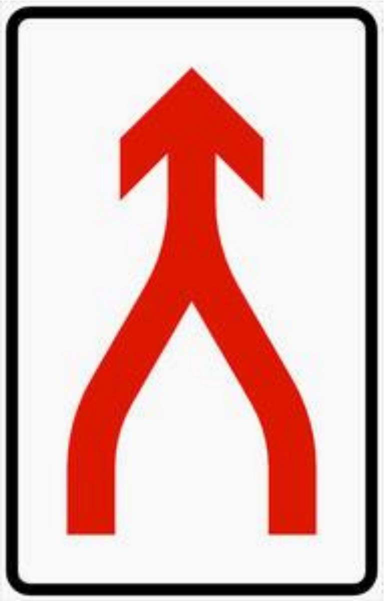
- 

No through road / Dead end

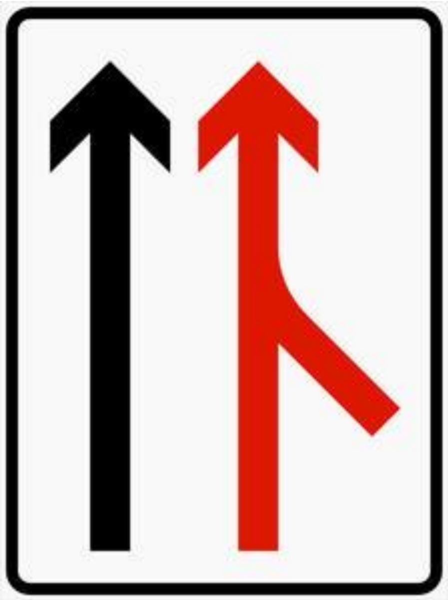
- 

Free choice of lane

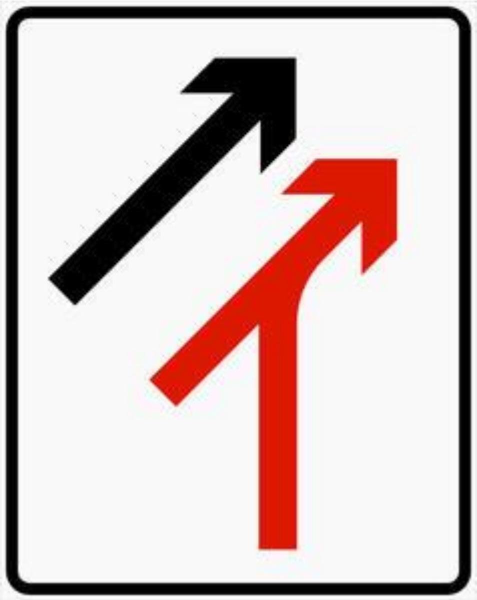
-

- 

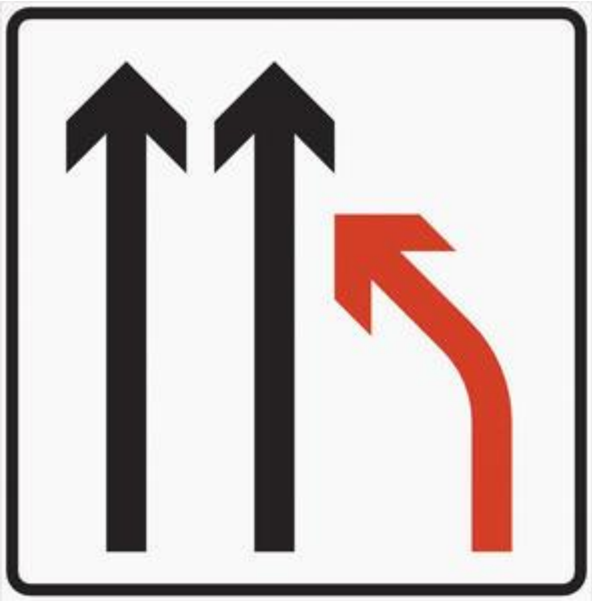
Merging lanes

- 

Acceleration lane

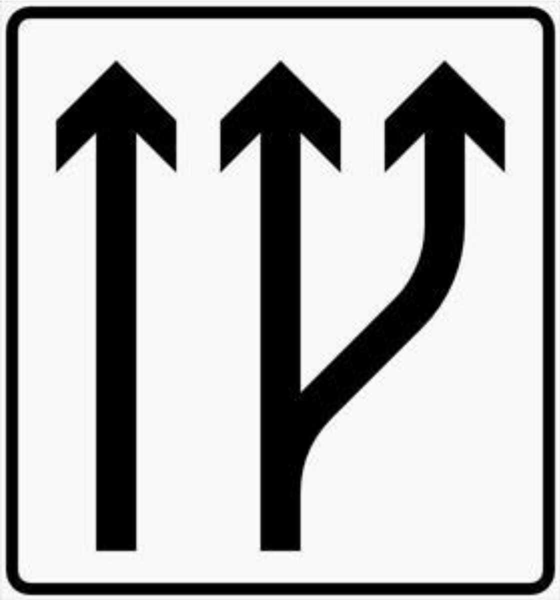
- 

Acceleration lane

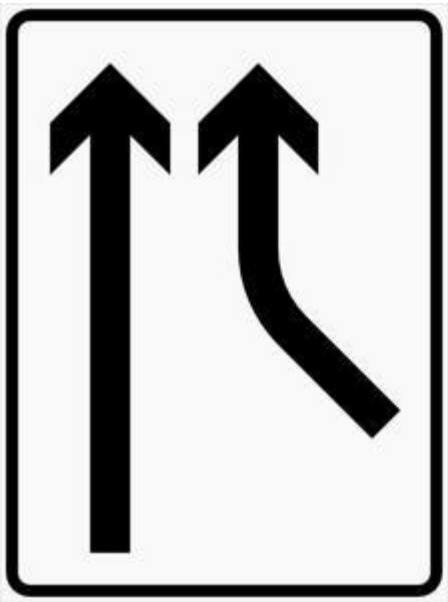
- 

End of lane

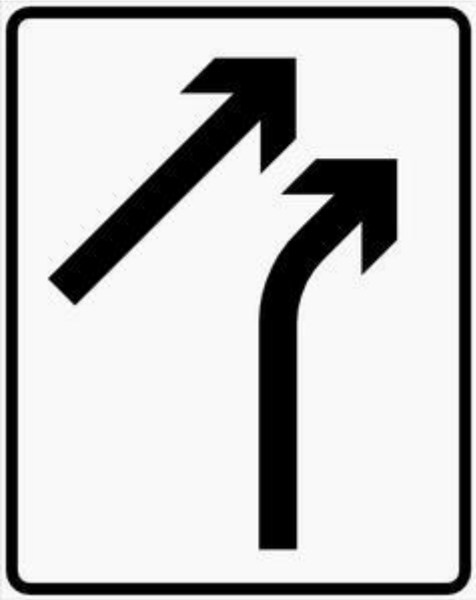
-

- 

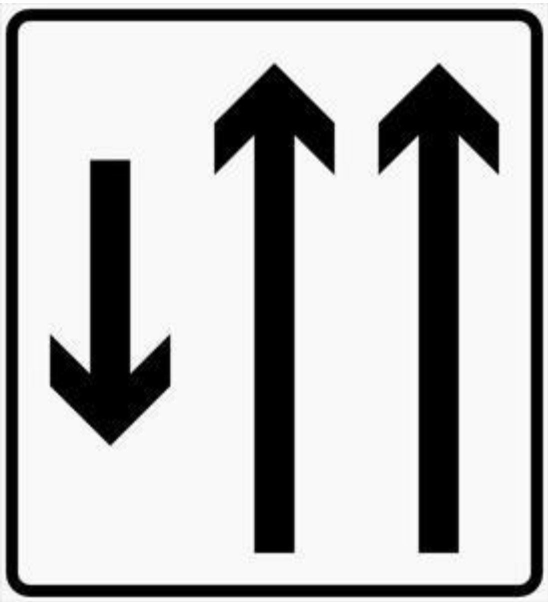
Lane begins

- 

Traffic joins road in separate lane

- 

  Traffic joins road in separate lane

- 

  Lane division

-

- 

Altered driving pattern

- 

Pedestrianised residential street / Living street

- 

End of pedestrianised residential street / living street

- 

Pedestrian street

- 

End of pedestrian street

- 

Parking

- 

Automatic speed control

- 

Automatic speed section control

- 

Video surveillance

-
- 

Information board

- 

Wrong way

- 

Emergency exit (for tunnels)

- 

Direction and distance to emergency exit (for tunnels)

Service information signs

- 

Radio channel

- 

First aid

- 

  Emergency telephone

- 

  Fire extinguisher

-

- 

Workshop

- 

Petrol station

- 

Toilet emptying facility

- 

Toilet

- 

Rest stop

- 

Rest stop with toilet

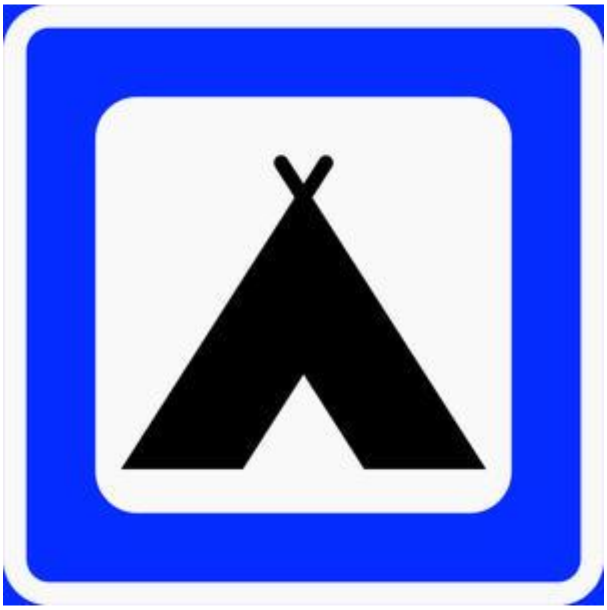
-

- 

Simple dining

- 

Dining

- 

  Camp site

- 

  Motor home site / Caravan park

-

- 

Camping park

- 

Hostel

- 

  Overnight accommodation

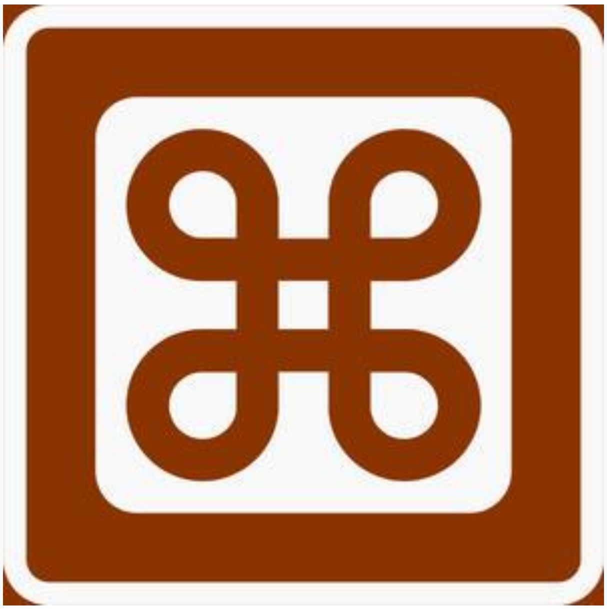
- 

  Information

-

- 

  Tourist information office

- 

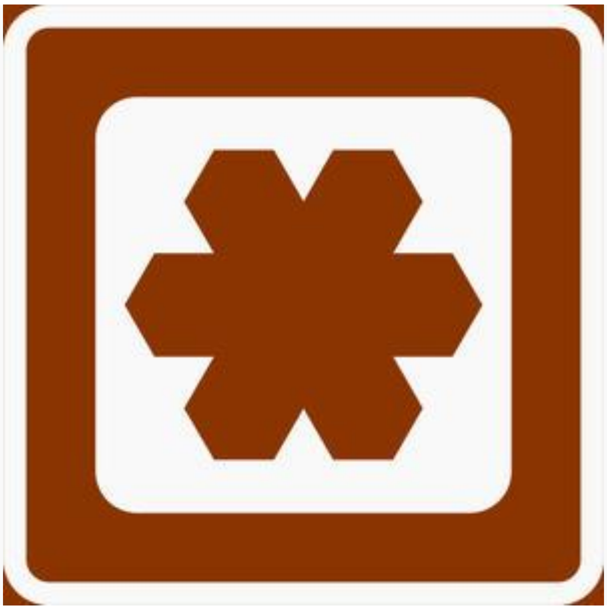
  Attraction

- 

Museum / Art gallery

- 

Viewpoint

- 

  Protected natural area

- 

  Bathing spot

- 

  Fishing area

- 

  Hiking trail

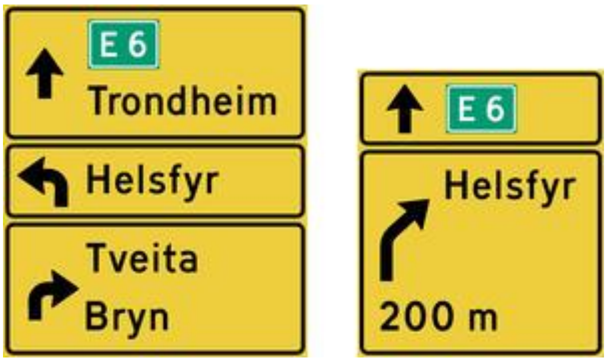
- 

Ski trail

- 

Traditional food and rural tourism

Direction signs

- 

Orientation board

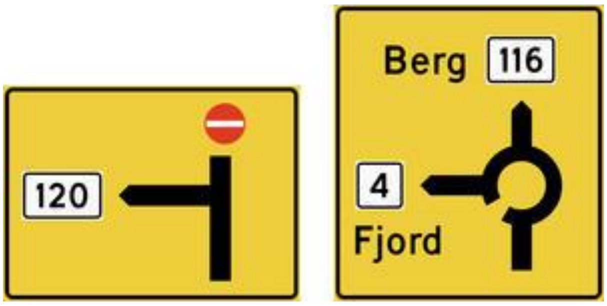
- 

Diagram orientation board

- 

Exit board

- 

Lane orientation board

- 

- 

Portal orientation board

- 

Board guide

- 

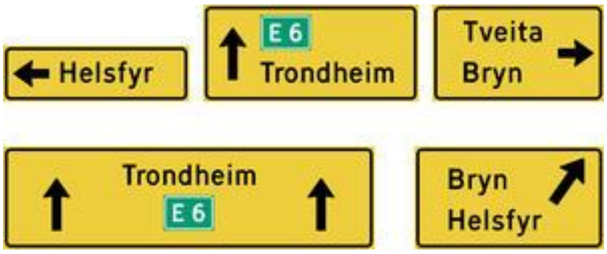
Regular destination guide

- 

Exit guide

- 

Lane guide

-
- 

Portal guide

- 

European route

- 

European route

- 

National road that is not a European route

- 

National road that is not a European route

-

- 

Numbered county road

- 

Numbered county road

- 

Ring road

- 

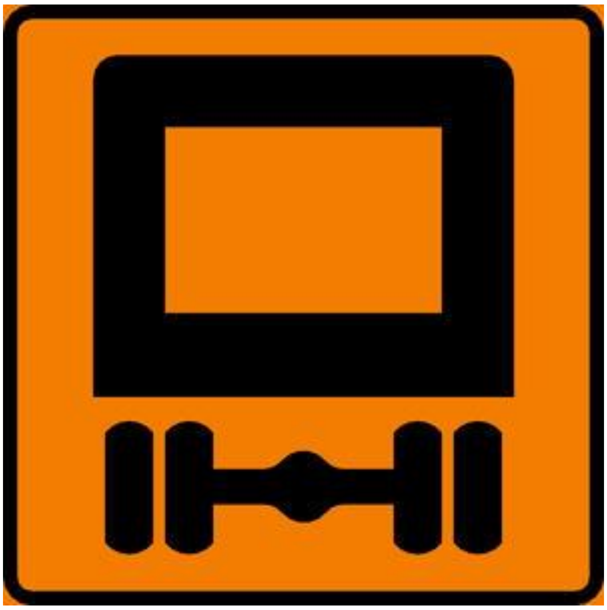
Ring road

- 

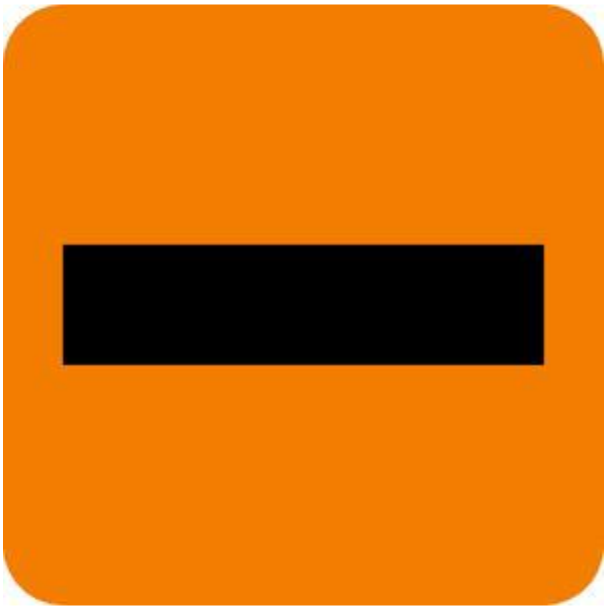
National tourist road

- 

Detour for large vehicles

- 

Route for vehicles transporting dangerous goods

- 

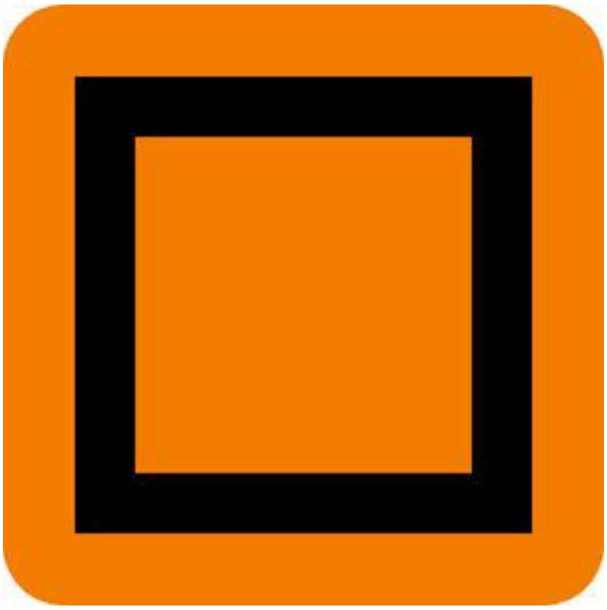
Miscellaneous detour

- 

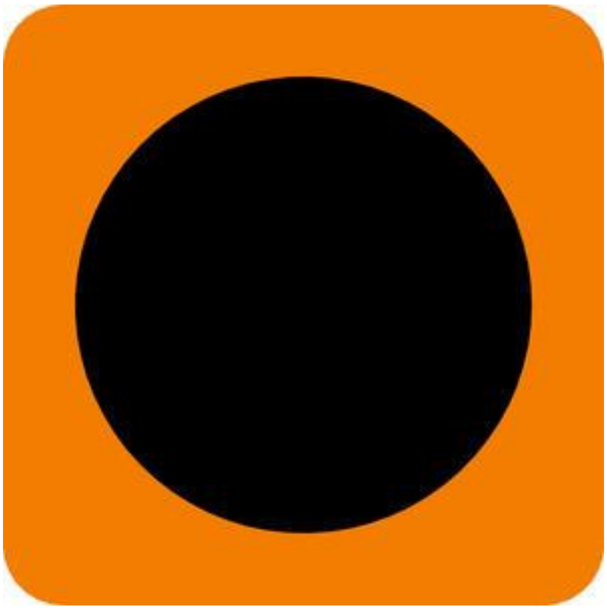
Miscellaneous detour

- 

Miscellaneous detour

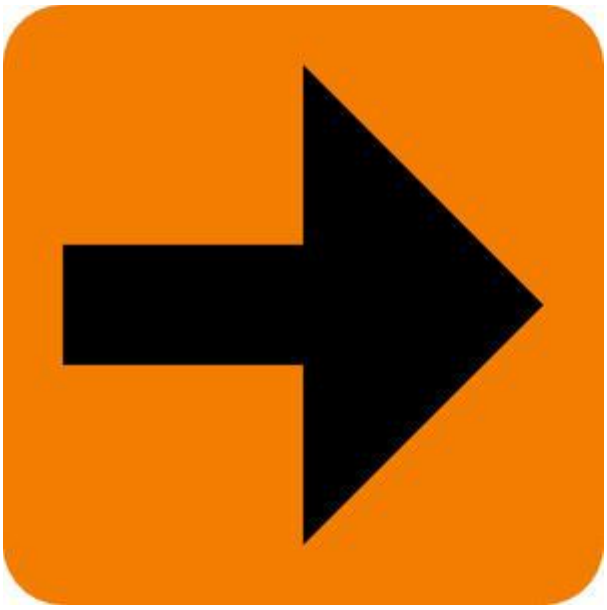
- 

Miscellaneous detour

- 

  Miscellaneous detour

-

- 

  Miscellaneous detour

- 

Junction number on multilane roads

- 

Junction number on multilane roads

- 

Junction number on two-lane roads

-
- 

Region name sign

- 

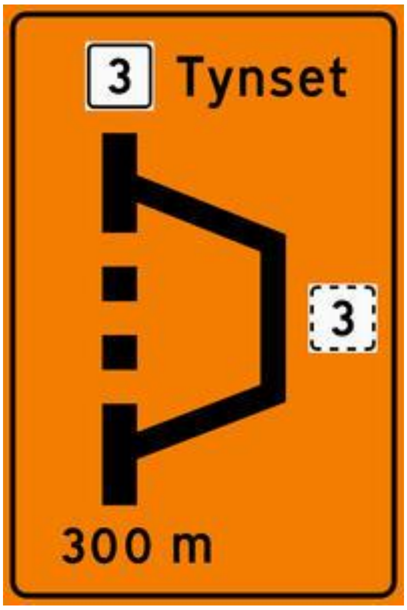
Street sign

- 

Collective guide sign

- 

Detour for certain vehicles

- 

Temporary detour

- 

End of temporary detour

-
- 

Guide for pedestrians

- 

Guide for bicycle route

- 

Board guide for bicycle route

- 

Bicycle route

-

- 

Distance sign for bicycle route

- 

Motorway

- 

Semi-motorway / Road for motor vehicles

- 

Toll road

- 

  Parking

-

- 

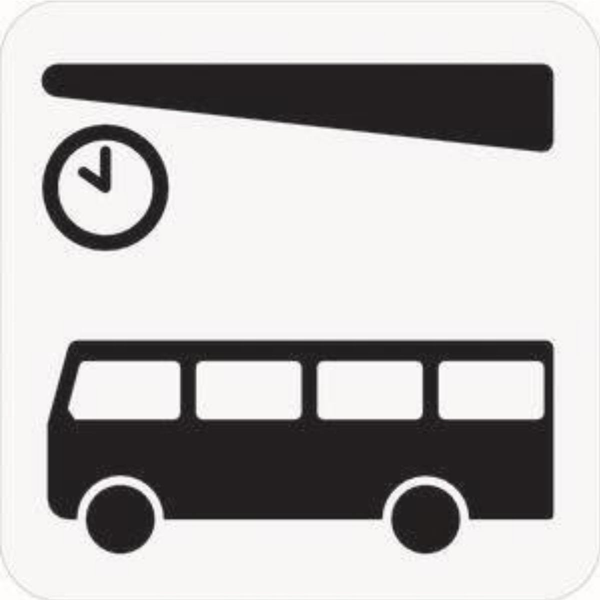
  Parking garage

- 

  Airport

- 

  Helipad

- 

  Bus station/terminal

- 

  Train station/terminal

- 

Ferry

- 

Chains

- 

Church

- 

Industrial area

- 

  Swimming hall

- 

  Alpine facility

-

- 

  Ski jumping hill

- 

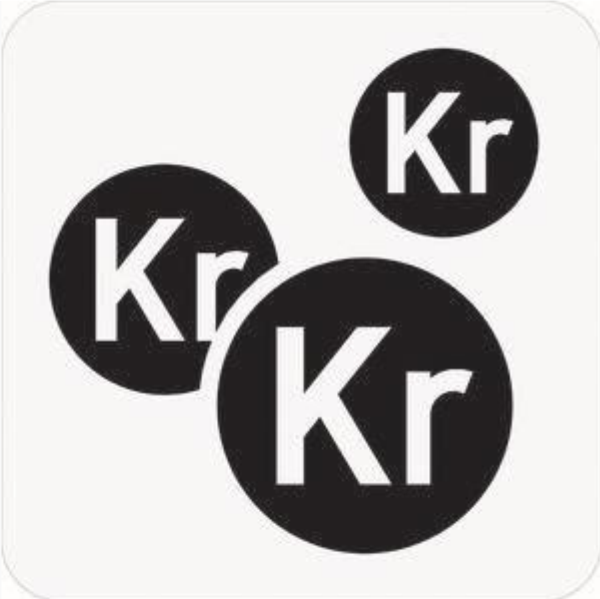
  Ski arena

- 

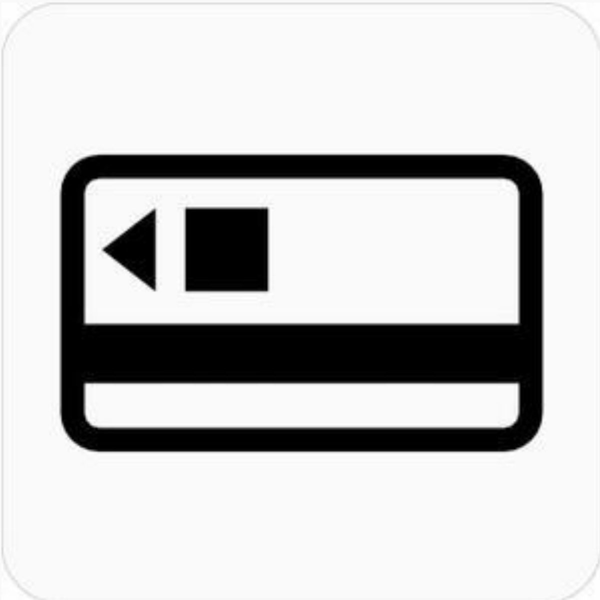
Payment with electronic toll tag

- 

Payment to attendant

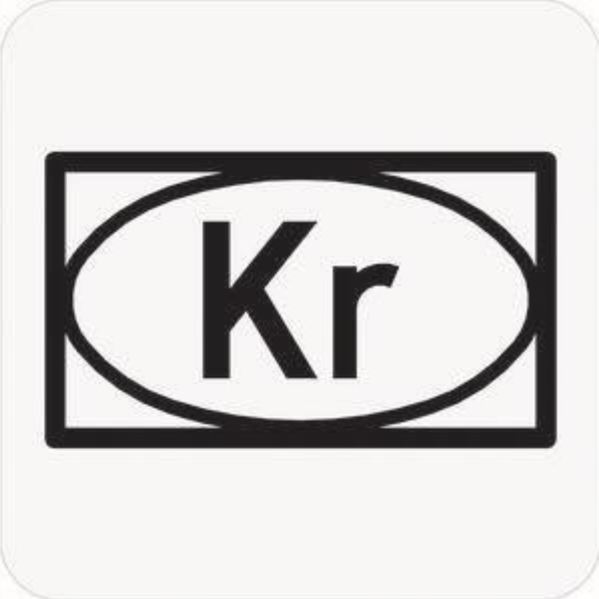
- 

Payment with coins to machine

- 

Payment with card to machine

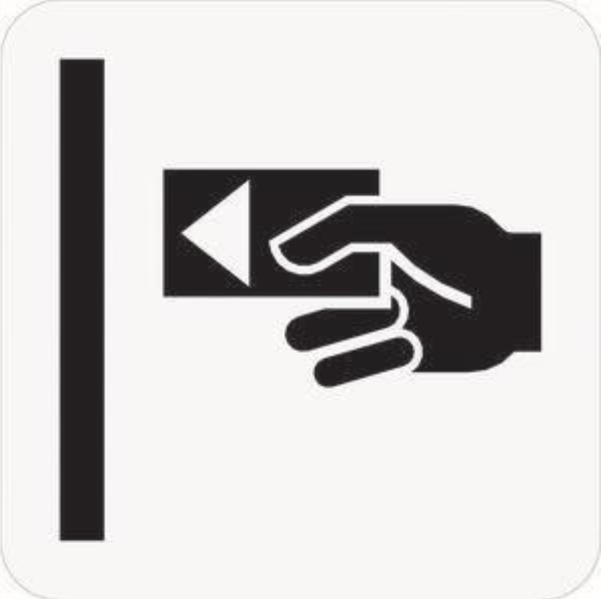
-

- 

Payment with banknotes to machine

- 

Remove ticket from closed payment system

- 

Use ticket in closed payment system

- 

Automatic toll road, no stopping required

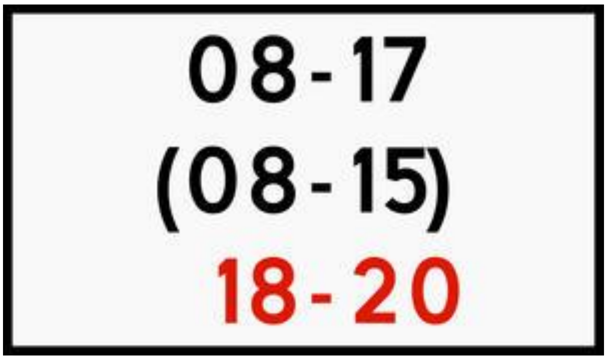
Supplementary signs

- 

Distance

- 

Stretch

- 

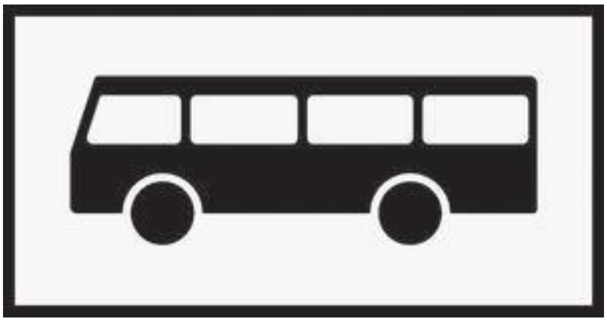
Time

- 

Car

- 

Lorry

- 

Bus

- 

Road train

- 

Trailer furnished for camping

- 

Bicycle

- 

Motorcycle and moped

- 

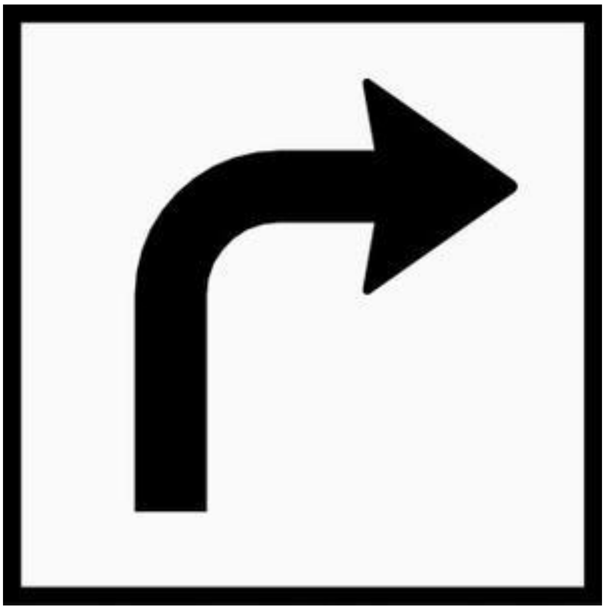
Driver with disabilities

- 

Vehicle furnished for camping

-
- 

Text

- 

Turn arrow

- 

Recommended speed

- 

Degree of incline

- 

Degree of incline

- 

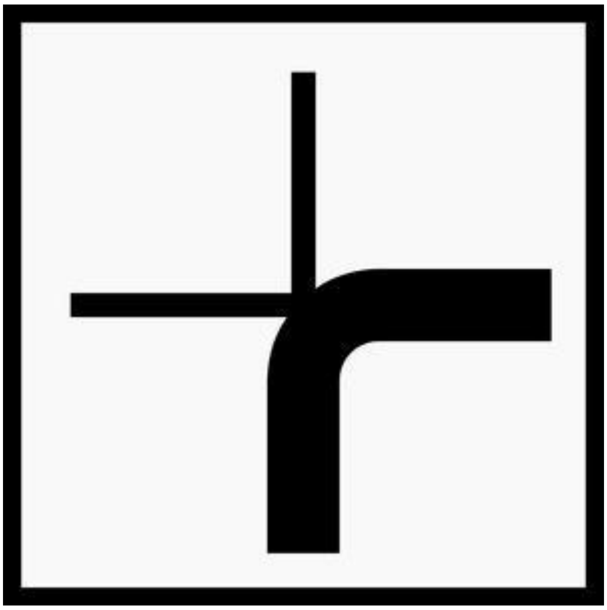
Actual available width

-
- 

Crossing lumber transport

- 

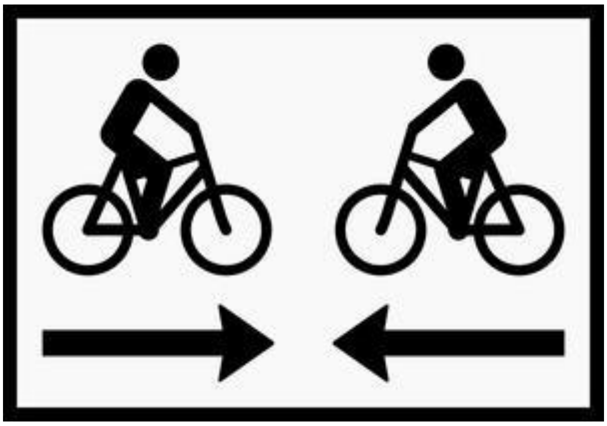
Particular danger of accidents

- 

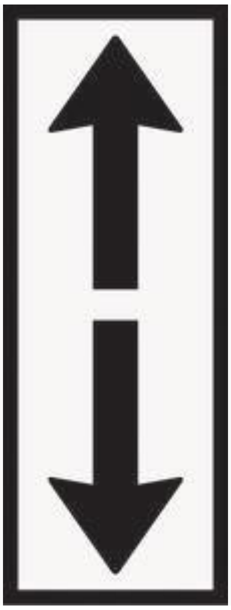
Course of priority road

- 

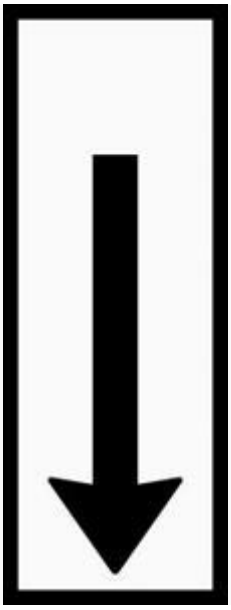
Advance warning of stop sign

- 

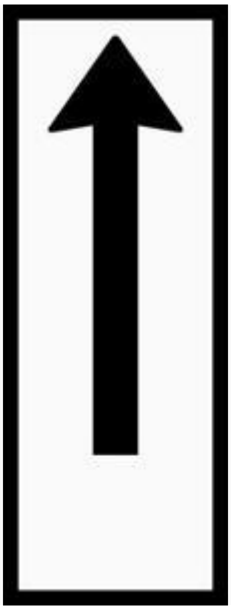
Bicycle traffic in both directions

- 

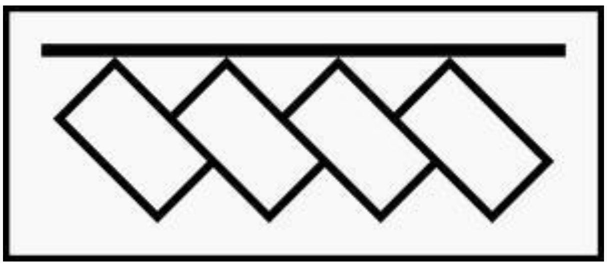
Stopping or parking regulation sign applies in both directions



Stopping or parking regulation sign applies in the direction of the arrow



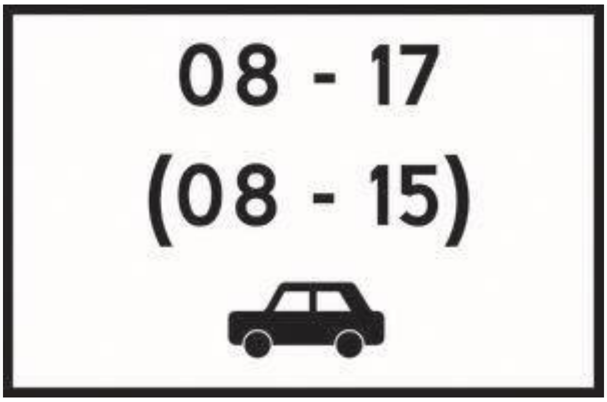
Stopping or parking regulation sign applies in the direction of the arrow



Arrangement for parked vehicles

-
- 

Parking board

- 

Combined regulation

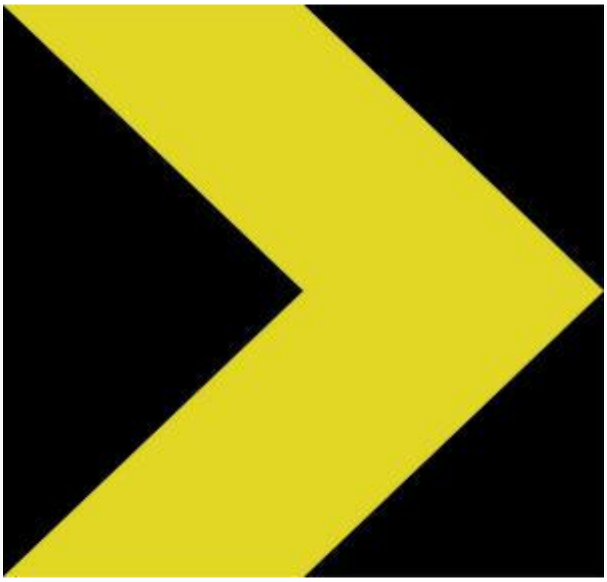
Marker signs

- 

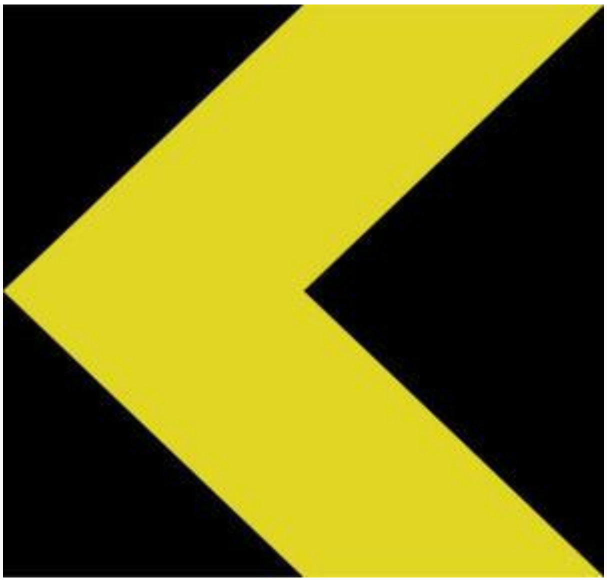
Background marker

- 

Background marker

- 

Directional marker



Directional marker

Obstacle marker — pass on the left

- 

Obstacle marker — pass on the right

- 

Obstacle marker — pass on either side

- 

Obstacle marker

Lanes diverge either side of the marker

Tunnel marker

- 

Tunnel marker

- 

Distance marker for tunnels

-

Edge marker post

Edge marker post

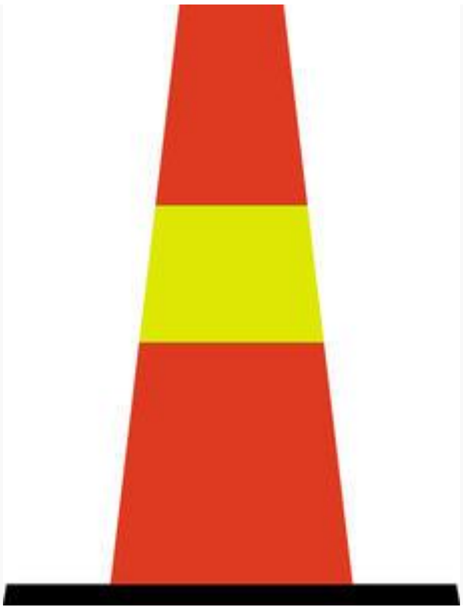
- 

  Edge marker post

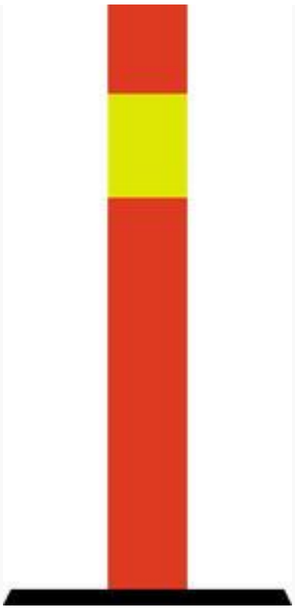
- 

  Barrier marker

-

- 

Traffic cone

- 

Traffic cylinder